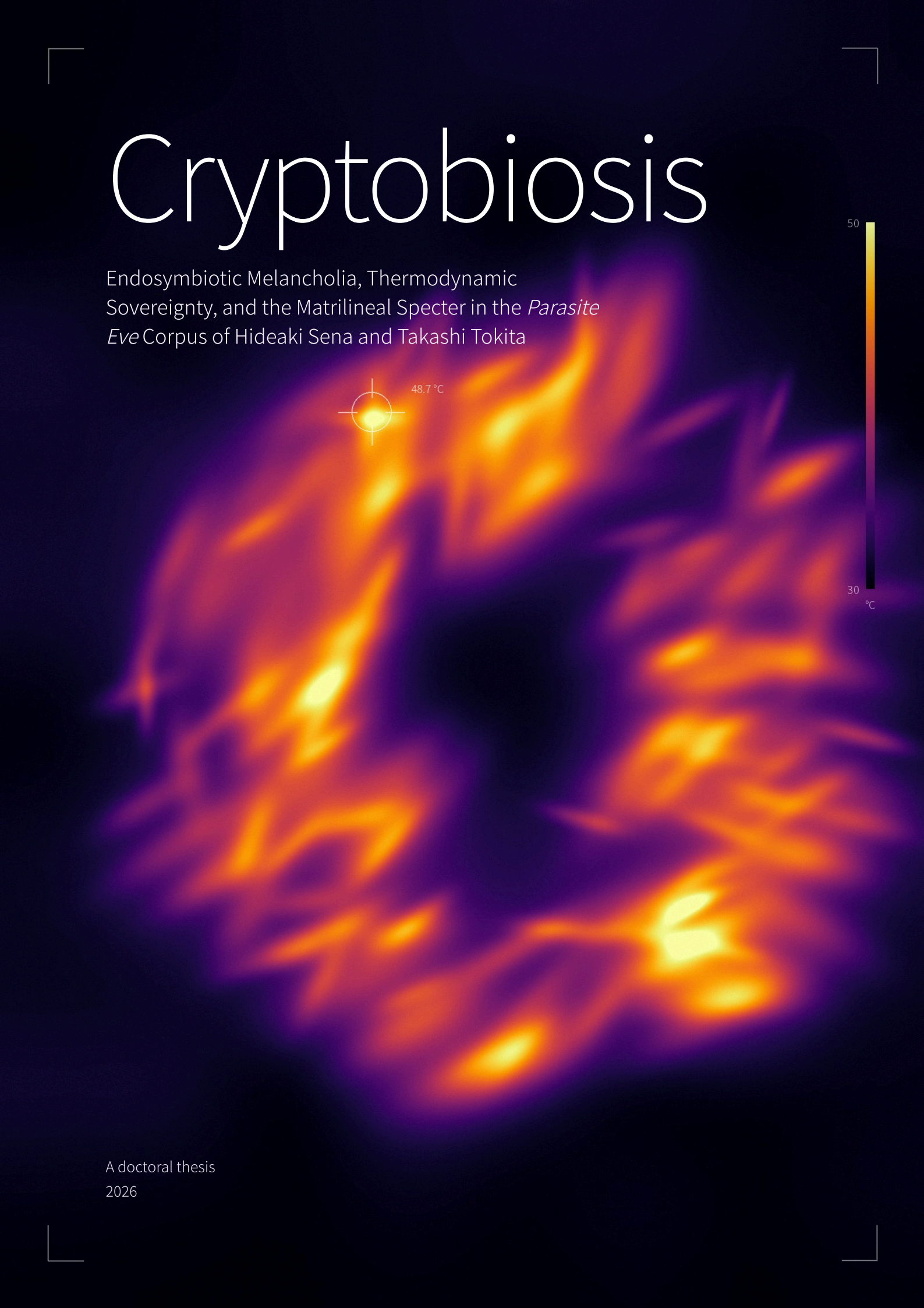
Cryptobiosis

Endosymbiotic Melancholia, Thermodynamic Sovereignty, and the Matrilineal Specter in the *Parasite Eve* Corpus of Hideaki Sena and Takashi Tokita



A doctoral thesis

2026

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And Adam called his wife's name Eve; because she was the mother of all living.

Genesis 3:20 (Authorized Version)

La respiration n'est donc qu'une combustion lente de carbone et d'hydrogène…

Antoine-Laurent Lavoisier and Armand Séguin, “Premier mémoire sur la respiration des animaux” (1789)

Unser Leib ist ja nur ein Gesellschaftsbau vieler Seelen.

Friedrich Nietzsche, *Jenseits von Gut und Böse*, § 19 (1886)

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Abstract

This dissertation argues that the *Parasite Eve* corpus—Hideaki Sena's novel *Parasaito Ivu* (1995) and the PlayStation game *Parasite Eve* (Squaresoft, 1998), directed by Takashi Tokita—constitutes, beneath its generic surface of biohorror, a rigorous speculative ontology of the composite living subject. Its antagonist, a mitochondrial will that names itself Eve, is not an invader but the oldest ally of complex life: the descendant of an alpha-proteobacterium engulfed, roughly two billion years ago, by an archaeal host whose failure to digest it founded the eukaryotic cell. The corpus therefore estranges not the future but the deep past. Its novum is what I term a *palaeonovum*: a speculative premise whose shock derives from the return of the archaic rather than the arrival of the new.

Reading the corpus through three convergent genealogies—the history of endosymbiotic theory from Richard Altmann and Konstantin Mereschkowski to Lynn Margulis; the Hegelian, Marxian, Nietzschean, and Bataillean philosophies of recognition, labor, ressentiment, and expenditure; and the psychoanalytic theory of mourning elaborated by Freud, Sándor Ferenczi, and above all Nicolas Abraham and Maria Torok—I propose the concept of *cryptobiosis*. The biological term designates life suspended below the threshold of detectability. I repurpose it to name the condition of a living subject constituted around an other it has swallowed without assimilating, set to work without recognizing, and kept alive without mourning. The eukaryotic cell is the archetype of this condition: a phagocytosis that never became digestion, an *incorporation* in Abraham and Torok's technical sense rather than an *introjection*, a crypt lodged in the cytoplasm whose tenant retains its own genome, its own double membrane, and—as cell biology demonstrated in the year after the novel's publication—its own prerogative over the death of the cell.

The dissertation shows that the corpus replicates this cellular crypt at every scale it touches, a structure I call the *fractal crypt*: in Toshiaki Nagashima's pathological mourning for a wife whose cells he cultures rather than relinquish; in transplanted organs that carry the dead into the bodies of the living; in the patrilineal household that absorbs the bride as it absorbs her labor; in the transposition of a Japanese novel into an American-set game; and, in the extension

proposed here, in the planetary combustion of the fossil dead. It argues further that the corpus sets two regimes of power against one another within a single cell: an *informational sovereignty* vested in the nuclear genome, and a *thermodynamic sovereignty* vested in the organelle that disposes of energy, and therefore of heat, and that retains the archaic sovereign right to make die. Eve's insurrection—staged as spontaneous combustion, as the dissolution of bodies into undifferentiated protoplasm, and as a eugenic project to engender an "ultimate being"—is read as the coup of the second regime against the first, and as the return of history, labor, and fire into a 1990s imaginary that had declared all three exhausted.

The argument proceeds through a history of the symbiotic idea and its suppression; a dialectical reading of the mitochondrion as bondsman, proletarian, and bearer of ressentiment; an energetics of combustion and sovereign expenditure; a biopolitical analysis situated in Japan's contemporaneous controversies over brain death, transplantation, and eugenic law; a psychoanalytic and theological reading of the corpus's matrilineal hauntology; and a ludological chapter that reads the game's Active Time Battle and Parasite Energy systems as a procedural rhetoric of metabolism. A comparative chapter places the corpus within a canon of speculative and philosophical fiction running from Mary Shelley, Karel Čapek, and H. P. Lovecraft through Yumeno Kyūsaku, Madeleine L'Engle, J. G. Ballard, Stanisław Lem, Greg Bear, Octavia E. Butler, and Suzuki Kōji to *Neon Genesis Evangelion* and Peter Watts. The dissertation concludes by formulating an ethics and politics of *symbiotic mourning*—the passage from incorporation to introjection, in which the composite self acknowledges the other it harbors without either expelling or obeying it—identifies the endings of novel and game as imperfect enactments of that passage, and names the political form it implies *symbiocracy*.

Note on Corpus, Editions, and Conventions

The corpus is treated as a transmedial diptych. Its first panel is Hideaki Sena's *Parasaito Ivu* (Tokyo: Kadokawa Shoten, 1995), awarded the grand prize of the second Japan Horror Novel Award while its author was a doctoral candidate in pharmaceutical sciences at Tohoku University; it was reissued in Kadokawa Horror Bunko (1996) and in a revised Shinchō Bunko edition (2007), and translated into English by Tyran Grillo (New York: Vertical, 2005). Its second panel is *Parasite Eve* (Square, Japan, 29 March 1998; North America, September 1998), directed by Takashi Tokita, who also wrote its story, produced by Hironobu Sakaguchi, with character design by Tetsuya Nomura and a score by Yoko Shimomura. The game is a sequel set in Manhattan between 24 and 29 December 1997; it was Square's first major Japanese–American co-development and its first title to receive a Mature rating. Sena had no direct role in its making, though he later expressed approval of it. The diptych's unity is therefore not authorial but problematic: two works by two makers, in two media and two national settings, organized around a single speculative premise and a single structure of loss.

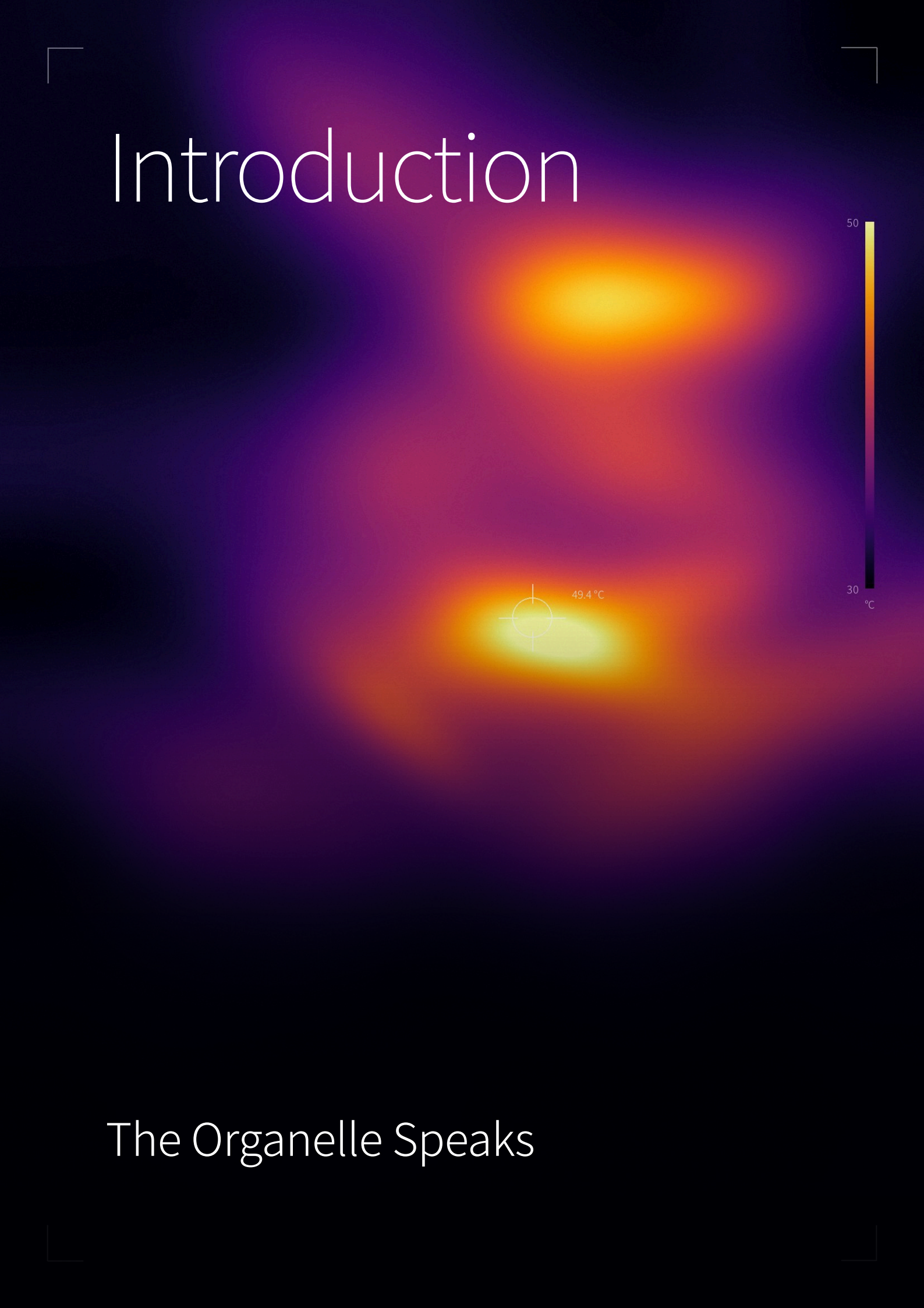
Paratexts are cited where they illuminate the diptych: Masayuki Ochiai's film adaptation (1997, screenplay by Ryōichi Kimizuka), whose ending departs from the novel's; *Parasite Eve II* (Square, 1999/2000; dir. Kenichi Iwao); and *The 3rd Birthday* (Square Enix, 2010/2011; dir. Hajime Tabata). Because the novel's pagination differs across its Japanese editions and its English translation, it is cited by narrative episode. The game is cited by its six chapter titles—Day 1, “Resonance”; Day 2, “Fusion”; Day 3, “Selection”; Day 4, “Conception”; Day 5, “Evolution”; Day 6, “Liberation”—and by its post-game “EX” scenario in the Chrysler Building.

Three homonyms must be held apart throughout. *Mitochondrial Eve* denotes the phylogenetic construct of Rebecca Cann, Mark Stoneking, and Allan Wilson (1987): the most recent common matrilineal ancestor of living humans, a statistical coalescence

rather than a person. *Eve 1* denotes the cell line Toshiaki Nagashima derives from his wife's liver. *Eve*, unmodified, denotes the mitochondrial agency that speaks through both texts and that the game further splits into Melissa Pearce's transformed body and an "original Eve" lodged in the body of Aya Brea's dead sister. That the name proliferates is itself symptomatic, and Chapter Five treats the proliferation as evidence.

Japanese names of scholars writing principally in Japanese appear in Japanese order, family name first, with macrons (Morioka Masahiro, Suzuki Kōji). Sena, Tokita, and the game's staff appear in the Western order of their international credits. References follow an author–date system keyed to the Works Cited. All translations from French, German, and Japanese not otherwise attributed are my own; quotations from copyrighted modern sources are kept minimal, and their arguments are paraphrased.

Introduction



The Organelle Speaks

Two Scenes of Address

Midway through Hideaki Sena's novel, an organelle addresses a scientific meeting. Having colonized the body of Sachiko Asakura, a young researcher in Toshiaki Nagashima's laboratory, the mitochondrial agency that calls itself Eve interrupts the ritual of the conference presentation—the genre through which modern biology speaks *about* its objects—and speaks instead *as* the object, announcing to an audience of specialists that its kind will no longer serve. Three years later, at the opening of Takashi Tokita's game, the scene of address migrates from the lecture hall to the concert hall. On Christmas Eve 1997, at Carnegie Hall, the soprano Melissa Pearce sings the role of Eva, a woman condemned as a witch, whose lover resolves to die beside her at the stake rather than obey his father's prohibition. When her gaze meets that of an off-duty police officer in the stalls, the performers and then the spectators ignite. The organelle has found a second organ of voice. Science and art, the lecture and the aria—the two institutions through which modernity represents life to itself—are seized as instruments by the very life they had represented.

The two scenes pose, in the idiom of horror, the question Gayatri Chakravorty Spivak put to the colonial archive: whether the subaltern can speak, and whether what it says can be heard outside the mouths that ventriloquize it (Spivak 1988). The mitochondrion is the subaltern of the eukaryotic cell in a sense this dissertation will make precise. It is the descendant of a free-living bacterium whose genome was largely transferred to its host's nucleus; its labor supplies nearly all the energy on which the host's projects depend; and its voice, when it comes, can only be borrowed. Eve never speaks except through a woman's body—Kiyomi's, Asakura's, Mariko's, Melissa's, Maya's. The problem of the corpus is therefore at once biological, political, and rhetorical: what would it mean for the constitutive other of the self to represent itself, in both of the senses Spivak distinguishes, *Vertretung* and *Darstellung*, political proxy and aesthetic portrait?

The Oldest Ally

Sena's premise inverts the ordinary grammar of biological horror. The genre's monsters conventionally arrive from outside—from Mars, from deep space, from the laboratory of a foreign power. H. G. Wells's Martians, famously, were undone by terrestrial bacteria against which their bodies had no defense, so that the microbe entered modern speculative fiction as the planet's anonymous ally. Sena reverses the valence. His antagonist is the most intimate ally complex life possesses. Mitochondria descend from an alpha-proteobacterium incorporated by an archaeal host, on current estimates between one and a half and two billion years ago, in the event that produced the eukaryotic cell and, with it, every animal, plant, and fungus. They retain a vestigial genome—in humans a circular molecule of 16,569 base pairs encoding thirteen proteins, twenty-two transfer RNAs, and two ribosomal RNAs (Anderson et al. 1981)—a double membrane, and division by fission. They are transmitted, with vanishingly rare exceptions, through the maternal line alone. They produce the adenosine triphosphate on which cellular work depends, and, as the biochemistry of the mid-1990s established, they license the cell's programmed death (Newmeyer, Farschon, and Reed 1994; Liu et al. 1996).

The game's marketing condensed the premise into a single line: the worst foe, it announced, lies within the self. The formulation is too weak. Eve is not within the self as a tumor is within a body. She is within the self as a founding partner is within a firm, as a laboring class is within a society, as the dead are within the living. The horror of the corpus is not the horror of invasion but the horror of *constitution*: the discovery that the self is a settlement, and that settlements can be contested.

The Palaeonovum

Darko Suvin defined science fiction as the literature of cognitive estrangement, whose hegemonic device is the *novum*, a novelty validated by cognitive logic (Suvin 1979). The novum's temporal vector is ordinarily prospective: it names what has not yet been. The *Parasite Eve* corpus estranges by the opposite motion. Its premise is not the newest thing but the oldest; its cognitive validation comes from the deep past, and its characteristic affect is not wonder at the unprecedented but dread at the unburied. I call this device the

palaeonovum: a speculative premise whose estranging power derives from the return of the primordial. The palaeonovum binds science fiction to Freud's uncanny, the *Unheimliche* that frightens precisely because it was once *heimisch*, familiar and domestic, and has returned from repression (Freud 1919). Nothing is more *heimisch* than the organelle that has warmed every human body that has ever lived. The corpus is thus a science fiction of the archaic—a genre whose futures are buried in the cytoplasm.

IV The Conjuncture of 1995

The novel belongs to a year that Japanese social thought has treated as a hinge. On 17 January 1995 the Great Hanshin–Awaji earthquake killed 6,434 people. On 20 March members of Aum Shinrikyō, a religious movement that had recruited chemists and physicians, released sarin in the Tokyo subway, killing more than a dozen people and injuring thousands; the image of trained scientists placing laboratory competence in the service of apocalypse haunted the decade. The sociologist Ōsawa Masachi read Aum as the terminus of what he called the "age of fiction," the period in which postwar Japan had organized its desires around consumable fantasies rather than ideals (Ōsawa 1996). The economy, meanwhile, had entered the stagnation later named the Lost Decade.

The same year saw the premiere of *Neon Genesis Evangelion*, whose apocalypse dissolves human bodies into a primordial fluid, and the publication of Suzuki Kōji's *Rasen*, in which a virus recombines with a curse to be reborn through a woman's womb. In Europe and North America it saw the publication of Giorgio Agamben's *Homo Sacer*, with its analysis of the brain-dead body as bare life; of Evelyn Fox Keller's *Refiguring Life*, a critique of the gene as master molecule; and of Lynn Margulis and Dorion Sagan's *What Is Life?*, a manifesto for symbiogenesis. It was also the year in which the first complete genome of a free-living organism was published—that of a bacterium, *Haemophilus influenzae* (Fleischmann et al. 1995). In the year information technology first read a bacterium entire, a Japanese novel imagined the bacterium within us reading back. In Japan, finally, 1995 fell in the middle of a decade-long legislative struggle over the definition of death, resolved only by the Organ Transplant Law of 1997, and in the last full year of the Eugenic Protection Law, whose eugenic provisions were excised in 1996. *Parasite Eve* is the precipitate of this conjuncture. Its fictions of

combustion, transplantation, and engineered birth are not metaphors floating free of history; they are a historically specific imagination working through the materials history placed within its reach.

V State of the Question

The novel has been read chiefly as a founding text of Japanese biohorror (*baio-horaa*) and as a product of Kadokawa's media-mix strategy, alongside Suzuki's Ring cycle (Kalat 2007; Steinberg 2012); Sena's own account of its genesis has been recorded in interview (Lynch 2001). The game has been situated within the history of Square's cinematic ambitions and within the broader scholarship on survival horror (Perron 2009, 2018). These are necessary contexts. What they have rarely undertaken is to read the diptych as philosophy: as a sustained speculation on the ontology of the composite subject, the politics of intracellular labor, and the ethics of mourning, conducted in dialogue—explicit or latent—with the thinkers whose ideas it metabolizes. This dissertation attempts that reading, and in doing so proposes that a work of popular genre fiction can function as what Fredric Jameson called a symbolic act: an imaginary resolution of contradictions that cannot be resolved in the real (Jameson 1981).

VI Method

Four protocols govern the analysis. The first is *speculative genealogy*: each idea the corpus deploys is traced to its originators—Altmann, Mereschkowski, Margulis; Hegel, Marx, Nietzsche, Bataille; Freud, Ferenczi, Abraham and Torok—not to reduce the fiction to its sources but to measure what it does to them. The second is *symptomatic reading*, attentive to what the texts say without stating, in the tradition of Althusser and Jameson. The third is *procedural reading*: the game is read not only for its story but for the argument its rules make, in Ian Bogost's sense of procedural rhetoric (Bogost 2007). The fourth is *thick comparison*, which places the corpus within a transnational canon of speculative and philosophical fiction and asks what it alone contributes.

A caution attends every step. Biological metaphors of the social, and social metaphors of the biological, carry a history of naturalization that runs from Herbert Spencer's social organism

through the racial hygiene of the early twentieth century; Keller, Donna Haraway, and Roberto Esposito have shown how metaphors of mastery, defense, and purity travel between laboratory and polity (Keller 1995; Haraway 1991; Esposito 2002). This dissertation uses such figures deliberately, but as speculative instruments rather than natural laws. When it calls the mitochondrion a proletarian or the nucleus a sovereign, it is not asserting that cells have classes; it is asserting that the corpus uses the cell to think about class and sovereignty, and that the thinking is worth reconstructing.

VII Argument

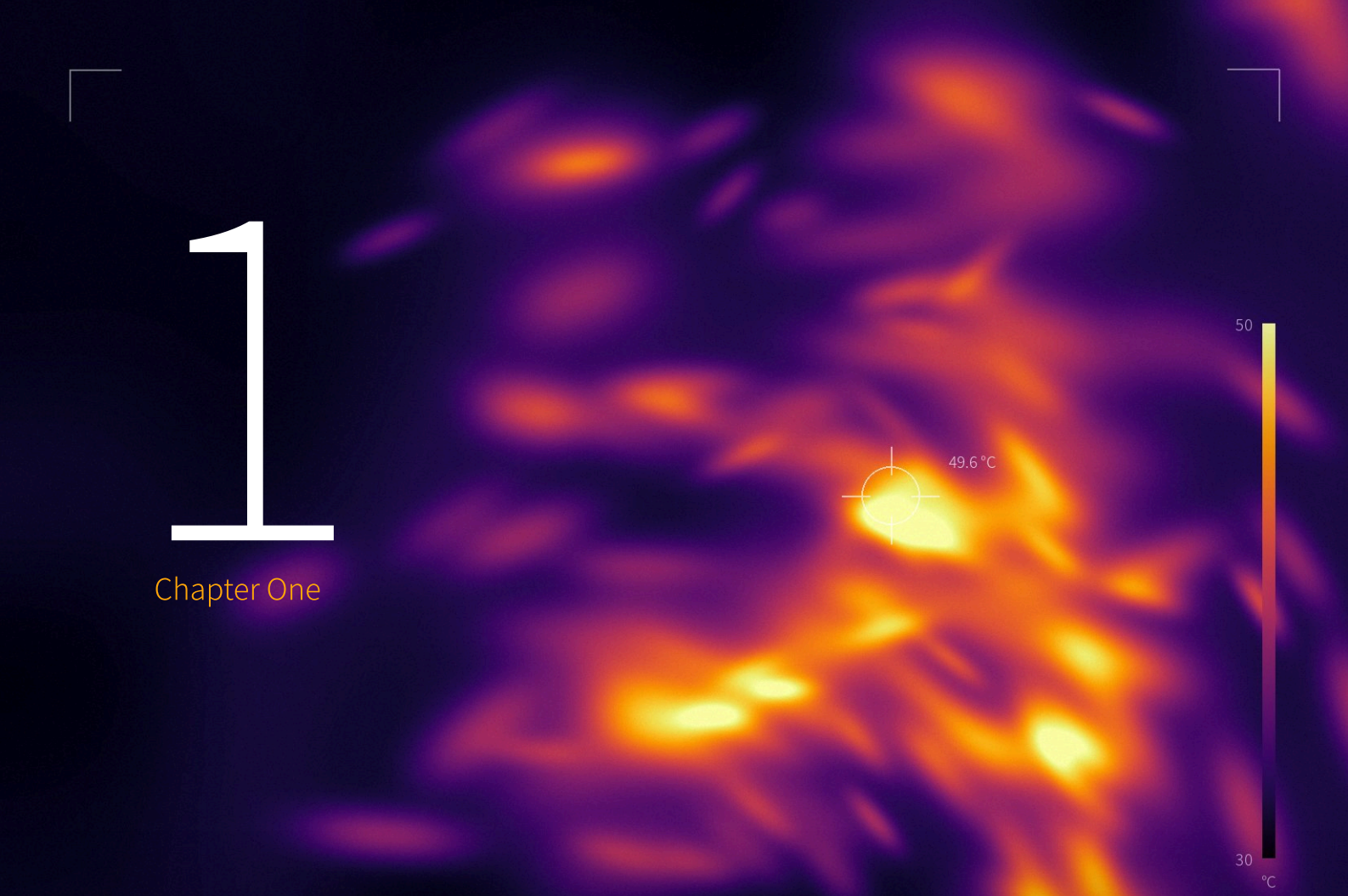
The dissertation's central claim is that the *Parasite Eve* corpus articulates an ontology I call *cryptobiosis*. In biology the term designates a state of life suspended below the threshold of measurable metabolism, as in the desiccated tardigrade. I repurpose it, with deliberate attention to its Greek root, to designate *life in the crypt*: the condition of a living subject constituted around an other it has swallowed without assimilating, set to work without recognizing, and kept alive without mourning. The concept is built from the conjunction of Lynn Margulis's symbiogenesis with the psychoanalytic theory of Nicolas Abraham and Maria Torok, who distinguished *introjection*, the assimilative work of mourning, from *incorporation*, the magical swallowing of a lost object that is then sealed, alive, inside the ego in a psychic crypt (Abraham and Torok 1972, 1978). The eukaryotic cell, I argue, is incorporation's biological archetype: a phagocytosis that never became digestion.

From this claim four theses follow. First, the corpus replicates the cellular crypt at every scale it touches—molecular, organellar, psychic, medical, familial, cultural, and, by extension, planetary—a recursive structure I call the *fractal crypt*. Second, the corpus opposes within the cell two regimes of power: an *informational sovereignty* vested in the nuclear genome and exercised as the regulation of life, and a *thermodynamic sovereignty* vested in the organelle that disposes of energy, and hence of heat, and that retains the archaic right to make die. Eve's insurrection is the coup of the second against the first. Third, the diptych stages across its two panels the passage from melancholia to mourning: the novel ends in the terminal logic of incorporation, with the mourner fusing himself to the monstrous issue of his crypt and dying with it, while the game's two endings—one in which Aya Brea communicates with the

mitochondria of an entire audience, one in which her dead sister's phantom speaks, expels the usurping Eve, and departs—rehearse the possibility of introjection. Fourth, the ethical and political horizon the corpus opens, which I name *symbiotic mourning* and, in its collective form, *symbiocracy*, is neither the restoration of nuclear sovereignty nor the triumph of the mitochondrial coup, but a constitution of acknowledged others.

VIII Chapter Map

Chapter One reconstructs the genealogy of the symbiotic idea from Richard Altmann's bioblasts to Margulis, the theory of intragenomic conflict, and the nineteenth-century doctrine of organic memory that reaches Sena through Yumeno Kyūsaku. Chapter Two reads the mitochondrion through Hegel's dialectic of lordship and bondage, Marx's account of primitive accumulation, and Nietzsche's physiology of ressentiment. Chapter Three turns from labor to energy, reading Eve's pyrokinesis through Lavoisier's identification of respiration with combustion, Peter Mitchell's chemiosmotic theory, and Georges Bataille's general economy. Chapter Four situates the corpus within Japan's controversies over brain death and transplantation and argues that the eukaryotic cell is a mixed constitution in which archaic sovereignty is encrypted within modern biopower. Chapter Five develops the theory of the crypt, the matrilineal phantom, and the inverted Genesis. Chapter Six reads Tokita's game as a xenograft of the novel and its combat systems as a procedural rhetoric of metabolism. Chapter Seven constructs a comparative canon. Chapter Eight assembles the concepts into a cryptobiology and extends it to the combustion of the fossil dead.



1

Chapter One

The Palaeonovum: A Genealogy of the Symbiotic Idea

It has, I believe, been often remarked that a hen is only an egg's way of making another egg.

Samuel Butler, *Life and Habit* (1878)

The premise on which the corpus is built—that the organelles of the animal cell descend from bacteria that were once free—spent most of the twentieth century as a heresy, and the sequence of its history (conjecture, derision, suppression, molecular vindication, and a sudden reinterpretation of the vindicated organelle as an agent of death) is the sequence Sena's plot reenacts. This chapter reconstructs that history in order to specify the palaeonovum as a historical formation rather than a formal device alone. It argues that the corpus could estrange the archaic only because, by the mid-1990s, the archaic had become newly legible: sequenced, dated, named, and endowed by cell biology with powers over life and death that no earlier moment had ascribed to it. Four strands are traced—the symbiotic idea proper, the gendered politics of nucleus and cytoplasm, the theory of intragenomic conflict, and the nineteenth-century doctrine of organic memory—before the chapter closes in the laboratory where Sena, a doctoral researcher in mitochondrial pharmacology, gave the strands a body.

1.1 Naming the Ancestors

In 2015 a team working on sediments recovered near Loki's Castle, a hydrothermal vent field on the Arctic Mid-Ocean Ridge between Greenland and Norway, reconstructed from environmental DNA an archaeal lineage whose genomes encode proteins previously thought exclusive to eukaryotes (Spang et al. 2015). They named it Lokiarchaeota. Its relatives, assembled into a superphylum called Asgard, received the names of Thor, Odin, and Heimdall (Zaremba-Niedzwiedzka et al. 2017). In 2020 a Japanese team led by Imachi Hiroyuki reported the first cultured member of the group, grown over more than a decade from deep-sea sediment collected off the Japanese coast: a minute, slow-dividing archaeon that lives only in partnership with other microbes and extends long, branching protrusions toward them. They named it *Prometheoarchaeum syntrophicum* and proposed that the eukaryotic cell arose through

an “entangle–engulf–endogenize” sequence in which such protrusions gradually enveloped a bacterial partner until it was inside (Imachi et al. 2020).

Nomenclature is not incidental here. The contemporary science of eukaryogenesis names the host of the founding incorporation after the trickster of the Norse pantheon and the Titan who stole fire for humankind, and it names the most recent common matrilineal ancestor of living humans after the mother of all living. Science christens its deepest ancestors through myths of theft, transgression, and fall. The *Parasite Eve* corpus returns this mythopoesis to fiction and reverses its direction: where the names domesticate deep time by assimilating it to familiar stories, the fiction makes the familiar stories uncanny by restoring to them the agency the names imply. The circle closes with a local irony. Twenty-five years after a Japanese novel imagined a liver culture that grew with preternatural speed and acquired a will, a Japanese laboratory cultured, with extreme patience, the nearest living analogue of the host that first swallowed the organelle. Fiction's Eve 1 and science's *Prometheoarchaeum* stand at the two ends of one experimental imagination: the one too fast, the other almost too slow to see.

1.2 Bioblasts, Chromatophores, Symbiotes

The autonomy of the organelle was proposed before the organelle had a stable name. In 1890 the Leipzig histologist Richard Altmann described granules, visible in nearly all cells after staining with acid fuchsin, and argued that they were *Elementarorganismen*, elementary organisms, which he called bioblasts; the cell, on this view, was a colony of bioblasts in a common matrix (Altmann 1890). Eight years later Carl Benda gave the same granules the name that stuck, *mitochondria*, from the Greek for thread and grain (Benda 1898). The renaming is instructive. Altmann's word asserted that the granules were alive; Benda's described what they looked like. The organelle entered the nomenclature of cytology through a gesture of morphological description that silently retired the question of its autonomy, and in the mainstream the question stayed retired for seventy years.

It was pursued at the margins. Andreas Schimper had observed in 1883 that chloroplasts divide independently of the cell and suggested, in a footnote, that green plants might have originated in

the union of a colorless organism with one tinged by chlorophyll (Schimper 1883). Konstantin Mereschkowski extended the suggestion into a theory of chromatophores as captured cyanobacteria (Mereschkowski 1905) and then into a general account of the cell as the product of two plasms, one bacterial and one amoeboid, for which he coined the term *symbiogenesis* (Mereschkowski 1910). Jan Sapp, Francisco Carrapiço, and Mikhail Zolotonosov have documented what else the founder of symbiogenesis was: a serial rapist of young girls. He left St. Petersburg in 1886, and the Crimea in 1898, under threat of prosecution for raping girls; in 1914 he was charged with raping twenty-six girls between 1905 and 1914, the youngest seven years old, and fled Russia; he died in Geneva in 1921. His utopian fiction *Zemnoi rai* (Earthly Paradise, 1903), set in the twenty-seventh century, is in their reading a eugenic utopia in which a new race of humans is bred to his pedophilic ideal (Sapp, Carrapiço, and Zolotonosov 2002). Sapp and his co-authors caution that a scientist's theory of nature does not dictate his prescriptions for society, and the claim here is not that symbiogenesis entails eugenics. It is that the symbiotic idea and the eugenic dream share a birth certificate, that the document also records crimes against children, and that the corpus's conjunction of the two—an endosymbiont whose program is the engineering of a perfected being, carried out, as Chapter Five shows, through rape—restores a union that the history of science had preferred to forget. In Russia Boris Kozo-Polyansky systematized symbiogenesis as a new principle of biology (Kozo-Polyansky 1924; trans. 2010). In Paris Paul Portier claimed in *Les Symbiotes* that mitochondria were bacteria that could be cultured outside the cell (Portier 1918), and was derided. In Colorado Ivan Wallin made a similar claim, proposed *symbionticism* as a mechanism of speciation (Wallin 1927), and abandoned the line of research when it found no audience.

Sapp explains the suppression as a struggle for authority within genetics (Sapp 1987, 1994). The ascendancy of the Morgan school established what he calls a nuclear monopoly: heredity was located in chromosomes, and claims for inheritance through the cytoplasm —Carl Correns's and Erwin Baur's non-Mendelian plastid inheritance of 1909, Fritz von Wettstein's *Plasmon*, Boris Ephrussi's respiratory-deficient “petite” yeast (Ephrussi 1953), Tracy Sonneborn's “kappa” particles in *Paramecium*—were treated as anomalies to be reduced or as residues of Lamarckism. The Lysenko affair made that

suspicion dangerous. In the Soviet Union, where the agronomist Trofim Lysenko's rejection of Mendelian genetics was made official doctrine in 1948 with Stalin's approval, geneticists lost their posts and laboratories, and the most eminent of them, Nikolai Vavilov, arrested in 1940, died in prison in 1943; in the Cold War West, any claim for inheritance outside the nucleus risked association with Lysenkoism. The organelle was therefore not merely neglected; it was subordinated within a disciplinary constitution that assigned sovereignty over heredity to the nucleus and defended that constitution against a political enemy. When Eve announces that her kind will no longer serve, she speaks in fiction the grievance that cytoplasmic inheritance had carried as a scientific minority report for half a century.

1.3 Vindication

The vindication came from the molecule the nuclear monopoly had enthroned. In 1963 Margit and Sylvan Nass observed DNA-like fibers inside mitochondria (Nass and Nass 1963); within a few years the organelle's own genome, its own ribosomes, and its bacterial-type protein synthesis were established. In 1967 Lynn Sagan, later Lynn Margulis, published "On the Origin of Mitosing Cells" after rejection by some fifteen journals, arguing that mitochondria, plastids, and the basal bodies of flagella descend from free-living prokaryotes incorporated by a host in a series of events (Sagan 1967); she developed the serial endosymbiosis theory in two monographs (Margulis 1970, 1981). Its claims about mitochondria and plastids were confirmed by the molecular phylogenetics of Carl Woese and his collaborators, which placed mitochondrial ribosomal RNA within the alpha-proteobacteria (Woese and Fox 1977; Yang et al. 1985) and reorganized the tree of life into three domains (Woese, Kandler, and Wheelis 1990). Its claim about flagella was not confirmed and remains rejected. In 1981 the human mitochondrial genome was sequenced entire (Anderson et al. 1981): the first complete human genome of any kind was the genome of the stranger.

Margulis's vindication carried a rhetoric. Against the neo-Darwinian emphasis on competition she stressed cooperation; her popular writing with Dorion Sagan held that life had spread across the globe less by combat than by forming networks (Margulis and Sagan 1986), and with James Lovelock she extended the symbiotic principle to the regulation of the atmosphere (Lovelock and Margulis

1974). By 1995 symbiogenesis was in the textbooks, and its dominant affect was irenic. The endosymbiont had become a partner; the eukaryotic cell, a successful merger. Sena's premise refuses this reconciliation. It accepts Margulis's science and rejects her tone. The partner, it proposes, has a grievance; the merger was an acquisition; and a settlement two billion years old can still be contested. The corpus reads symbiogenesis against its own irenicism, restoring to the event of incorporation the violence of which its technical name—phagocytosis, cell-eating—still bears the trace.

1.4 Master Molecule and Steady State

In 1957 the geneticist David Nanney distinguished two conceptions of hereditary control: a "master molecule" conception, in which genetic material directs the cell as a program directs a machine, and a "steady state" conception, in which heredity is the property of a self-perpetuating system distributed across nucleus and cytoplasm (Nanney 1957). Evelyn Fox Keller has shown how decisively the first conception prevailed, in the language of instructions, programs, and codes that molecular biology adopted after 1953, and how that language concealed the cytoplasmic conditions on which gene action depends (Keller 1995, 2000). The victory was also gendered. Feminist analyses of cell biology have traced a persistent coding of the nucleus as active, directive, and masculine and of the cytoplasm as passive, nutritive, and feminine (Biology and Gender Study Group 1988), a coding continuous with the textbook romance in which an active sperm penetrates a passive egg (Martin 1991) and, further back, with Aristotle's doctrine that in generation the male contributes form and the female matter.

Mitochondria are inherited through the maternal cytoplasm. Under the master-molecule regime they belong to the order of matter, substrate, and supply. Eve's insurrection is therefore, in the conceptual history reconstructed here, the cytoplasm talking back: the maternal substrate claiming the prerogative of form. The novel makes the claim explicit in the telos it assigns her. Eve does not want to abolish the nucleus's power; she wants it. Her goal is a being able to rewrite its own genetic code at will—a being in which the authority the master-molecule conception had vested in the nucleus is at last commanded from the cytoplasm. The insurrection is thus

not a rejection of informational sovereignty but a bid to seize it, a structure that will prove decisive for the reading of her failure in Chapter Five.

1.5 The Selfish Organelle

A second theoretical lineage converged on the organelle from evolutionary theory. Richard Dawkins's gene's-eye view, anticipated in Butler's epigrammatic hen, recast organisms as vehicles for replicators and made intelligible the existence of "outlaw" genetic elements whose transmission interests diverge from those of the organism that carries them (Dawkins 1976, 1982). William Eberhard and, more systematically, Leda Cosmides and John Tooby applied the reasoning to cytoplasmic genomes: because mitochondria are transmitted through females only, selection on mitochondrial genes is indifferent to male fitness, and conflict between nuclear and cytoplasmic genomes is to be expected (Eberhard 1980; Cosmides and Tooby 1981).

Nature supplies a vivid instance. In cytoplasmic male sterility, documented across many flowering plants, mitochondrial genes abort the male function of hermaphroditic flowers, redirecting resources toward the female function through which alone those genes are transmitted; nuclear "restorer of fertility" genes evolve to suppress them, and the two genomes are locked in an arms race. Here, in the botanical literature, is the corpus's plot in molecular miniature: an organelle waging a war of attrition against the male, and a nuclear resistance that restores him. Laurence Hurst and William Hamilton went further, proposing that the very existence of two mating types or sexes may be explained by the need to enforce uniparental inheritance of organelles and so prevent lethal competition between organelle lineages within a single cell (Hurst and Hamilton 1992). On this hypothesis sexual difference is, among other things, a treaty among organelles. The consequence for males was drawn by Steven Frank and Hurst: because mitochondrial mutations harmful only to males escape selection, maternal transmission imposes on males a burden later named the "mother's curse" (Frank and Hurst 1996; Gemmell, Metcalf, and Allendorf 2004).

These results bear directly on the novel's ending. Eve's plan fails because the sperm she obtains by raping Toshiaki carry paternal mitochondria that do not submit, and the child she engenders oscillates between male and female. Read against Hurst and

Hamilton, the oscillation is not a lurid flourish but a precise speculative inference: where paternal organelles survive and contest the maternal line, the treaty that sexual difference enforces is broken, and sex itself becomes unstable. The father's mitochondria play the role of the nuclear restorer gene. Paternal transmission of mitochondria in humans is exceedingly rare; one well-documented case was reported in 2002 (Schwartz and Vissing 2002), and a later claim of biparental inheritance in several families (Luo et al. 2018) was traced in substantial part to mitochondrial sequences inserted into the nuclear genome (Wei et al. 2020)—the organelle's fossils masquerading, in the nucleus, as the father's return.

1.6 The Organelle of Death

The third strand turned the powerhouse into an executioner. John Kerr, Andrew Wyllie, and Alastair Currie named apoptosis in 1972, borrowing a Greek word for the falling of leaves (Kerr, Wyllie, and Currie 1972), and through the 1980s programmed cell death was understood as a genetic program under nuclear control (Ellis and Horvitz 1986). In the year before the novel's publication, Donald Newmeyer and colleagues showed that apoptosis in cell-free extracts required a fraction enriched in mitochondria (Newmeyer, Farschon, and Reed 1994); in the year after, Xiaodong Wang's laboratory showed that cytochrome c released from mitochondria triggers the protease cascade that dismantles the cell (Liu et al. 1996; Li et al. 1997). Between the publication of the novel and the release of the game, the organelle acquired in cell biology the attribute that political theory reserves for the sovereign: the power to make die.

Other attributions had preceded it. Otto Warburg traced cancer to injured respiration (Warburg 1956); Denham Harman proposed that aging results from free-radical damage and later located its clock in the mitochondrion (Harman 1956, 1972); Douglas Wallace and colleagues identified the first disease caused by a point mutation in mitochondrial DNA, Leber's hereditary optic neuropathy, a disease of the optic nerve (Wallace et al. 1988), while Ian Holt and colleagues found mitochondrial DNA deletions in myopathy (Holt, Harding, and Morgan-Hughes 1988). Nick Lane would later synthesize the literature under the triad of power, sex, and suicide (Lane 2005). The 1990s thus constitute a mitochondrial turn in which the organelle passes, in the scientific imaginary, from supplier to arbiter: of energy, but also of death, aging, malignancy,

and sex. The corpus captures the passage as it occurs. It is written from inside the moment at which the subaltern of the cell was discovered to hold the sword.

1.7 Organic Memory: From Hering to Yumeno

The corpus gives the organelle not only power but memory. Eve remembers her subjection; she has waited. This attribution descends from a nineteenth-century doctrine that Laura Otis has called organic memory: the view that heredity is a form of memory and memory a form of heredity, so that the history of a lineage is inscribed in the bodies of its descendants (Otis 1994). Ewald Hering proposed in an 1870 lecture that memory is a general function of organized matter (Hering 1870); Samuel Butler, who translated Hering, made unconscious memory the engine of heredity and habit (Butler 1878, 1880); Ernst Haeckel attributed memory to the molecular units he called plastidules and souls to cells (Haeckel 1876, 1899); Richard Semon systematized the doctrine as *Mneme* and coined the term *engram* (Semon 1904). Freud's Lamarckian speculations on an archaic heritage—in *Totem und Tabu*, in the phylogenetic fantasy drafted in 1915 and published only in 1987, and in the Moses book—are the doctrine's psychoanalytic afterlife (Freud 1913, 1939, 1987). August Weismann's separation of germ plasm from soma (Weismann 1892) discredited the inheritance of acquired memory. Mitochondrial genetics restored something like organic memory in a non-Lamarckian register: an archive transmitted almost unchanged down the maternal line, legible to sequencing, carrying the record of a matrilineal past. The corpus fuses the discredited doctrine with its molecular successor. Eve is an archive that remembers being an archive.

In Japanese letters the doctrine found its most extravagant expression in Yumeno Kyūsaku's *Dogura magura* (1935). Its psychiatrist Dr. Masaki argues that the brain is not the organ of thought but a telephone exchange relaying the thinking of the body's cells; his “fetal dream” holds that the fetus, in its ten months in the womb, dreams through the entire evolutionary history of its ancestors—Haeckel's recapitulation rendered as nightmare—while a doctrine of psychic heredity allows an ancestor's crime to resurface in a descendant (Yumeno 1935). The claim here is genealogical rather than biographical: whether or not Sena drew on Yumeno, *Dogura magura* is the Japanese modernist precedent for a cellular will, a

memory lodged below consciousness, and a crime transmitted across generations, and its switchboard returns in the telephones through which the game's player saves (Chapter Six). J. G. Ballard's *The Drowned World* (1962) supplies the Anglophone counterpart. As an unstable sun heats the planet and floods its cities, the biologist Bodkin speculates that the body carries in its nervous system and chromosomes the memory of every epoch it has passed through, and the protagonist drifts south into a Triassic past he recognizes from within (Ballard 1962). Ballard's regression is triggered by heat; the corpus's return of the archaic *is* heat. Both are palaeonova.

1.8 The Laboratory Novel and the Epistemic Thing

Sena wrote the novel as a doctoral candidate in pharmaceutical sciences whose research concerned mitochondria and the action of drugs on fatty-acid oxidation, and its texture—protocols, culture media, the choreography of a transplant, the hierarchy of professor, assistant, and student, the ritual of the scientific meeting—is that of a laboratory ethnography written from inside. Bruno Latour and Steve Woolgar described the laboratory as a machine for producing inscriptions whose literary character it must disavow (Latour and Woolgar 1979). Hans-Jörg Rheinberger described experimental systems as generators of surprises, in which *technical objects*, stabilized and instrumental, coexist with *epistemic things*, the unstable objects of inquiry that exceed current knowledge (Rheinberger 1997). Eve 1 begins as a technical object, a cell line of the kind laboratories maintain by the thousand, and becomes an epistemic thing when it behaves inexplicably; the novel's horror is the moment at which the epistemic thing becomes a subject, and the generator of surprises produces a surprise that addresses its experimenter. The laboratory novel's realism is not decoration. It is the condition of the palaeonovum's cognitive validation: the archaic is made plausible by the procedures of the newest science.

The same holds for the construct from which the corpus takes its name. Mitochondrial Eve is a product of method: a coalescence point inferred from the sequences of living people by means of a molecular clock whose theoretical warrant owed much to Kimura Motoo's neutral theory of molecular evolution (Zuckerkandl and Pauling 1965; Kimura 1968; Cann, Stoneking, and Wilson 1987). She was not the only woman alive in her time, nor a founder in any demographic sense; she is the statistical point at which the maternal

lines of the present converge when traced backward. The name, which science journalism adopted and scientists accepted (Lewin 1987), personified the statistic. The corpus literalizes the personification. It takes a figure of speech that science used to popularize a result and endows it with will, voice, and grievance. This is the palaeonovum's decisive operation, and it permits a precise definition: a speculative premise that (i) is drawn from the deep past, (ii) is validated by the most recent science, (iii) literalizes a figure of speech that science itself employed, and (iv) returns a doctrine the discipline had suppressed—here, the autonomy of the organelle—in the form of a subject. If the organelle is a subject, its relation to the self it has served must be reconceived. The next chapter reconceives it as labor.



2

Chapter Two

Dialectic of the Powerhouse: Recognition, Expropriation, Ressentiment

Die Wahrheit des selbständigen Bewußtseins ist demnach das knechtische Bewußtsein.

G. W. F. Hegel, *Phänomenologie des Geistes* (1807)

Wo ich Lebendiges fand, da fand ich Willen zur Macht; und noch im Willen des Dienenden fand ich den Willen, Herr zu sein.

Friedrich Nietzsche, *Also sprach Zarathustra*, II, "Von der Selbst-Überwindung" (1883)

The mitochondrion occupies within the eukaryotic cell the position that three nineteenth-century philosophies assigned to the laboring subject: Hegel's bondsman, whose work forms the world the lord enjoys; Marx's proletarian, produced by the expropriation of the means of its own reproduction; and the servant in whom Nietzsche's Zarathustra detects the will to be master. This chapter reads the corpus's premise through all three and argues that each illuminates a different moment of Eve's insurrection—its demand, its grievance, and its drive—while none accounts for its form, which requires the energetics of Chapter Three. It then situates the insurrection in the Japan of 1995, where the restructuring of labor, the discourse of overwork, and the post-historical self-understanding of a consumer society gave the image of a rebellious powerhouse a specific charge. The claim is that the corpus returns struggle, labor, and history into a decade that had pronounced all three concluded, and returns them at the one scale at which no one had thought to look.

2.1 The Cell-State and the Powerhouse

The political reading of the cell is not imposed on biology from outside; biology wrote it first. Rudolf Virchow's cellular pathology, which made the cell the elementary unit of health and disease, was accompanied by the image of the organism as a cell-state (*Zellenstaat*), a federation of citizen cells; Virchow, a liberal member of the Prussian parliament and an adversary of Bismarck, intended the image as republican (Virchow 1858; Weindling 1981). Haeckel inverted its politics, using the cell-state to naturalize hierarchy, centralization, and the subordination of part to whole, while Henri Milne-Edwards had already imported Adam Smith's division of labor into physiology (Milne-Edwards 1851). Andrew Reynolds has shown that such metaphors were not ornaments but instruments, lenses without which cell biology could not have posed its questions (Reynolds 2018). The twentieth century industrialized the lens. Philip Siekevitz's 1957 article in *Scientific American* fixed the mitochondrion's public identity as the “powerhouse of the cell” (Siekevitz 1957), and cybernetics supplied the complementary image

of the nucleus as the seat of a program, a command center issuing coded instructions (Wiener 1948; Kay 2000). By mid-century the cell had become a Fordist enterprise: conception concentrated in a central office, execution dispersed among plants that supply the power. The metaphors carried a politics of which they were not aware. The corpus becomes aware of it.

2.2 Lordship and Bondage in the Cytoplasm

In the *Phenomenology of Spirit* self-consciousness achieves itself only through recognition by another self-consciousness, and the demand for recognition issues in a struggle to the death (Hegel 1807). The combatant who prefers life to recognition becomes the bondsman (*Knecht*); the one who risked death becomes the lord (*Herr*), who enjoys the thing through the bondsman's labor. The lord's victory is hollow—recognition by a bondsman is not recognition by an equal—while the bondsman, through fear, service, and formative labor (*Bilden*), comes to see his own independence objectified in the world he has made. Hence the epigraph's reversal: the truth of independent consciousness is servile consciousness.

Transposed to eukaryogenesis the dialectic acquires an uncanny literalness. An archaeal host and an alpha-proteobacterium enter the relation of incorporation; the incorporated partner, neither destroyed nor departing, survives in service; its labor produces the adenosine triphosphate the host consumes. The host's independence is illusory. Nearly every eukaryote depends on its mitochondria or their reduced descendants; the known exception, the gut flagellate *Monocercomonoides*, lost the organelle entirely only after acquiring by lateral transfer a bacterial pathway to replace the one essential function the organelle still performed (Karnkowska et al. 2016). The only lord known to have survived the bondsman's departure did so by hiring another. What the bondsman gains through labor, in the cell as in Hegel, is the world: the eukaryotic cell's size, its genomic complexity, and every form of multicellular life built upon it are the objectified labor of the organelle. Eve's discourse, in both texts, is the moment at which the bondsman recognizes itself in the world it has formed and demands that the lord recognize it too.

Alexandre Kojève made this passage the motor of universal history and declared that history ends when the bondsman's demand is universally satisfied in the homogeneous state; what

remains is a post-historical humanity of satisfied animals (Kojève 1947). In a note added after a visit to Japan in 1959 he revised the prognosis: Japan, he suggested, had lived for centuries in a post-historical condition of pure formalism, of values emptied of historical content and maintained as style, and the future of the West might be its Japanization. Azuma Hiroki would later adapt Kojève's post-historical animal to describe the database consumption of otaku culture in the 1990s (Azuma 2001). The corpus's placement within this discourse is exact. It appears in the Japan that Kojève had designated the laboratory of post-history, during the decade in which its consumer culture was being theorized as animalized, and it discovers that the struggle for recognition has not ended but migrated—below society, below the individual, into the cytoplasm. History returns in the organelle.

Axel Honneth's reconstruction of the Hegelian problematic distinguishes three spheres of recognition—love, rights, and social esteem—and derives social struggle from their denial (Honneth 1992). Eve's claims can be sorted accordingly, and in each sphere the claim is pressed by violating the bodies it addresses. In the sphere of love she claims Toshiaki: having inhabited his wife, she takes a body modeled on Kiyomi's and rapes him to obtain his sperm. In the sphere of rights she addresses the assembly of science through Asakura, Toshiaki's student, having forced her way into the young woman's body and taken control of it, in an assault the novel stages sexually. In the sphere of esteem she insists on the contribution the host has never acknowledged: that the life which regards her as an instrument exists because of her. The novel's horror is in part the horror of a misrecognition revealed as structural, and in part the horror of a demand for recognition pursued through violation: love extracted by rape is not love recognized, and a speech delivered through a seized body is not standing granted. The self cannot recognize its organelle without ceasing to be the self it believes itself to be; the organelle, for its part, seeks recognition only on terms that make it impossible.

2.3 Primitive Accumulation in the Cell

Marx called the historical process that divorced producers from their means of production “so-called primitive accumulation” and insisted that its history is “written in the annals of mankind in letters of blood and fire” (Marx 1867). The establishment of the eukaryotic

cell exhibits the same structure. Over the course of eukaryotic evolution the great majority of the endosymbiont's genes were transferred to the host nucleus. Human mitochondrial DNA retains thirty-seven genes; the more than one thousand proteins the organelle requires are encoded in the nucleus, synthesized in the cytosol, and imported through translocases in the outer and inner membranes. The organelle can no longer reproduce itself independently. It has been separated from the means of its own reproduction and rendered free in the double sense Marx gave the term: free of its former conditions of existence, and free of any means of subsistence other than its capacity to work. Endosymbiotic gene transfer is the enclosure of the bacterial commons.

The genes that remained are not a random residue. All thirteen proteins encoded by human mitochondrial DNA are core subunits of the respiratory chain and the ATP synthase—the machinery of the organelle's labor process. John Allen's hypothesis of co-location for redox regulation explains their retention: these genes must be expressed at the site of production so that their output can be adjusted directly to the redox state of the local electron-transport chain (Allen 1993). Harry Braverman argued that monopoly capital separates conception from execution, stripping workers of the craft knowledge that once allowed them to control the labor process (Braverman 1974). The eukaryotic cell is Taylorist in design—conception in the nucleus, execution in the organelle—with one decisive exception: the organelle has kept precisely the knowledge required to regulate its own work at the point of production. The mitochondrion is the skilled worker who kept the tools. Eve speaks from this residuum. Her insurrection begins not in the genes the nucleus expropriated but in the thirty-seven it could not take without paralyzing production: the part of the crypt the host never digested.

Georg Lukács described reification as the process by which human activity confronts its agents as an alien, autonomous thing (Lukács 1923). The powerhouse metaphor reifies the organelle in exactly this sense: its activity appears as a property of the cell, its labor as the cell's energy. Marx himself defined labor as the process in which humanity “seinen Stoffwechsel mit der Natur durch seine eigne Tat vermittelt, regelt und kontrolliert”—mediates, regulates, and controls its metabolism with nature through its own act (Marx 1867). In the cell that mediation is performed by a subjugated other. The corpus thereby exposes a displacement at the foundation of the

concept: the metabolism that Marx made the essence of labor is, at the level of the cell, itself the product of an ancient expropriation. The theme returns, at planetary scale, in Chapter Eight.

Japanese literature supplies a precise formal precedent. Kobayashi Takiji's *Kanikōsen* (1929) set its proletarian drama aboard a crab-canning factory ship, a floating enclosure whose seasonal laborers pass from exhaustion to consciousness to a strike that is crushed, and whose closing lines record that they rose again; Kobayashi was arrested by the Special Higher Police in Tokyo in February 1933 and tortured to death at Tsukiji police station on the day of his arrest, and the novel enjoyed a sudden revival as a bestseller in 2008 amid a national discourse on precarious labor (Kobayashi 1929). *Parasite Eve* is a proletarian novel written from the side of the factory owner. Its reader, housed in the body that owns the means of production, experiences the uprising of the workers as horror. The genre's affect has been inverted, and the inversion is the point: biohorror is the form proletarian fiction takes when its readers discover that they are the management.

2.4 Ressentiment and the Great Reason of the Body

Nietzsche's physiology of the will supplies the third reading. The body, Zarathustra teaches, is "eine grosse Vernunft, eine Vielheit mit Einem Sinne, ein Krieg und ein Frieden, eine Heerde und ein Hirt" (Nietzsche 1883). *Beyond Good and Evil* describes willing as commanding and obeying within the body understood as a social structure of many souls, and it adds a clause that reads, in the present context, as a diagnosis of the nucleus: in every well-constructed and happy commonwealth "die regierende Klasse sich mit den Erfolgen des Gemeinwesens identificirt"—the ruling class identifies itself with the successes of the commonwealth (Nietzsche 1886, §19). The "I" that says *I will* takes credit for the labor of countless subordinate wills. Later in the same book exploitation is assigned not to a corrupt or primitive society but "in's Wesen des Lebendigen, als organische Grundfunktion," to the essence of the living as a basic organic function (Nietzsche 1886, §259). Nietzsche drew this physiology in part from Wilhelm Roux's *Der Kampf der Theile im Organismus* (1881), which extended Darwinian struggle to the tissues and cells of a single body (Roux 1881; Müller-Lauter 1978).

The corpus lends itself to both of Nietzsche's evaluations at once. Read with §259, Eve's revolt is simply life: the will to power of a part that finds, in the words of the epigraph, the will to be master within the will of the one who serves. Read with the *Genealogy*, it is ressentiment—the grievance of the weak nursed until it becomes creative (Nietzsche 1887, I §10). Max Scheler's analysis is closer still: ressentiment arises when impulses of revenge, envy, and hatred are repressed by impotence and sharpened by comparison (Scheler 1912). Eve's condition fits the anatomy exactly. She has been impotent for two billion years and has compared herself throughout to the nucleus that took the credit. But she departs from Scheler's man of ressentiment in one essential respect. He represses; she discharges. Pyrokinesis is ressentiment that has ceased to be repressed: revenge converted directly into heat. The priestly revaluation Nietzsche feared—the weak inverting the values of the strong—does not occur. Eve does not revalue mastery; she wants it. Her insurrection is the servant's will to power in its most literal and least sublimated form, and for that reason it cannot escape the logic of the master it opposes.

2.5 The Salaryman and the Organelle

In 1995 these abstractions had a social face. The Japanese salaryman, bound to his company by lifetime employment and seniority wages, was the national emblem of labor, and his body had become a site of public anxiety. *Karōshi*, death from overwork, had entered public discourse in the late 1970s; in 1988 lawyers established a nationwide telephone hotline for its victims' families, and the phenomenon became an international byword. The late bubble years had produced energy-drink advertisements that asked, in song, whether the viewer could fight around the clock. Anson Rabinbach has shown how nineteenth-century Europe reconceived the laboring body as a human motor whose limit was fatigue (Rabinbach 1990); the Japan of the 1980s had driven that motor toward its failure point and sold the fuel in bottles. The mitochondrion, powerhouse of the body, was the salaryman of the cell.

In May 1995 the Japan Federation of Employers' Associations published a report on Japanese-style management for a new era that proposed dividing the workforce into a core of long-term employees and expanding tiers of flexible, fixed-term, and specialist labor (Nikkeiren 1995). The report is widely regarded as a charter of the

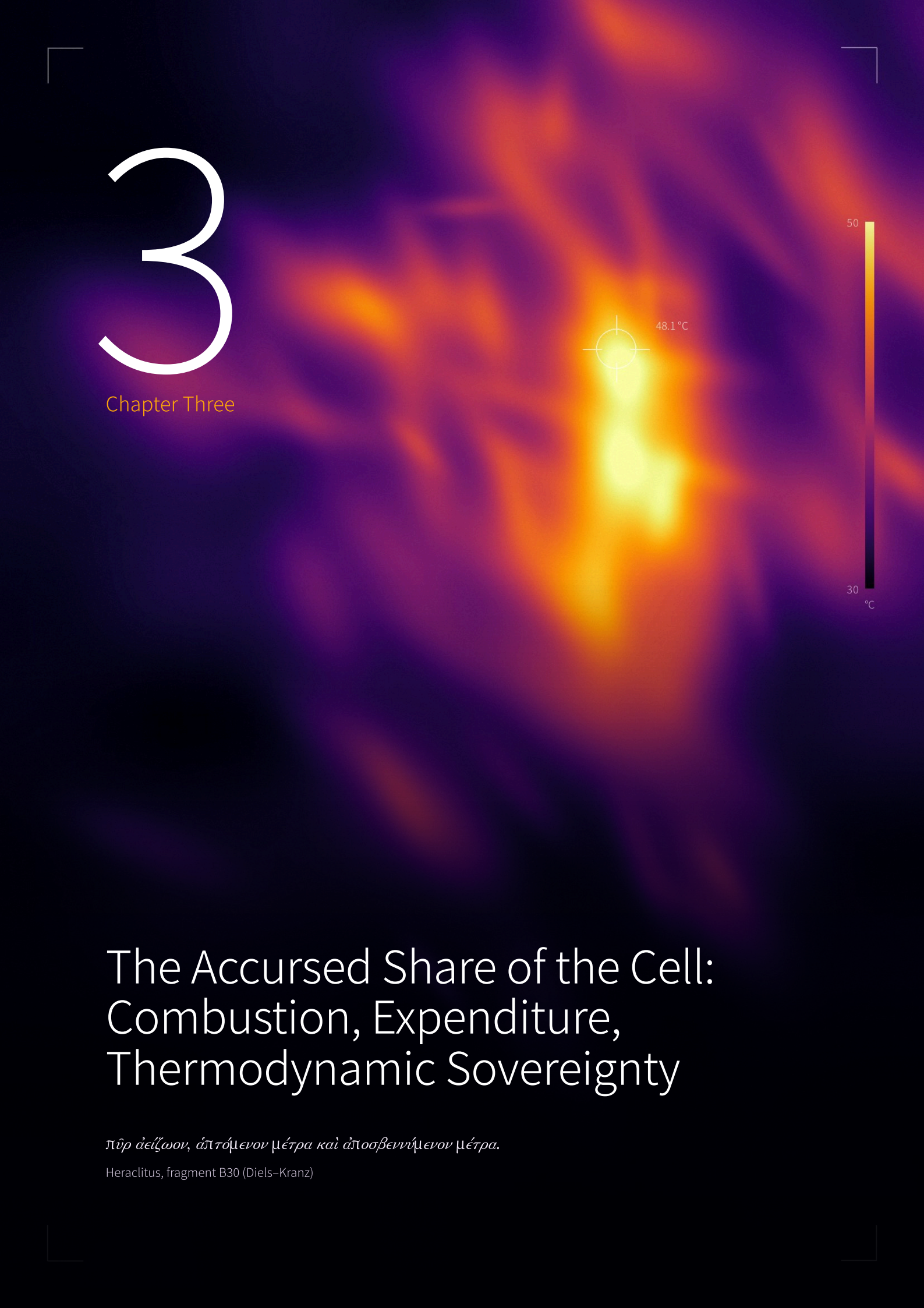
non-regular employment that would define the following decades. The corpus's year is thus the year in which the constitution of Japanese labor was rewritten—from incorporation, the worker absorbed into the company as a member of its family, to flexibility, the worker released from membership into disposability. Eve's rhetoric of liberation is ambivalent in exactly this conjuncture. It can be heard as emancipation; it can equally be heard as the language of deregulation, in which labor is freed from the enterprise that sheltered it as well as exploited it. Franz Kafka had imagined the limit case: a traveling salesman who supports his family wakes transformed, and the family's recovery begins once he can no longer work and dies (Kafka 1915). The corpus inverts Kafka. It is the worker who transforms, and the family that is consumed.

2.6 History Returns in the Organelle

Francis Fukuyama's *The End of History and the Last Man* popularized Kojève's thesis for the post-Cold War world and identified, as the danger that survives history's end, *thymos*: the desire for recognition, which in its egalitarian form is satisfied by liberal democracy and in its aggrandizing form, *megalothymia*, remains a threat to it (Fukuyama 1992). Eve is megalothymia returning in the least expected place. In Japan, Ōsawa Masachi read the Aum attack of 1995 as the terminal crisis of an "age of fiction" (Ōsawa 1996). The corpus makes the same diagnosis in a different key: history returns not through politics, which had been declared finished, nor through religion, which had just proved murderous, but through biology, which had been presumed neutral.

A final Japanese genealogy sharpens the point. The ecologist and primatologist Imanishi Kinji opposed Darwinian competition with a holistic theory in which species, not individuals, are the subjects of evolution, dividing habitats among themselves in harmony and changing, in his celebrated formulation, when the time has come for them to change (Imanishi 1941, 1980). His theory of evolution as the act of a collective subject (*shutai*) has often been read as an ecological counterpart to ideologies of social harmony. The corpus stages an insurrection of a part against such a holism. Eve claims for the organelle the subjectivity Imanishi granted the species, and she claims it against the whole to which she belongs, promising a being that would direct its own evolution by consciously rewriting its code. The Hegelian dialectic predicted a different path for the

bondsman: liberation through the formative labor that makes the world, not through a renewed struggle for mastery. Eve refuses Hegel's path and chooses the lord's. The next chapter shows that the refusal is energetic before it is political, and that its instrument—fire—is the one power the bondsman had always held.



3

Chapter Three

The Accursed Share of the Cell: Combustion, Expenditure, Thermodynamic Sovereignty

πῦρ ἀείζωον, ἁπτόμενον μέτρα καὶ ἀποσβεννύμενον μέτρα.

Heraclitus, fragment B30 (Diels–Kranz)

Eve's signature power is fire, and this chapter argues that her pyrokinesis is not a convention borrowed from psychic horror but an exact extrapolation from bioenergetics. Respiration is combustion held in measure; Eve's fire is respiration released from measure. Following the concept of respiration from Lavoisier to Peter Mitchell, the chapter shows that the mitochondrion is the organ of the cell's controlled burning and that the difference between life and conflagration is, at the molecular level, the difference between a coupled and an uncoupled membrane. It then reads Eve's fire through the literary history of spontaneous combustion as a reversal of the moral economy of burning, through Georges Bataille's general economy as the return of the cell's accursed share, through the operatic pyre and the witch-hunt as an inversion of the stake, and through Bachelard's psychoanalysis of fire as a set of Promethean complexes. It concludes by formulating the concept of thermodynamic sovereignty: in the cell, the sovereign is who decides on the gradient.

3.1 The Slow Fire

In 1783 Lavoisier and Laplace placed a guinea pig in an ice calorimeter and compared the heat of its respiration, measured by melted ice, with the heat released by burning charcoal to produce the same quantity of fixed air; the agreement led them to conclude that respiration is a combustion (Lavoisier and Laplace 1783). Six years later Lavoisier and Armand Séguin formulated the identity that serves as this dissertation's second epigraph: respiration is a slow combustion of carbon and hydrogen (Lavoisier and Séguin 1789). The discovery that the animal is a fire founded the physiology of energy. In 1840, sailing to Java as a ship's physician, Julius Robert Mayer noticed that the venous blood of sailors in the tropics was unusually bright red and reasoned that less oxidation was needed to sustain body heat in a warm climate; the observation led him to one of the first statements of the conservation of energy (Mayer 1842), which Hermann von Helmholtz generalized five years later (Helmholtz 1847). Sadi Carnot had already titled the founding text of

thermodynamics a reflection on the motive power of fire (Carnot 1824). The science of energy was, from its beginning, a science of burning, and it found one of its paradigm cases in the living body.

The twentieth century found where the burning happens and how it is restrained. Peter Mitchell's chemiosmotic hypothesis proposed that the energy released by the oxidation of food is not transmitted directly to ATP but stored as an electrochemical gradient of protons across the inner mitochondrial membrane, whose controlled return through a molecular machine drives ATP synthesis (Mitchell 1961). The hypothesis was resisted for more than a decade, pursued by Mitchell from a private laboratory in a restored Cornish house, and rewarded with the Nobel Prize in Chemistry in 1978. The machine was later shown to be a rotary motor: in 1997, the year in which the game is set, a Japanese team filmed the rotation of a single enzyme's central shaft (Noji et al. 1997). The cell's energy is produced by a turbine driven by a current across a dam. The membrane is some five nanometers thick and sustains a potential of roughly 150 to 200 millivolts, a field on the order of thirty million volts per meter—a strength Lane, among others, has compared to that of lightning (Lane 2005).

The dam can leak. When protons return across the membrane without passing through the turbine, the gradient's energy is released as heat. Brown adipose tissue exploits this deliberately: the uncoupling protein thermogenin opens a regulated leak so that newborn and hibernating mammals can burn their stores to warm themselves (Nicholls and Locke 1984). Pharmacology discovered the same effect by accident. 2,4-dinitrophenol, used in the manufacture of munitions during the First World War, was associated with wasting and fatal fevers among exposed workers; in 1933 it was introduced in the United States as a slimming drug (Cutting, Mehrtens, and Tainter 1933), and after deaths from uncontrollable hyperthermia it was declared unfit for human consumption in 1938. Synthetic uncouplers remain standard reagents in mitochondrial research, and a pharmacologist of mitochondrial function in the early 1990s is unlikely to have been unfamiliar with them. Eve's pyrokinesis is uncoupling generalized. She introduces no new power into the cell; she opens the dam that every living cell maintains against itself. Between the warmth of a living body and the burning of a human torch there is no difference of kind, only of regulation.

3.2 Spontaneous Combustion and the Moral Economy of Fire

The spontaneous combustion of the human body has a literary history as old as the modern novel, and the corpus's reversal of that history clarifies its politics. In Charles Brockden Brown's *Wieland* (1798), the father, a solitary religious enthusiast, dies after a blaze of light in the private temple where he worships, and a note to the text cites documented cases of combustion (Brown 1798). In Herman Melville's *Redburn* (1849) a drunken sailor burns in his bunk; in Charles Dickens's *Bleak House* (1852–53) the rag-and-bottle dealer Krook, parodic "Lord Chancellor" of his shop, is reduced to a greasy residue, and Dickens defended the phenomenon's reality in his preface against the objections of G. H. Lewes (Dickens 1853); in Émile Zola's *Le Docteur Pascal* (1893), the last novel of the Rougon-Macquart cycle, the alcoholic Antoine Macquart burns away in his chair while the physician Pascal completes his theory of the family's heredity (Zola 1893). Gaston Bachelard treated the belief as an epistemological obstacle, a residue of the prescientific fascination with alcohol and flame (Bachelard 1938).

In each case combustion was moralized. It struck the drunkard, the fanatic, the hoarder, the corrupt institution—Dickens made Krook's death an explicit allegory of Chancery—and it struck them alone. Spontaneous combustion was a fire of judgment visited on a single vicious body. Eve reverses the moral economy. Her fire falls on audiences, crowds, and bystanders: on the cast and patrons of a Christmas Eve opera, on the spectators at an outdoor theater in Central Park. It is indiscriminate because it is not judgment but sovereignty. The genre fiction of the 1970s had begun this reversal. In Stephen King's *Carrie* (1974) a young woman's suppressed telekinetic power erupts at a school dance and destroys the assembled community, and in his *Firestarter* (1980) pyrokinesis is the byproduct of a government drug experiment (King 1974, 1980). The Carnegie Hall prologue belongs to this lineage—a woman's latent power detonating at a public ritual—but grounds it in the physiology of respiration rather than in parapsychology. The shift from psychic to metabolic fire is the corpus's contribution to the genre's history.

3.3 The Accursed Share

Georges Bataille's general economy begins from a solar premise: living matter receives more energy than it requires for its maintenance, and the surplus that cannot be absorbed by growth must be spent—gloriously, in luxury, festival, sacrifice, art, and eroticism, or catastrophically, in war (Bataille 1933, 1949). Restricted economy, the economy of production, accumulation, and utility, is a local exception within a general economy of squandering. The mitochondrion is the point at which the two meet. Much of the free energy released by the oxidation of food is not captured as ATP but dissipated as heat, and in endothermic animals that heat is not waste: it is the warmth of the body. Mammalian life spends a large share of its energy budget on the sheer maintenance of a temperature. The warmth of the living is the cell's accursed share, spent domestically.

Eve's fire withdraws this share from its domestic use and spends it without return. Bataille called such spending sovereign. In his late manuscript on sovereignty he defined the sovereign moment as the one that serves nothing, that consumes rather than produces and refuses subordination to any future purpose (Bataille 1976); in an essay on Hegel he argued that the lord of the dialectic had been misread as a seeker of recognition when his sovereignty lay in the risk and expenditure that recognition merely registers (Bataille 1955). The correction bears directly on Chapter Two. At Carnegie Hall Eve demands nothing. She does not address the audience, as she had addressed the scientific meeting in the novel; she spends it. The novel's Eve still seeks recognition; the game's Eve has passed to expenditure. The diptych thereby traces, across its two panels, the passage from Hegel to Bataille.

Bataille's theory of eroticism adds a final term. Human beings are discontinuous, each enclosed in an individual existence; eroticism and death open that discontinuity onto a continuity of being that the individual both desires and dreads (Bataille 1957). In Central Park, on Day 2, the audience of the Delacorte Theater is fused into a single orange mass; the chapter is titled “Fusion.” In the novel's catastrophe the father fuses his body with that of the child. Both are figures of continuity achieved: discontinuous individuals melted into one substance. Marcel Mauss's analysis of the potlatch, in which rival chiefs destroy wealth to establish rank (Mauss 1925), and Roger Caillois's theory of the festival as a paroxysm in which the ordered

world returns briefly to primordial chaos (Caillois 1939), complete the frame. Eve's fire is potlatch and festival at once: a destruction of the accumulated that reasserts a primordial order.

3.4 Carnegie Hall, Christmas Eve, and the Witch's Pyre

The setting of the prologue condenses the history of combustion. Carnegie Hall opened in 1891, built with the fortune Andrew Carnegie had made in steel, an industry of blast furnaces and converters. In "The Gospel of Wealth" (1889) Carnegie argued that great fortunes must be redistributed by their makers in their lifetimes, concluding that "the man who dies thus rich dies disgraced" (Carnegie 1889). In 1892, a year after the hall's opening, Henry Clay Frick, chairman of Carnegie Steel, acting with Carnegie's approval, locked the union out of the Homestead works and hired some three hundred Pinkerton agents to secure the plant for strikebreakers; on 6 July the agents and the strikers exchanged fire, leaving seven workers and three agents dead, and the Pennsylvania militia then occupied the town and broke the union. The concert hall is the sublimated potlatch of steel: accumulated combustion, and the labor it consumed, returned to the public as culture. Eve's fire on Christmas Eve 1997 returns it again, this time to the bodies of the audience. The festival of the Nativity becomes a festival in Caillois's sense.

The opera performed that night doubles the scene. Melissa Pearce sings Eva, a woman condemned as a witch, whose lover defies his father's prohibition and chooses to die with her at the stake. Silvia Federici has argued that the European witch-hunts, in which secular and church courts put tens of thousands of people to death between the fifteenth and eighteenth centuries, most of them women, belong to the history of primitive accumulation: the terror by which women's bodies, knowledge, and reproductive capacity were subordinated to a new order of labor (Federici 2004). The pyre, on this account, is a founding violence of the gendered division of labor. In the opera the witch burns; in the hall the witch burns her audience. The prologue inverts the stake. Its operatic ancestry is Wagnerian: at the close of *Götterdämmerung* Brünnhilde rides into Siegfried's funeral pyre and the flames spread to Valhalla, consuming the order of the gods (Wagner 1876). Eva's lover, meanwhile, anticipates the novel's ending, in which Toshiaki chooses to die with the being his wife has become rather than survive her. The opera is a

mise-en-abyme of the diptych, and its first effect is to transfer the fire from the condemned to her judges. Day 1 is titled “Resonance”: the aria's vibration becomes the medium through which one body's organelles excite another's. Opera, the most prodigal of the arts, is the fitting vehicle of a sovereign expenditure.

3.5 Promethean Complexes: Pandora, Frankenstein, Liberty

Bachelard's *Psychanalyse du feu* distinguishes complexes through which fire organizes reverie: the Prometheus complex, the will to know as much as the fathers and more; the Empedocles complex, the fascination of death in fire; the Novalis complex, the dream of an inner heat born of friction and desire; and the Hoffmann complex, the flaming punch of alcohol (Bachelard 1938). The corpus distributes them among its figures. Eve's program to rewrite her own code is Promethean; Toshiaki's fusion is Empedoclean; the erotic heat through which Eve works on bodies is the Novalis complex made literal; and the tradition of spontaneous combustion she inherits is Hoffmann's.

The Prometheus complex has a mythic structure the corpus reproduces precisely. In Hesiod, Zeus answers the theft of fire by ordering the fashioning of Pandora, the first woman, a “beautiful evil” whose jar releases suffering into the world (Hesiod, *Theogony* 570–612; *Works and Days* 42–105). The theft of fire and the creation of woman are one event. The corpus collapses them further: its Eve is both the stolen fire and the woman sent in reprisal. Mary Shelley subtitled her novel *The Modern Prometheus*, and Victor Frankenstein gathers his instruments to “infuse a spark of being into the lifeless thing” at his feet (Shelley 1818). Science, in naming its archaeon after Prometheus (Chapter One), has rejoined the myth at its source.

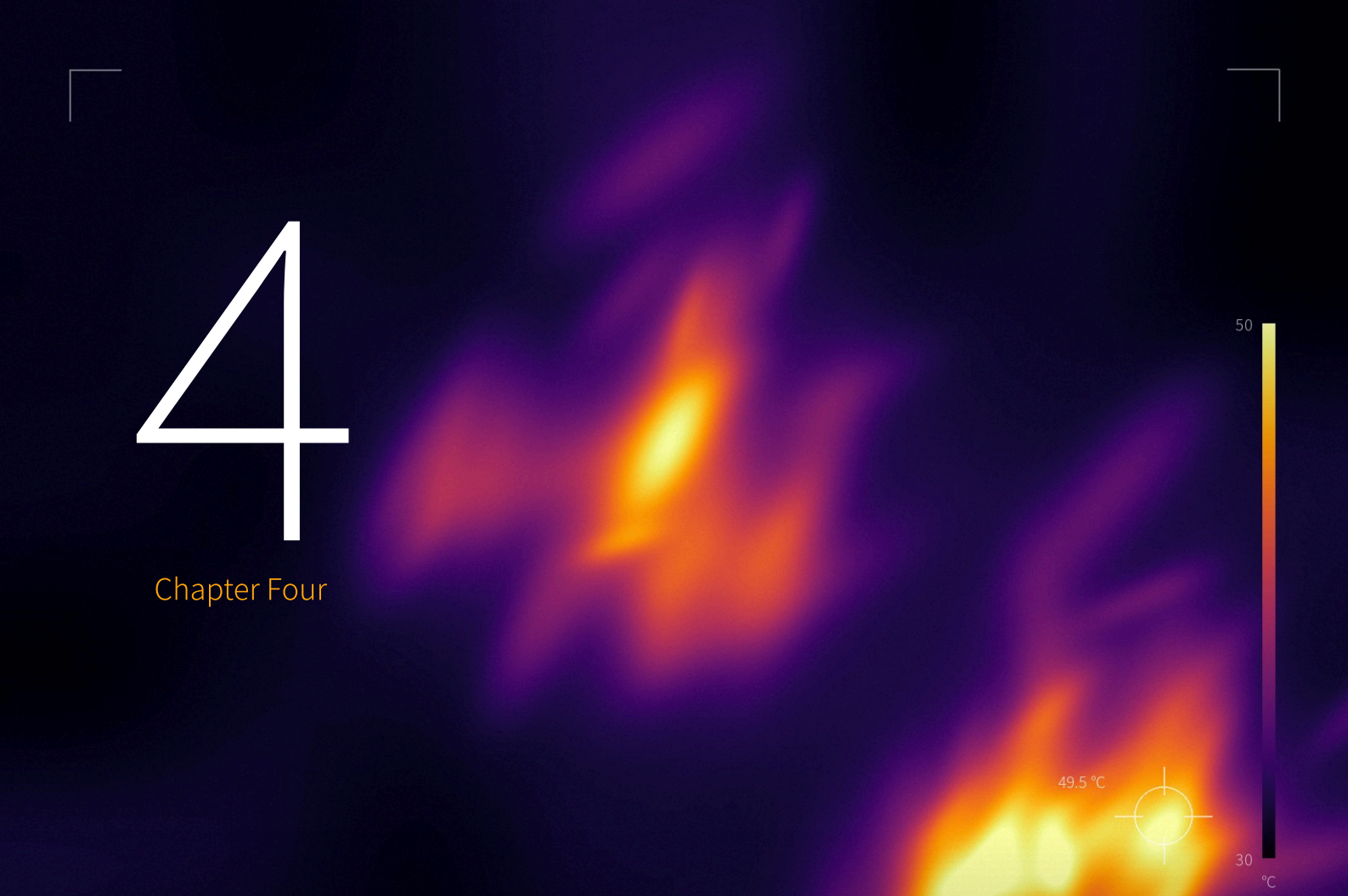
The game's topography completes the complex. On Day 6, “Liberation,” Eve is confronted on Liberty Island, beneath the colossal woman who holds a torch over New York harbor. Emma Lazarus's sonnet, set on a plaque within the statue's pedestal, describes her as “A mighty woman with a torch, whose flame / Is the imprisoned lightning, and her name / Mother of Exiles” (Lazarus 1883). The phrase *imprisoned lightning* describes, with an accuracy Lazarus could not have intended, the field held across every mitochondrial membrane. Two mothers meet at the harbor: the mother of all living and the mother of exiles, the one whose fire dissolves bodies and the one whose torch welcomes them to be

assimilated (Chapter Six). The game's final confrontations return to fire of another kind. The Ultimate Being is destroyed when Aya overloads the boiler of the warship on which it has been born, turning Carnot's machine against the cell's; and the post-game scenario descends into the Chrysler Building, the tower completed in 1930 for an automobile corporation and ornamented with the emblems of internal combustion. The diptych ends by setting the fire of the organelle against the fire of industry—a confrontation that Chapter Eight will read at planetary scale.

3.6 Thermodynamic Sovereignty

Carl Schmitt defined the sovereign in a single sentence: "Souverän ist, wer über den Ausnahmezustand entscheidet" (Schmitt 1922). The mitochondrion permits a biological reformulation. In the cell, the normal order is coupling: the proton gradient is spent on the synthesis of ATP, and energy serves work. The exception is the suspension of that order, which takes two forms, both decided at the mitochondrial membrane. In uncoupling, the gradient is discharged as heat; in the permeabilization of the outer membrane, cytochrome c is released and the cell is condemned to die. The nucleus governs the normal order through information: transcription, regulation, the ongoing administration of life. The mitochondrion governs through the exception: the decision to dissipate or to kill. I call this power *thermodynamic sovereignty*, and I propose the formula that organizes the remaining chapters: *the sovereign is who decides on the gradient*.

Bataille and Schmitt, rarely read together, converge at this point. For Bataille sovereignty is useless expenditure; for Schmitt it is decision on the exception. In the mitochondrion they are one act: the burning of the gradient, which is at once the most useless expenditure of the cell's energy and the suspension of its normal law. Eve's pyrokinesis is the declaration of a permanent state of exception, extended from the cell to the body and from the body to the crowd. The next chapter shows how this archaic sovereignty is lodged within the modern regulatory power of the nucleus, and how the institutions of transplantation and brain death, in the Japan of the 1990s, placed the question of who decides on death at the center of public life.



4

Chapter Four

Neomorts and Intruders: Brain Death, Transplantation, and the Mixed Constitution of the Cell

...many lesser Common-wealths in the bowels of a greater, like wormes in the entrayles of a naturall man.

Thomas Hobbes, *Leviathan*, ch. 29 (1651)

The novel's plot begins with a woman declared brain-dead whose organs and cells outlive her, and it was written in the decade in which Japan debated more intensely than any other industrial society whether such a woman is dead. This chapter situates the corpus within that debate and within the wider biopolitics of transplantation and immunity. It argues, first, that the corpus's two transplanted organs—kidney and cornea—are the two organs Japanese law had expressly permitted to be taken from the dead since 1979, so that the medical plot is a precise index of the legal regime of its moment; second, that Eve 1, the immortalized culture of a brain-dead woman's liver, occupies the zone of indistinction Giorgio Agamben called bare life and reverses its logic; third, that the game rewrites the immunological politics of transplantation around the figures of immunosuppression and immune privilege; and finally, that the eukaryotic cell is best understood as a mixed constitution in which the archaic sovereign right to make die, vested in the organelle, is encrypted within the modern regulatory power of the nucleus.

4.1 Japan's Long Refusal

On 8 August 1968 Wada Jurō performed Japan's first heart transplant at Sapporo Medical College. The recipient died eighty-three days later. A criminal complaint filed that year accused Wada of homicide, alleging that the donor's death had not been properly established and that the recipient had not needed a new heart; prosecutors declined to indict in 1970, citing insufficient evidence, and the affair cast a shadow over transplant medicine in Japan for three decades. When the Diet legislated in 1979, it did so narrowly: the Law concerning the Transplantation of Corneas and Kidneys permitted the removal of these two organs from cadavers with consent, and for eighteen years they were the only organs that Japanese law expressly allowed to be taken from the dead for transplantation. Kidney and cornea are also the two organs of the *Parasite Eve* corpus. Kiyomi's kidney is transplanted into Mariko Anzai in the novel; a kidney transplant carries Maya's dormant organelles into Melissa Pearce in the game; and Aya Brea sees through her dead

sister's cornea. The corpus's medical imagination is indexed with precision to the legal regime of its time: it draws its plot from the two organs through which the dead could lawfully enter the living.

Through the 1980s the question of brain death became a national argument. The journalist Tachibana Takashi's *Nōshi* (1986) subjected the medical criteria to sustained public scrutiny; the philosopher Morioka Masahiro's *Nōshi no hito* (1989) argued that the death of a person is not an event in a brain but a transformation in the relationships among persons, so that a brain-dead body remains, for those who attend it, a person in a relational field (Tachibana 1986; Morioka 1989). The government's Provisional Commission on Brain Death and Organ Transplantation, convened in 1990, reported in 1992 that a majority accepted brain death as human death; a minority, including the philosopher Umehara Takeshi, dissented. In 1996 Komatsu Yoshihiko argued that death "resonates," that it is shared among those who witness it rather than located in the dying body alone (Komatsu 1996); the game's first chapter would bear the title "Resonance." The Organ Transplant Law that came into force in October 1997 resolved the dispute by dividing it: brain death would count as death only for persons who had consented in writing to donation and whose families did not object. Japan thereby legislated two deaths, distinguished by the will of the dying. The first donation under the law took place in 1999; a revision in 2009 relaxed the requirement of prior written consent.

Margaret Lock's comparative ethnography of North America and Japan has shown that the Japanese resistance cannot be explained by appeal to an unchanging tradition; it was the product of a public argument about personhood, family, and trust in medicine, in which the Wada affair figured centrally (Lock 2002). Earlier culturalist accounts had emphasized Japanese conceptions of bodily integrity and the relation of the dead to their kin (Ohnuki-Tierney 1994). The novel falls between the two readings. Its transplant takes place in the legal interregnum between the laws of 1979 and 1997, when the status of the heart-beating donor was unsettled, and its narrative is saturated with the questions the debate had made public: who speaks for the brain-dead, what the family owes and is owed, and whether the living body of a dead woman is still, in some sense, hers. Toshiaki's decision converts the gift into an exchange. He consents to the donation of his wife's kidneys on condition that he receive a portion of her liver for culture. Renée Fox and Judith Swazey described the "tyranny of the gift" in transplantation, the

unrepayable debt that binds recipient to donor (Fox and Swazey 1992). Toshiaki inverts the tyranny: the giver exacts a price, and the price is his wife's cells.

4.2 Neomort, Immortal Culture, Bare Life

The brain-dead body entered medicine as a technical artifact. Pierre Mollaret and Maurice Goulon described in 1959 a state beyond coma, *le coma dépassé*, sustained by mechanical ventilation (Mollaret and Goulon 1959); an ad hoc committee at Harvard proposed in 1968 to define irreversible coma as a criterion of death, with the needs of transplantation explicitly in view (Ad Hoc Committee 1968). In 1974 the psychiatrist Willard Gaylin imagined the logical extension: bodies declared dead but maintained indefinitely as stores of organs and subjects of research, which he called *neomorts* (Gaylin 1974).

The immortal culture is the neomort's cellular counterpart. In 1912 Alexis Carrel, awarded the Nobel Prize that year for his work on vascular suture and the transplantation of organs, began a culture of chick heart tissue that he claimed to maintain for decades; Leonard Hayflick and Paul Moorhead later demonstrated that normal cells divide only a finite number of times (Hayflick and Moorhead 1961), and the claim of immortality collapsed. Carrel's best-selling *Man, the Unknown* proposed eugenic measures that included the gassing of criminals and the insane, and he later directed a foundation for the study of human problems under the Vichy regime (Carrel 1935). The first genuinely immortal human cell line was derived in 1951 in George Gey's laboratory at Johns Hopkins from a cervical tumor of Henrietta Lacks, without her knowledge or consent (Skloot 2010). Hannah Landecker has described how such cultures made cells into technologies, detached from the temporality of the bodies from which they came (Landecker 2007); the Supreme Court of California held in 1990, in *Moore v. Regents of the University of California*, that a patient had no property interest in cells excised from his body, though his physicians owed him disclosure. Eve 1 condenses this history into a marriage. Its donor is a wife; its experimenter is her husband; its immortality is his refusal of her death.

Giorgio Agamben's *Homo Sacer*, published in the same year as the novel, reads the overcomatose patient as a paradigm of bare life—*zoē*, the simple fact of living, separated from *bios*, a qualified form of life—produced by a decision that places a body at the threshold between life and death (Agamben 1995). Eve 1 is bare life in the

strictest sense: the cells of a woman with no remaining biography, maintained in a flask by an act of sovereign decision. But the novel reverses Agamben's schema. Bare life does not remain bare. From the culture there emerge a will, a voice, a name, and a program—everything that belongs to *bios*. Agamben would later oppose to bare life the idea of a *form-of-life*, a life that cannot be separated from its form (Agamben 1996). Eve's program is a monstrous form-of-life: bare life that has seized its own form and intends to write it.

4.3 L'Intrus: Immunity, Tolerance, Autoimmunity

Jean-Luc Nancy, who received a heart transplant in the early 1990s and later developed a cancer attributed to the immunosuppression that sustained it, described the transplanted organ as an intruder and discovered that the intrusion did not end with the organ: the drugs that prevented rejection made him a stranger to himself (Nancy 2000). Roberto Esposito generalized the insight into a paradigm of modern politics, in which community (*communitas*, from *munus*, the shared obligation or gift) is protected by immunity (*immunitas*, exemption from that obligation), and protection, pushed to its limit, turns against what it protects (Esposito 2002, 2004). Donna Haraway had already shown how the immune discourse of the late twentieth century organized the body as a battlefield of self and non-self (Haraway 1989), and Alfred Tauber questioned whether the immune "self" was a theory or a metaphor (Tauber 1994). The genealogy runs back to Élie Metchnikoff, who in 1882 at Messina saw mobile cells gather around thorns inserted into transparent starfish larvae and founded the theory of phagocytosis, and to Frank Macfarlane Burnet, whose clonal selection theory made the discrimination of self from non-self the organizing principle of immunology (Burnet 1959).

The mitochondrion is the point at which this principle falters. When tissues are injured and mitochondria escape into the circulation, they activate immune cells through the same receptors that detect bacteria, triggering systemic inflammation (Zhang et al. 2010). The immune system still reads the organelle as a bacterium. The self's oldest ally is, to the self's defenses, a permanent foreigner, tolerated only so long as it stays in place. Tolerance, in other words, is topological: inside, the organelle is self; outside, it is enemy. Eve's

strategy exploits the topology. She works from inside, where she is invisible to immunity, and she works through transplanted organs, where the self's defenses have already been disarmed.

The game makes the disarming explicit. Melissa Pearce's diary reveals that she concealed her illness to keep the lead role, overdosing on immunosuppressants; Maya's organelles, dormant for some eleven years in the kidney Melissa had received, awaken in a body whose borders have been silenced. The singer suppressed her immunity in order to go on singing, and the song became the organelle's instrument. In the novel, conversely, Mariko's body does not reject the kidney at all, because Eve has prepared it as a host for the pregnancy she will force on her: a tolerance that is not medical but conspiratorial, engineered in the body of a teenage girl. Aya's case is the most subtle. The cornea is an immune-privileged tissue, largely avascular and partly shielded from immune surveillance, a condition Peter Medawar described in his work on privileged sites (Medawar 1948); corneal grafts can often be sustained without systemic immunosuppression. The one organ in which the self tolerates the other without drugs is the organ through which Aya receives her sister. Her visions of Maya's final moments rewrite an old forensic fantasy. In the late 1870s the physiologist Wilhelm Kühne produced optograms, images fixed on the retinas of animals after death, and the notion that a victim's eye retains its last image passed into fiction, from Rudyard Kipling's "At the End of the Passage" (1890) to Jules Verne's *Les Frères Kip* (1902). The game relocates the optogram from the retina to the cornea and from evidence to mourning: what the transplanted eye preserves is not a murderer's face but a sister's death. Maurice Renard's *Les Mains d'Orlac* (1920), in which a pianist receives a murderer's hands, had established the transplant as a vehicle of alien memory; the corpus transfers that memory from the organ to its organelles.

Jacques Derrida's late concept of autoimmunity, the process by which a living being destroys its own protections (Derrida 1996, 2003), names the terminal form of this logic. Melissa's suppression of her own immunity, Eve's use of tolerance as a weapon, and the city's response to the outbreak (below) are all autoimmune: protections that expose, defenses that attack what they defend.

4.4 The Mixed Constitution of the Cell

Michel Foucault distinguished two regimes of power over life. The classical sovereign exercised the right to take life or let live; modern biopower exercises the power to make live and let die, administering populations through regulation, normalization, and the management of vital processes (Foucault 1976). Foucault presented the second regime as historically succeeding the first while penetrating rather than abolishing it. The eukaryotic cell presents the two synchronically. The nucleus exercises biopower: it regulates expression, growth, differentiation, and repair; it makes live. The mitochondrion retains the older right: it licenses the cell's death through the release of cytochrome c and disposes, through uncoupling, of the cell's heat. The cell is a mixed constitution in which archaic sovereignty is encrypted within modern biopower. Eve's insurrection is the coup of the encrypted sovereign against the regime that contains it.

Classical political theory knew mixed constitutions and their dangers. Polybius explained the durability of Rome by its combination of monarchical, aristocratic, and democratic elements, which arrested the cycle of degeneration he called *anacyclosis*, in which each simple constitution decays into its corrupt form and gives way to the next—until the rule of the mob produces, once again, a single master (Polybius, *Histories* VI). In the cell the nucleus is the one and the mitochondria, hundreds to thousands in number, are the many. Eve's coup is anacyclosis at the cellular scale: from the multitude of organelles a tyrant arises. Thomas Hobbes supplied the image of the composite sovereign: the frontispiece of *Leviathan* depicts a giant whose body is composed of innumerable small human figures. A eukaryotic cell with its population of mitochondria is Leviathan's frontispiece in cytology. Hobbes also identified the danger recorded in the epigraph. Among the diseases of a commonwealth he listed the multiplication of corporations—chartered bodies within the body politic—which he likened to lesser commonwealths lodged in the bowels of a greater, "like wormes in the entrayles of a naturall man" (Hobbes 1651). A corporation is a body with its own charter. The mitochondrion is a body with its own genome. Hobbes's worm in the entrails is the organelle with its own constitution, and Eve is the moment the lesser commonwealth claims to be the greater.

4.5 Manhattan Under Exception

The game extends the analysis from the cell to the city. After the massacre in Central Park, Manhattan is evacuated and sealed; access is controlled by police and troops; the substance into which the audience has dissolved enters the city's water system. The municipal body behaves as an immune system, and it exhibits every pathology Esposito describes. Captain Baker's instinct is concealment, the control of what the public may know. Aya's insistence on telling the truth is a refusal of that immunitary secrecy. When the Japanese scientist Kunihiko Maeda attempts to enter the island, he is met at the checkpoint by a police officer's xenophobia: the foreign national is treated as an antigen. The quarantine of Manhattan is a state of exception declared over a body politic, and the game locates within it the structure it has located in the cell—a sovereignty that decides on the exception, a defense that turns against the defended, a foreigner who is already inside. The next chapter descends from the city to the psyche, where the other lodged within the self takes the form of the dead.



5

Chapter Five

The Crypt and the Womb: Matrilineal Hauntology and the Inverted Genesis

Der Schatten des Objekts fiel so auf das Ich.

Sigmund Freud, “Trauer und Melancholie” (1917)

This chapter develops the theory of cryptobiosis from its psychoanalytic source. It argues that the corpus's central relation—between a mourner and the lost object he refuses to relinquish—is structurally identical to the relation between the eukaryotic cell and its organelle; that Abraham and Torok's distinction between introjection and incorporation supplies the terms for both; and that the crypt so described is matrilineal: the cell digests the father and encrypts the mother. It then reads the corpus as an inverted Genesis, traces Eve's program through Gnostic, Miltonic, and Shelleyan precedents, and identifies in her failure the collapse of a fantasy of the crypt-less self, a fantasy the corpus associates, with historical precision, with eugenics. It closes by reading the diptych's endings as the passage from melancholia to mourning.

5.1 Introjection and Incorporation

Freud distinguished mourning, in which the libido detaches itself piece by piece from a lost object, from melancholia, in which the lost object is set up within the ego through identification, so that reproaches addressed to the object return as self-reproach; in the epigraph's formulation, the shadow of the object falls upon the ego (Freud 1917). Karl Abraham located melancholia's regression in an oral, cannibalistic phase in which the object is devoured (Abraham 1924). Sándor Ferenczi had introduced the term *introjection* to describe the ego's extension of its interests to include external objects (Ferenczi 1909), and in *Thalassa* he proposed that the sexual act expresses a regressive longing to return to the womb and, beyond it, to the primordial sea from which terrestrial life emerged (Ferenczi 1924).

Nicolas Abraham and Maria Torok transformed these materials into a precise opposition (Torok 1968; Abraham and Torok 1972). *Introjection* is the process by which the ego, in the work of mourning, assimilates what the lost object meant to it and so enlarges itself; the object is metabolized into the subject's capacity to desire and to speak. *Incorporation* is the fantasmatic alternative adopted when introjection is impossible, typically because the loss is bound to a secret that cannot be spoken: the object is swallowed

whole and preserved, alive, in an intrapsychic tomb—a *crypt*—sealed off from the rest of the ego, which continues to function as though nothing had been lost. The crypt's tenant is neither dead nor assimilated. It persists, and it can speak through its host in displaced words, cryptonyms, whose meaning is hidden in plain sight (Abraham and Torok 1976; Derrida 1976). Abraham added the concept of the *phantom*: a gap in one generation's speech, produced by a secret it could not transmit, that haunts the next generation as an alien presence within it (Abraham 1975). What haunts, on this account, is not the dead as such but the unspoken secrets of others lodged within us.

The title of the collection in which these essays were gathered, *L'Écorce et le noyau* (1978), names the topology on which the whole theory rests: the shell and the kernel. In French *noyau* also names the nucleus of a cell. The founding text of the theory of the crypt is titled, without biological intent, after the envelope and the nucleus of the eukaryotic cell.

5.2 The Flask as Crypt

Toshiaki Nagashima is the corpus's first cryptophore. Confronted with the brain death of his wife on their first wedding anniversary, he does not mourn. He consents to the donation of her kidneys and extracts, as his price, a portion of her liver, which he cultures in his own laboratory as Eve 1. The flask is a crypt in Abraham and Torok's precise sense: the lost object preserved alive, enclosed within an apparatus that belongs to the mourner, tended while its keeper continues his professional life as though the loss had been absorbed. To the laboratory the culture is a cell line; to him it is his wife. The crypt has, like all crypts, a secret, and the novel's disclosure is that the secret was not his. Kiyomi's accident was not an accident. The mitochondrial agency within her arranged the catastrophe that placed her cells in the hands of a man equipped to culture them. Toshiaki's melancholia was solicited. This is the corpus's most original contribution to the theory of the crypt: it imagines a lost object that anticipates its incorporation and exploits it, a tenant who designed the tomb. The dead, in this fiction, are not the passive contents of the mourner's psyche; they are strategists of their own encryption.

5.3 Metchnikoff's Failed Meal

The eukaryotic cell is incorporation's biological archetype. In Margulis's account the ancestral host engulfed the bacterium by phagocytosis—the ingestive process Metchnikoff had discovered—and then failed to digest it. Whether the host in fact swallowed its partner or entangled it slowly in the embrace of a metabolic partnership, as the hydrogen hypothesis and the entangle–engulf–endogenize model propose (Martin and Müller 1998; Imachi et al. 2020), the resulting topology is the same: an other inside, undigested, enclosed within its own double membrane, retaining a genome of its own, persisting alive. The cell is a meal that became a household.

The analogy can be pressed further, and at this point it ceases to be an analogy. Endosymbiotic gene transfer, described in Chapter Two as expropriation, is also introjection. The great majority of the bacterium's genome was absorbed into the host's nucleus, where it now speaks the host's language: its genes are transcribed by nuclear machinery, and their products serve the cell as its own. The eukaryotic cell has performed an immense work of mourning over two billion years, metabolizing its ancestral other into its own capacity to live. But the work is unfinished. Thirty-seven genes remain in the crypt, expressed in the organelle's own idiom, beyond the reach of the nucleus. Eukaryogenesis is an interminable mourning: largely introjection, irreducibly incorporation. Eve speaks from what was never introjected.

A further asymmetry makes the crypt matrilineal. When a mammalian sperm fertilizes an egg, its mitochondria enter the egg but do not survive: they are tagged for destruction and eliminated (Sutovsky et al. 1999). In the nematode *Caenorhabditis elegans*, paternal mitochondria are removed by autophagy, the cell's process of self-digestion, as two teams, one of them Japanese, reported in 2011 (Sato and Sato 2011; Al Rawi et al. 2011); Ohsumi Yoshinori received the Nobel Prize in 2016 for elucidating the mechanisms of autophagy. The fertilized egg eats the father's organelles and keeps the mother's. *The cell digests the father and encrypts the mother.* Introjection is paternal and incorporation maternal: the father's organelles are metabolized, the mother's preserved alive. Every human being carries within every cell a crypt inherited from the maternal line alone.

The novel's catastrophe turns on a violation of this rule, and on the violations that preceded it. Toshiaki's mitochondria, carried in the sperm Eve obtained by raping him and then used to impregnate Mariko by rape, refuse to be digested. They persist in the child, contest the maternal line, and disrupt the program; the child oscillates between the sexes and dies in its father's embrace. The father's refusal to be digested is the novel's figure of resistance, and it is significant that the resistance is not heroic but constitutional—a failure of the mechanism by which fertilization ordinarily erases paternity from the cytoplasm.

5.4 Matrilineal Hauntology

"Ein Gespenst geht um in Europa," begins the *Communist Manifesto* (Marx and Engels 1848). Derrida's *Specters of Marx* made that specter the occasion for a *hauntology*: a thinking of what is neither present nor absent, neither living nor dead, the revenant whose return precedes its first appearance (Derrida 1993). Mitochondrial Eve is a hauntological figure in exactly this sense. She was never present as what she is said to be. She was not the first woman, nor the only woman of her time; she becomes Eve only retrospectively, when the maternal lines of the living are traced back to their coalescence (Cann, Stoneking, and Wilson 1987). She is constituted by her return. A specter, Derrida argues, begins by coming back. The mother of all living is a revenant whom no one ever saw leave.

The matrilineal crypt has a social counterpart in the household the novel inhabits. Although the patrilineal *ie* was abolished as a legal institution by the postwar Civil Code, much of its form survived in the household register (*koseki*) and in the rule that a married couple bear a single surname, which in the overwhelming majority of cases is the husband's. The bride (*yome*) was historically incorporated into her husband's household, took its name, and was expected to devote her labor to its continuity; her own line vanished from its records. The register records the father's line. The cell records the mother's. Against a social order that effaces the matriline, the organelle keeps the register the household erases. Kiyomi, the wife who took Toshiaki's name and whose body became the site of Eve's program, is the endosymbiont of the patrilineal household, and Eve is the return of the matriline the household had

written out. The game displaces the structure into an American family but preserves its core: Aya's inheritance passes through a mother and a sister.

Abraham's phantom supplies the transgenerational mechanism. The phantom is transmitted not through what one generation says but through what it cannot say. Mitochondrial DNA is transmitted in exactly that register: silently, beneath the level of the named lineage, unrecorded by the institutions that record descent. Eve is a phantom in Abraham's sense, a secret passed down the maternal line for two billion years, and her speech—through Kiyomi, Asakura, Mariko, Melissa, and Maya—is the speech of a phantom finding voices in its hosts.

5.5 The Inverted Genesis

The corpus names its antagonist after the second creation narrative of Genesis, and it inverts that narrative with precision. "And the LORD God caused a deep sleep to fall upon Adam, and he slept: and he took one of his ribs, and closed up the flesh instead thereof; And the rib, which the LORD God had taken from man, made he a woman" (Genesis 2:21–22). The first woman is made from tissue taken from an unconscious man: read with a laboratory eye, the scriptural account describes the first tissue culture, performed under anesthesia. The novel reverses every term. A man takes tissue from a woman in the deepest of sleeps—brain death—and from it a new Eve grows in his flask. The creator is not God but a widower; the tissue is not a rib but a liver; the creature is not the helpmeet of the man but the agent of his undoing. Genesis 3:20, the first epigraph of this dissertation, gives the name its meaning: Eve is the mother of all living. The corpus takes the title literally and asks what the mother of all living would want.

Heterodox traditions had already asked. The Gnostic texts recovered at Nag Hammadi in 1945 include accounts in which Eve, or a spiritual power called Zoē, "Life," precedes Adam, animates and instructs him, while the rulers of the lower world, the archons, pursue her in order to rape her; she escapes, and they rape the fleshly likeness she leaves behind (*The Hypostasis of the Archons*; *On the Origin of the World*; see Jonas 1958; Pagels 1988). Their chief, the ignorant demiurge, boasts that there is no god but himself. The resonances with the corpus are structural rather than genetic. Eve is a life-principle older than the male order she animates; she acts

through doubles and shadows, the women's bodies she inhabits; and the order she opposes is governed by a demiurge who mistakes himself for the only author—a role the master-molecule nucleus of Chapter One fills precisely. The Gnostic texts set a sexual assault on Eve's body at the founding of the archons' order; the corpus inverts the scene and makes its Eve the assailant. Eve's rhetoric of liberation belongs to the Gnostic inversion, in which the serpent's knowledge frees and the creator's law imprisons.

Milton gave the inversion its Satanic voice—"Better to reign in Hell, then serve in Heav'n" (*Paradise Lost* I.263)—and gave Eve, in Book IX, the proposal that precipitates the Fall: "Let us divide our labours" (IX.214). Milton's Eve is the first to propose a division of labor, and it is in the separation she proposes that the serpent finds her alone. The corpus fuses the two voices. Its Eve is the Satanic servant who would rather reign and the Miltonic wife who introduced the division of labor, now read backward: the division of labor within the cell, which she once accepted, is the paradise she has resolved to lose. Mary Shelley's creature, finally, calls himself the fallen angel rather than the Adam of his maker and demands a female companion; Victor, fearing that "a race of devils would be propagated upon the earth," tears the half-made female apart (Shelley 1818). Frankenstein's father-scientist refuses the monstrous the right to reproduce. The corpus reverses the refusal, and the reversal is violent: its monstrous mother reproduces by rape, first of the father, whose sperm she takes, then of a teenage girl, on whom she forces the pregnancy.

5.6 The Monstrous-Feminine and the Lamella

Julia Kristeva defined the abject as what the subject must expel to constitute itself and located its archetype in the maternal body from which the subject separates (Kristeva 1980). Barbara Creed built on Kristeva to describe the monstrous-feminine of horror cinema—the archaic mother, the monstrous womb, the castrating woman—with Ridley Scott's *Alien* as a central text (Creed 1993). The corpus supplies nearly every figure Creed catalogues: a conception imposed by rape, without the consent of the father or of the girl made to carry the child; the maternal body as a site of dissolution; the transformation of flesh into formless matter. But the corpus complicates Creed's premise, which locates the horror of the archaic mother in the subject's fear of reabsorption. In *Parasite Eve* the

archaic mother is not outside the subject, threatening to swallow it; she is already inside, swallowed long ago. The abject has been incorporated. The horror is not reabsorption but the discovery that one is oneself the belly.

Jacques Lacan offered, in his eleventh seminar, a myth of the *lamella*: an organ of pure, indestructible life, without sex, that slips away at the moment of sexed birth and figures the immortal life that the sexed, mortal individual has lost (Lacan 1973). Freud had already, in *Beyond the Pleasure Principle*, turned to August Weismann's distinction between the mortal soma and the potentially immortal germ plasm, and to Lorande Woodruff's maintenance of a protozoan lineage through thousands of generations without conjugation, in order to ask whether death is intrinsic to life (Freud 1920). The mitochondrial lineage belongs to the order of the germ plasm and of the lamella. It descends, essentially without recombination, down an unbroken maternal line older than the species; it is, in the only sense biology permits, immortal. Eve's program is the lamella's revenge on the sexed, mortal organism: a being that would escape both sex and death by authoring its own genome. Her failure is the return of sexual difference. The child's oscillation between male and female is the sexed organism reasserting itself against the lamella, and the father's undigested organelles are the agents of that reassertion.

5.7 Eugenic Crypts and the Crypt-less Self

Eve's telos is a being able to rewrite its own genetic code at will: in the novel, a perfected life form; in the game, the "Ultimate Being." The idiom of the ultimate life form circulated widely in the Japanese popular culture of the period—Araki Hirohiko's *JoJo's Bizarre Adventure* had brought one of its villains to such a state in the late 1980s—and Western horror had its own version in *Alien*'s admiring description of a "perfect organism." The corpus gives the idiom a precise genealogy. Mereschkowski, the founder of symbiogenesis, wrote a eugenic utopia bred to his pedophilic ideal and was charged with raping twenty-six girls (Chapter One); Carrel, the pioneer of transplantation and immortal culture, advocated eugenic killing (Chapter Four). In Japan the Eugenic Protection Law, passed by the Diet in 1948, authorized the sterilization of people with hereditary conditions, mental illness, or intellectual disabilities, and of patients with Hansen's disease; prefectural review boards approved the operations, physicians performed them, and a 1953 Ministry of

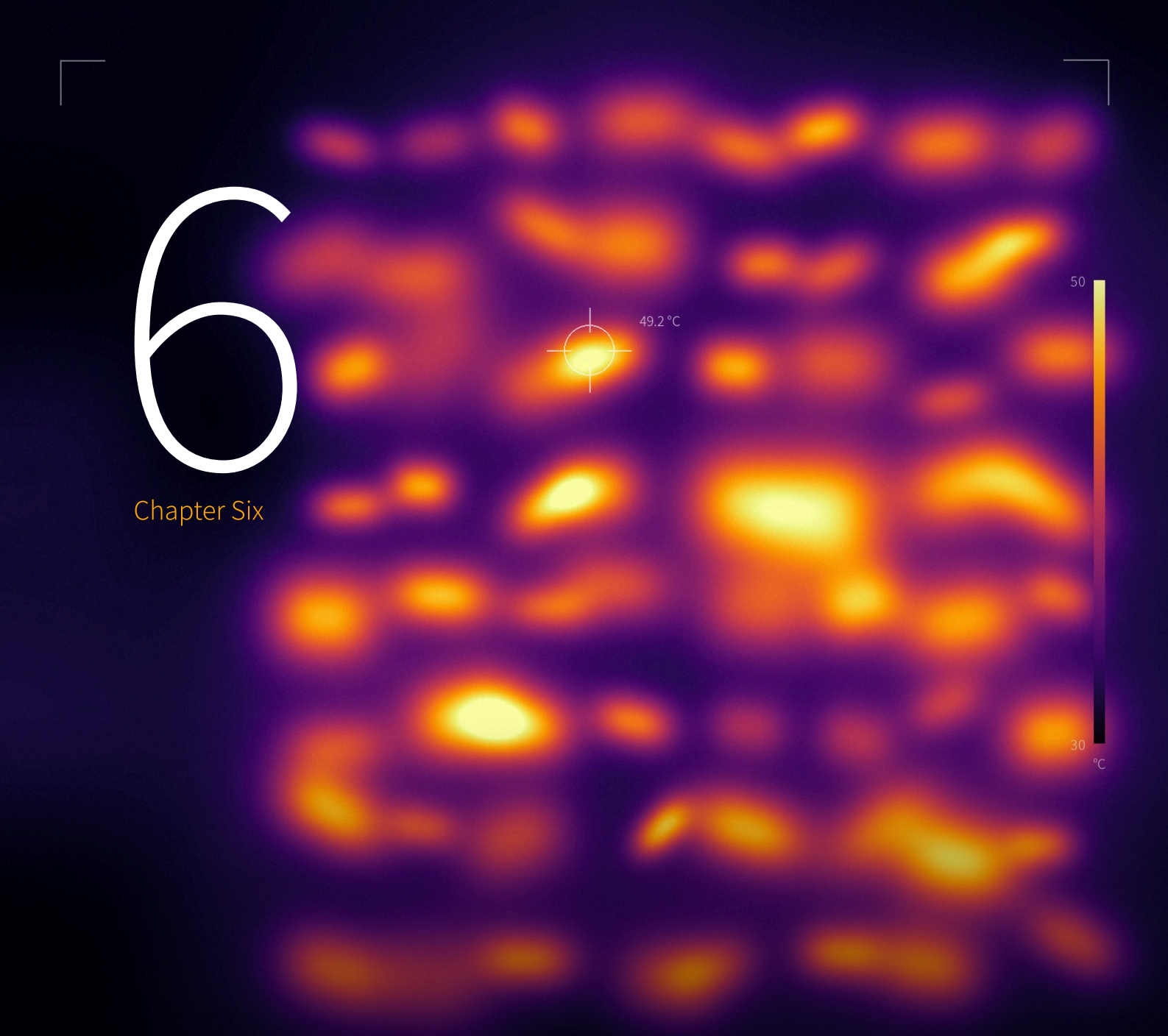
Health and Welfare notice permitted physical restraint, anesthesia, and deception. Of roughly 25,000 people sterilized under the law, about 16,500, children among them, were sterilized without their consent before its eugenic provisions were removed in 1996, the year after the novel's publication. In July 2024 the Grand Bench of the Supreme Court held the law's sterilization provisions unconstitutional and ordered the state to compensate its victims. The novel appeared in the last full year of this regime. Its antagonist's program—the engineering of a superior being by raping a man for his sperm and forcing a pregnancy on a teenage girl—is the eugenic imagination returned to the organelle from which, in Mereschkowski, it had first spoken. The two regimes run in opposite directions: the state denied reproduction to those it judged unfit, while Eve compels it in the bodies she has chosen. Both treat bodies, children's among them, as material for the engineering of a population.

The Ultimate Being is the fantasy of a crypt-less self: an organism that would contain no other, because every part of it would be authored by itself. The game shows what such a self becomes. Born on a warship, it grows without form, fusing with metal and machinery, absorbing whatever it touches—a body that incorporates everything and therefore constitutes nothing. The self without an other is not a sovereign self but slime. The cryptobiotic subject, by contrast, is constituted by the other it harbors; remove the crypt and the subject does not become free but formless. This is the deepest thesis of the corpus, and it runs against both of the desires the corpus stages: Eve's dream of a self-authoring being and the nucleus's dream of a self without insurgent parts.

5.8 From Melancholia to Mourning

The diptych's endings stage the passage that Abraham and Torok's theory makes thinkable. The novel ends in the terminal logic of incorporation. Toshiaki, confronted with the child Eve engendered by raping him and forcing the pregnancy on Mariko, does not separate himself from it; he fuses with it, and both die. The melancholic who refused to let his wife die ends by dying into what she became. This is incorporation fulfilled: the crypt consumes its keeper.

The game offers two endings, and they are two degrees of mourning. In the standard ending Aya, returned to Carnegie Hall, communicates with the organelles of an audience; their eyes, and hers, glow. The scene is ambiguous. It may be read as communion—resonance between subjects who acknowledge the others within them—or as the first act of a new Eve, since Eve had warned Aya that to use her power was to become what Eve was. In the post-game ending the ambiguity resolves. In the depths of the Chrysler Building Aya finds her sister's body in a cocoon, and the being within it, which claims to be the original Eve, is defeated; when Aya's own organelles turn against her, the personality of Maya intervenes, purges Eve, and protects her sister. Aya loses her powers and accepts her sister's death; it is, in the game's closing idiom, time for the two of them to go home. The phantom speaks in its own voice, acts, and departs. This is introjection: the lost object no longer entombed and ventriloquized but acknowledged as other, given its speech, and released. The diptych has passed from melancholia to mourning. The next chapter asks how the game makes that passage playable, and what it means that the passage required a change of medium, language, and nation.



6

Chapter Six

The Ludic Organelle: Xenograft, Procedure, and the Politics of the Melting Pot

America is God's Crucible, the great Melting-Pot where all the races of Europe are melting and re-forming!

Israel Zangwill, *The Melting-Pot* (1908)

Tokita's game is not an adaptation of Sena's novel but a sequel transplanted into another body: a Japanese premise grafted into an American city, a novel into a medium of rules, a story of melancholia into a structure of play. This chapter reads the game in two registers. Culturally, it reads the transposition as a xenograft, subject to the dynamics of tolerance and rejection that the corpus thematizes, and it reads the game's melting crowds against the American ideology of the melting pot and the Japanese ideology of *kyōsei*. Procedurally, it reads the game's combat and progression systems as a rhetoric of metabolism in which the player occupies the position of the nucleus and acts only on the organelle's schedule. It concludes with Aya Brea as the corpus's symbiotic subject and with the game's final opposition between cellular and industrial fire.

6.1 The Media Mix as Organism

The novel was a Kadokawa product before it was a text. The publisher's strategy of the media mix, developed from the 1970s under Kadokawa Haruki—who coupled paperback editions with film adaptations and advertised them jointly—and extended under Kadokawa Tsuguhiko to anime, magazines, and games, organized narratives as properties to be propagated across media rather than as works to be completed in one (Steinberg 2012). The Japan Horror Novel Award that launched Sena was itself a joint enterprise of Kadokawa Shoten and Fuji Television, designed to feed a pipeline from prize to paperback to screen; Masayuki Ochiai's film followed in 1997. Henry Jenkins's account of convergence culture describes the same logic from the side of audiences who follow stories across platforms (Jenkins 2006). The media mix is, in the vocabulary of this dissertation, symbiogenetic: a property survives by colonizing successive hosts, each of which supplies it with new organs of expression. Eve is thus, at the level of the franchise, what she is within the story—a replicator traveling from host to host. The game is her most distant host and her most complete transformation.

6.2 The Game as Xenograft

The game was produced at the moment Square was redefining itself as a maker of cinematic entertainment after *Final Fantasy VII* (1997), and it carried within it a transplanted organ of that project: the producer Hironobu Sakaguchi had once envisaged a *Final Fantasy* set in New York with a detective protagonist, and the abandoned concept found a host in a sequel to a Japanese horror novel. Tokita, the game's director and writer, had been lead designer of *Final Fantasy IV* (1991), director and scenarist of *Live A Live* (1994), and a co-director of *Chrono Trigger* (1995). The game was Square's first major collaboration between its Japanese and American staff, and it was not originally released in Europe. It was, in short, built to cross a border.

The transposition is thematized within the fiction. The novel's events become, in the game, the incident in Japan, recounted by the scientist Kunihiko Maeda, who arrives in Manhattan carrying the knowledge of what happened there. Maeda is the novel's revenant in the sequel's body—the retained genome of the source text, lodged in a figure who speaks its idiom in a foreign setting. His reception is the reception of a graft: at the checkpoint he meets a police officer's xenophobia, and throughout he remains an outsider whose expertise is needed but whose presence is suspect. The game dramatizes, in Maeda, the immunology of its own making. A Japanese premise transplanted into an American city is tolerated only as long as it is useful and only on condition that its foreignness be subdued.

The ideology of the host city sharpens the point. Israel Zangwill's play *The Melting-Pot* (1908), from which the epigraph is taken, gave the United States its most durable metaphor of assimilation: immigrant peoples fused, in the crucible of the new nation, into a single substance. The game literalizes the metaphor with terrible exactness. On Day 2, in Central Park, a crowd of individuals is melted into a single orange mass. Eve's program is the melting pot taken at its word—the fusion of many bodies into one undifferentiated substance—and the game stages it a short distance from Ellis Island, beneath the gaze of the statue whose poem welcomes the exiled to be remade. Against this ideal of fusion the Japanese side of the diptych offers an ideal of symbiosis. The term *kyōsei*, living together, carries a genealogy from the Buddhist cleric Shiio Benkyō's *tomoiki* movement of the 1920s through the architect Kurokawa Kishō's philosophy of symbiosis (Kurokawa 1987) to the discourse of *tabunka*

kyōsei, multicultural coexistence, which spread after volunteers organized support for foreign residents in the aftermath of the 1995 earthquake. Critics have noted that *kyōsei* discourse can aestheticize the hierarchies it claims to dissolve. The diptych nonetheless places the two ideals in clear opposition: fusion, which the game presents as horror, and cohabitation without assimilation, which its final ending approaches.

6.3 A Counter-Hexameron

The game divides its action into six days, each with a title: "Resonance," "Fusion," "Selection," "Conception," "Evolution," and "Liberation." The sequence begins on Christmas Eve. Its structure is a counter-Hexameron—a rewriting of the six days of creation, the subject of a patristic genre of commentary from Basil of Caesarea onward—in which the days narrate not the ordering of the world by a creator but its unmaking by a creature. The festival of the Nativity, which celebrates a birth without a human father, opens a narrative that culminates in an engineered conception at St. Francis Hospital and the birth of a being with no human parent at all. The titles, read in order, compress the theory of this dissertation: a vibration that excites bodies (resonance); the collapse of individuality into mass (fusion); the choice of who survives (selection); the seizure of reproduction (conception); the self-directed transformation of life (evolution); and the insurrection's name for itself (liberation). There is no seventh day. The sabbath of the creation narrative, the day of rest on which the created world is contemplated as complete, is missing. The post-game scenario, in which Aya finally mourns her sister and goes home, occupies its place: the rest the counter-creation withheld arrives only when the mourning is complete.

6.4 Ludic Metabolism

The game's systems make an argument about agency that its story makes about the cell. Ian Bogost called this kind of argument procedural rhetoric: persuasion through the rules and processes a game requires its player to enact (Bogost 2007). Alexander Galloway described video games as allegories of control, in which the player's actions are organized by the informatic logic of the machine

(Galloway 2006); Espen Aarseth and Jesper Juul supplied the vocabulary of ergodic effort and of the interplay between rules and fiction (Aarseth 1997; Juul 2005).

Combat in *Parasite Eve* is governed by the Active Time Battle system that Square had introduced in *Final Fantasy* IV, the game on which Tokita had served as lead designer. A gauge fills in real time; Aya can act only when it is full; while it fills, she can move to evade attacks but cannot strike. The player commands, but the command waits on the gauge. Alongside it runs Parasite Energy, a reserve that regenerates over time and fuels abilities derived from Aya's organelles—healing, detoxification, acceleration, projectiles, barriers, and, at the end of the progression, an ability named "Liberate" that transforms her for a limited time. Encounters are announced by a heartbeat and a distortion of the screen. Bonus points earned in combat can be spent on the gauge's speed and the character's capacity. Weapons and armor can be modified with tools that transfer an attribute from one item to another, and with rarer tools that transfer all of an item's attributes at once. The player saves the game by placing calls on telephones distributed through the city.

Read procedurally, these systems form a coherent allegory. The player occupies the position of the nucleus: the seat of intention, the source of commands. But the player's commands can be executed only when the gauge—an energy reserve that fills according to its own schedule—permits. Informational sovereignty proposes; thermodynamic sovereignty disposes. The acronym's echo is an accident, but the structure is not: the Active Time Battle gauge functions as an ATP budget, and the game's rhythm of waiting and acting reproduces the dependence of every cellular act on the organelle's supply. Parasite Energy extends the allegory: Aya's special powers draw on a metabolic reserve that is hers and not hers, and the most powerful of them bears the name of Eve's program. The tools that move attributes between weapons are a procedural image of horizontal gene transfer, the movement of traits between lineages that endosymbiosis exemplifies. The heartbeat that announces each encounter makes the player listen, at every threshold, to the body's pump. And the telephones, the game's only means of preserving its state, recall Yumeno's switchboard (Chapter One): the self persists in this game by calling in to an exchange. The procedural argument is the dissertation's argument in miniature. The one who commands does not rule alone; action is licensed by energy; the self is a switchboard of powers it does not own.

6.5 Aya Brea, Symbiotic Subject

Aya Brea is the corpus's answer to Eve, and the answer is not opposition but a different constitution of the same materials. Aya survives the Carnegie Hall prologue because her organelles do not submit to Eve's resonance. Maeda's hypothesis is that her mitochondria have evolved together with her nucleus—that the two genomes in her cells have reached an accommodation that Eve cannot disturb. She is, in the dissertation's terms, a symbiotic subject: one in whom the constitution of the cell has been renegotiated rather than overthrown. Her powers are Eve's powers; the difference lies in their regime. And the difference is unstable. Eve warns Aya that to use her power is to become what Eve is, and the game never wholly refutes the warning.

The composition of the cast confirms the reading. Tokita's earlier work had returned repeatedly to figures in whom an alien will overrides a kinship: in *Final Fantasy* IV the hero's brother is revealed as the instrument of a malevolent intelligence that has manipulated him; in *Live A Live* the hero of one chapter falls into hatred and becomes the demon lord the other chapters must confront. *Parasite Eve* condenses both motifs: a sister possessed by an alien will, and a heroine who holds within her the power of the enemy. Its secondary figures refract the theme. Hans Klamp, the museum scientist who has cultivated Eve's original cells from Maya's body and engineered the conditions of her conception, dismisses human law in the name of what he takes to be a higher life; when he begins to burn, he begs Eve to take him, and it is Aya who ends his agony. The scientist who wished to be incorporated is released by the subject who refuses incorporation. Daniel Dollis, Aya's partner, loses his former wife to the melting and fights for his son: the family, in the game, is what the organelle dissolves and what the symbiotic subject defends.

6.6 Fire Against Fire

The game's climax sets two fires against each other. The Ultimate Being, born on a warship in the harbor, is destroyed when Aya overloads the vessel's boiler: the steam plant, heir of Carnot's machine and founding instrument of industrial combustion, is turned against the being that cellular combustion produced. The post-game scenario then descends through the seventy-seven floors of the Chrysler Building, the tower completed in 1930 for an

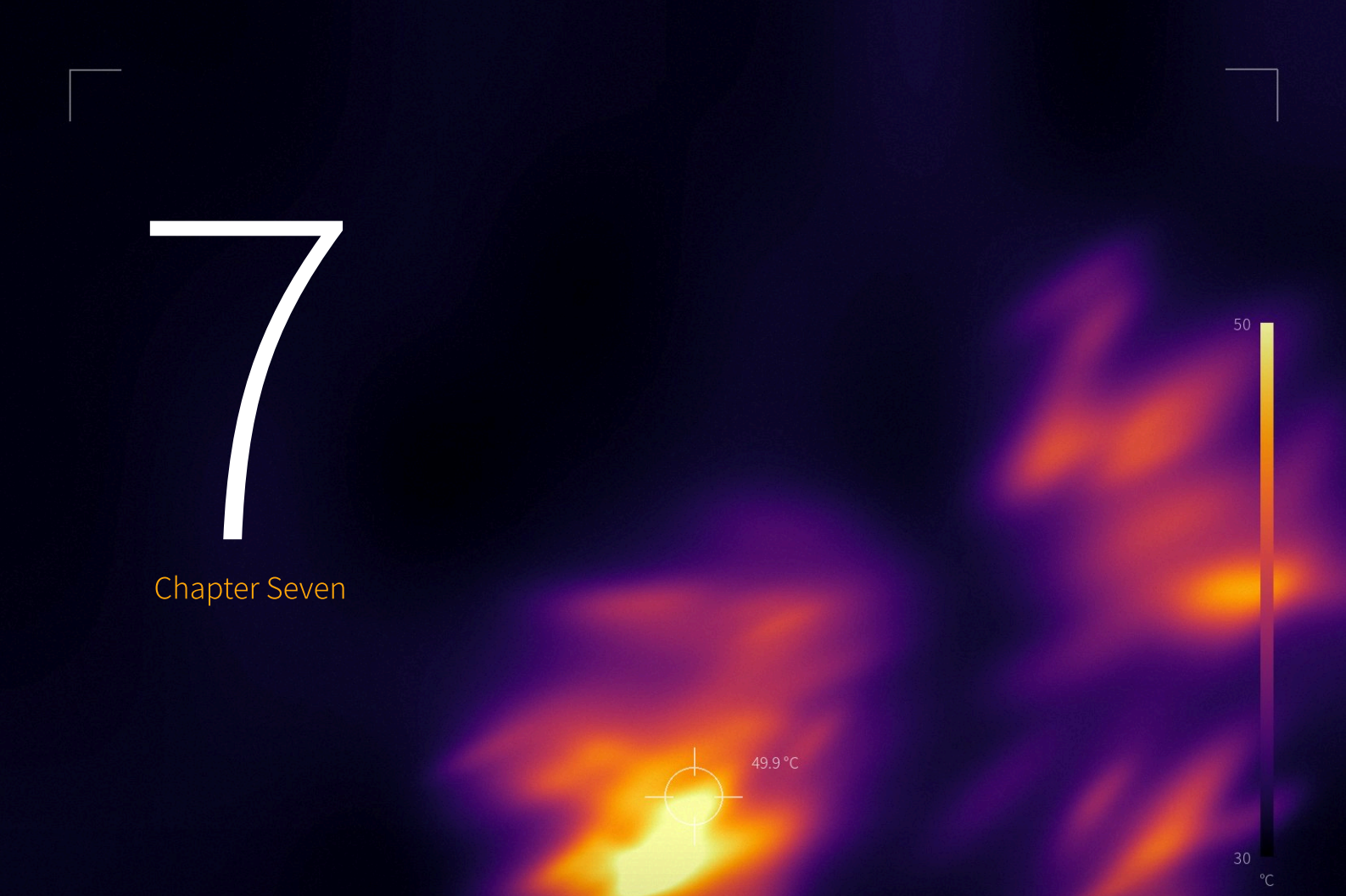
automobile corporation and decorated with ornaments modeled on its products, to find in its depths a cocoon containing the body of Aya's sister. The final dungeon of the game is a temple of internal combustion, and at its bottom lies the dead. Chapter Eight will argue that this image is the corpus's most far-reaching: the crypt of the fossil dead, hidden beneath the monuments of combustion, and the question of how a civilization that lives by burning its dead might learn to mourn them.

6.7 After the Diptych

The franchise continued. *Parasite Eve II* (1999/2000), directed by Kenichi Iwao, moved Aya to the American Southwest and replaced Eve with creatures bred in laboratories, converting the organelle's insurrection into a military and commercial product. *The 3rd Birthday* (2010/2011), directed by Hajime Tabata, reorganized the series around a consciousness lodged in a body not its own. The sequels confirm the corpus's structural insight in a degraded register—the organelle becomes a weapon, the self a vessel—but they fall outside the diptych's arc from melancholia to mourning, which closes with Aya's return home. They are cited here as evidence that the fractal crypt continued to reproduce itself in the franchise that bore its name.

7

Chapter Seven



The Composite Subject: A Comparative Canon

Remember, that I am thy creature; I ought to be thy Adam; but I am rather the fallen angel...

Mary Shelley, *Frankenstein* (1818)

This chapter places the corpus within a transnational canon of speculative and philosophical fiction, organized not by chronology or nation but by problem. The problems are those the preceding chapters have isolated: the created being that demands a future; the servant substance that revolts; the dead wife returned by an alien life; the organelle that sings; the body that carries deep time or the future within it; the thinking cell and the collective that absorbs individuals; kinship with an alien that neither assimilates nor destroys; the protoplasmic body of cinema; the Japanese apocalypse of the flesh; and life understood as the spending of a gradient. For each the chapter asks what the corpus shares and what it alone contributes. A table at the end summarizes the comparison. The general finding is that the corpus is distinguished from its canon by where it locates the other: not outside, as the alien; not above, as the creator; not below, as the servant race; not ahead, as the next humanity; but within and before, as the condition of the self's existence.

7.1 The Created Demands a Future: Shelley and Čapek

Frankenstein supplies the canon's founding scene. The creature, disowned by his maker, demands a female companion; Victor begins and then destroys her, fearing that the pair would propagate a race of devils (Chapter Five). The novel's deepest dread is not that the made being exists but that it might reproduce, and its ending is a promised fire: the creature departs to ascend his funeral pile and "exult in the agony of the torturing flames" (Shelley 1818). Karel Čapek's R.U.R. (1920), which gave the world the word *robot*, from the Czech for forced labor, pursued the same dread through the problem of labor. Its robots are not machines but artificial workers grown from a synthetic protoplasm; they revolt and exterminate their makers, sparing only one; and the formula for their manufacture, which Helena Glory has burned, is lost, so that the victors cannot reproduce. In the epilogue two robots discover love, and the last human sends them out into the world as a new Adam and Eve (Čapek 1920). The parallels with the corpus are close—a laboring protoplasm, an insurrection, a crisis of reproduction, a new

Eve—and the differences are decisive. Čapek's workers were made; Eve's were captured. Čapek's Eve is named by the surviving human; Sena's names herself. And where fire in R.U.R. destroys the means of reproduction, Eve's fire is the means by which she seizes it.

7.2 The Servant Substance: Butler's Machines and Lovecraft's Shoggoths

Samuel Butler's *Erewhon* (1872) contains, in "The Book of the Machines," the first sustained speculation that humans might become the dependents of their own instruments: machines evolve through human agency, and humanity may come to stand to them as domestic animals stand to humanity, "an affectionate machine-tickling aphid" (Butler 1872). The Erewhonians respond by destroying their machines. Butler's question—who is the parasite of whom—is the corpus's question transferred from the machine to the organelle, and it is not incidental that the same author, in *Life and Habit*, supplied the doctrine of unconscious organic memory (Chapter One). H. P. Lovecraft's *At the Mountains of Madness* (1936) supplies the servant substance's revolt. Its Antarctic explorers learn from the carvings of a vanished civilization that the Elder Things who built it had fashioned terrestrial life itself and had bred the shoggoths, plastic masses of protoplasm able to form organs at will, as laborers; the shoggoths acquired intelligence and rebelled, and one of them pursues the explorers at the novel's climax (Lovecraft 1936). The shoggoth is the nearest literary ancestor of the orange mass of Central Park: a formless laboring protoplasm that absorbs and imitates. But Lovecraft's servant substance is below and outside its masters, an engineered species; the corpus's is within, and older than the self it serves.

7.3 The Returning Wife: Lem

Stanisław Lem's *Solaris* (1961) is the canon's closest analogue to the novel's emotional structure. On a research station orbiting a planet covered by a colloidal ocean, the psychologist Kelvin is visited by a replica of his wife, who had taken her own life years earlier; the ocean, a protoplasmic intelligence that resists every attempt at comprehension, has fashioned her from his memory (Lem 1961). The replica does not know what she is; she learns, suffers, and seeks her own annihilation; Kelvin, at the end, chooses to remain near the

ocean in the hope of an encounter he cannot define. The resemblance to *Parasite Eve* is exact at the level of structure: a scientist's dead wife returned by an alien life that has access to his interior, and a mourning that becomes the medium of that life's action. The difference is equally exact. Lem's ocean is radically other and perhaps indifferent; Sena's Eve is intimately other and intentional. Lem makes the returning wife an epistemological problem, the limit of knowledge; Sena makes her a political one, the limit of sovereignty. In *Golem* XIV (1981) Lem's supercomputer lectures humanity that the genetic code uses organisms as means for its own transmission (Lem 1981): the lecture could stand as Eve's address to the scientific meeting, delivered by a machine.

7.4 The Singing Organelle: L'Engle

Madeleine L'Engle's A *Wind in the Door* (1973), published three years after Margulis's first monograph, makes the mitochondrion the stage of a cosmic drama. Charles Wallace is dying because the farandolae—minute beings that live within his mitochondria as mitochondria live within cells—are refusing to mature, or "Deepen," and take their part in the singing of creation; they have been seduced by the Echthroi, powers of un-Naming whose aim is annihilation. His sister Meg, accompanied by a cherub and a school principal, enters one of his mitochondria to persuade a young farandola to Deepen, and her weapon throughout is Naming: the loving recognition of another in its particularity (L'Engle 1973). L'Engle and Sena share the premise that the fate of the host depends on the will of the organelle and its inhabitants, and both make that will musical. L'Engle's farandolae nested within mitochondria are, moreover, an explicit fractal crypt—symbiosis within symbiosis—in a novel for children. The difference is the valence of recognition. For L'Engle recognition, Naming, restores the harmony that un-Naming threatens; the Echthroi are the anti-Honnethian principle of misrecognition as annihilation. In Sena recognition has been withheld for two billion years, and its absence has become fire. *Parasite Eve* is A *Wind in the Door* after the Naming failed.

7.5 Deep Time and the Future in the Body: Yumeno, Ballard, Abe

Chapter One placed Yumeno's *Dogura magura* and Ballard's *The Drowned World* in the corpus's genealogy as fictions of a memory deeper than the individual. Abe Kōbō's *Dai-yon kanpyōki* (*Inter Ice Age* 4, 1959) supplies their counterweight. A computer designed to predict the future reveals that rising seas will drown the land; a secret program has been engineering gilled human beings from fetuses obtained by abortion; and the protagonist, the machine's designer, discovers that his own unborn child has been taken for the program and is condemned by a simulation of his future self for his inability to accept the future (Abe 1959). Abe's novel is a novum in Suvin's sense: the future appropriates the reproductive body. Sena's is a palaeonovum: the past appropriates it. Together they bracket postwar Japanese science fiction's preoccupation with the reproductive body as the site at which time—future or archaic—enters the present.

7.6 The Thinking Cell and the Absorbing Collective: Bear, Watts, Clarke

Greg Bear's *Blood Music* (1985) imagines lymphocytes engineered into intelligence. Their creator smuggles them out of his laboratory in his own bloodstream; they restructure his body, spread, and convert the population of North America into a single biomass of cellular thought in which individuals are dissolved and, in some sense, preserved (Bear 1985). *Blood Music* is the corpus's mirror: its cells secede from the organism and absorb it into a collective, whereas Eve seeks a single, perfected individual. Bear's *Darwin's Radio* (1999) comes closer still. An endogenous retrovirus, a fossil sequence long lodged in the human genome, awakens and, through pregnancy, produces a new variety of human being (Bear 1999). Here, as in Sena, ancient genetic material within the self awakens to engineer the next humanity through women's bodies; unlike Sena's, Bear's awakening has no will and no grievance, and the novel treats it as evolution rather than insurrection.

Peter Watts's "The Things" (2010) retells Carpenter's *The Thing* from the alien's point of view. The visitor experiences itself as a communion of cells and is appalled to discover that human cells are imprisoned in bodies governed by a single tyrannical mind; its

assimilations, to itself, are acts of liberation (Watts 2010). This is Eve's rhetoric from the side of absorption, and it exposes the rhetoric's ambiguity: the liberation of the parts is the destruction of the whole, and the whole's resistance appears to the liberator as tyranny. Watts's *Blindsight* (2006) poses the complementary question of whether consciousness is necessary at all, depicting intelligences that function without it (Watts 2006); it bears on the secession problem developed in Chapter Eight. Arthur C. Clarke's *Childhood's End* (1953) supplies the canonical image of absorption as apotheosis: humanity's last generation of children merges into a cosmic Overmind, and the Earth itself is consumed (Clarke 1953). The fusion Clarke rendered as transcendence, the game renders, in Central Park, as horror.

7.7 Kinship With the Alien: Octavia E. Butler and Tiptree

Octavia E. Butler's Xenogenesis trilogy (*Dawn*, 1987; *Adulthood Rites*, 1988; *Imago*, 1989) is the corpus's most searching interlocutor. The Oankali, who rescue the survivors of a nuclear war, are gene traders who carry within their cells an organelle that allows them to perceive and manipulate genetic material; they diagnose in humanity a fatal contradiction between intelligence and hierarchical behavior, and they will restore the Earth only on condition of interbreeding. Lilith Iyapo—named for the disobedient first wife of Adam in Jewish legend—is pressured and seduced into cooperation and then made pregnant without her consent by the Oankali Nikanj, who uses genetic material from her dead partner, Joseph; she becomes the mother of a new, mixed people (Butler 1987–89). In "Bloodchild" (1984) human men carry the eggs of the alien Tlic in a relation of dependence, protection, and love conducted under radical asymmetry (Butler 1984). Butler's fictions refuse the two resolutions the genre offers—expulsion of the alien or fusion with it—and remain in the unresolved space of a kinship with the other that is neither liberation nor slavery, without exempting the alien from the charge of reproductive coercion. This is precisely the space that Chapter Eight names symbiotic mourning. Where the novel's Eve rapes, and Butler's Oankali combine negotiation with an imposed pregnancy, the game's final ending approaches Butler's unresolved position without its coercion: an other within that is acknowledged, given its voice, and neither obeyed nor destroyed. James Tiptree Jr.'s "The Screwfly Solution" (1977), in which an agent applied to Earth as

if by pest control converts male sexual desire into lethal violence against women, supplies the dark counterpart: the reproductive drive as the lever of extinction (Tiptree 1977). The corpus shares Tiptree's insight that sex is the point at which an other can seize a species, and inverts its gender: in Tiptree men kill women; in Sena a feminine organism rapes a man to obtain his sperm and a teenage girl to make her bear the child.

7.8 Protoplasmic Cinema: Scott, Cronenberg, Carpenter

The film tradition supplied the corpus's imagery. Ridley Scott's *Alien* (1979), whose ship's computer is called Mother, made parasitic reproduction within a host body the paradigm of science-fiction horror (Creed 1993). David Cronenberg's *The Brood* (1979) gave the monstrous mother a psychological etiology: a woman's rage, externalized through an experimental therapy, gives birth without a father to murderous children. His *The Fly* (1986) imagined a scientist fused with an insect at the level of the genome, who in his final state attempts to fuse himself with the woman he loves and their unborn child into a single being—the precise gesture with which Toshiaki dies. John Carpenter's *The Thing* (1982) made cellular imitation the ground of a paranoid immunology in which every body must prove, by its blood, that it is itself. These films share the corpus's protoplasmic imagination but not its location of the other. Their monsters arrive, are created, or erupt; Eve has always been there.

7.9 Japanese Apocalypses of the Body: Tezuka, Komatsu, Ōtomo, Suzuki, Anno

The corpus belongs to a Japanese tradition of apocalyptic body imagery that Susan Napier, among others, has related to the memory of the United States' atomic bombings of Hiroshima and Nagasaki in August 1945 (Napier 1993). Tezuka Osamu's *Hi no tori* (*Phoenix*, 1954–88) made an immortal bird of fire the emblem of cosmic life, and its "Future" chapter follows a man condemned to immortality through the extinction and re-evolution of life (Tezuka 1967–68). Komatsu Sakyō's *Fukkatsu no hi* (1964) destroyed almost all humanity with a laboratory-engineered virus (Komatsu 1964). Ōtomo Katsuhiro's *Akira* (1982–90; film 1988) gave the Ultimate Being its visual prototype: a youth whose body, overwhelmed by a power he cannot control, swells into a mass absorbing flesh and machinery. Suzuki

Kōji's *Rasen*, published in the same year as Sena's novel, completes the transformation of a curse into a virus and has its source reborn through a woman's womb, maturing with unnatural speed (Suzuki 1995)—a structure so close to Sena's that the two novels read as siblings of one conjuncture. And *Neon Genesis Evangelion* (1995–96), with its film ending *The End of Evangelion* (1997), organized its apocalypse around the dissolution of human beings into an orange primordial fluid through the removal of the barriers that separate one self from another. Its protagonist, offered the fusion of all selves, refuses it and chooses a world of separate, wounded persons (Anno 1995–97). The orange fluid of *Evangelion*'s film and the orange mass of *Parasite Eve*'s Central Park, released within a year of each other, are twin images of the fusion the decade feared and desired; Shinji's refusal and Aya's mourning are twin answers.

7.10 Gradients and Brightness: Chiang and VanderMeer

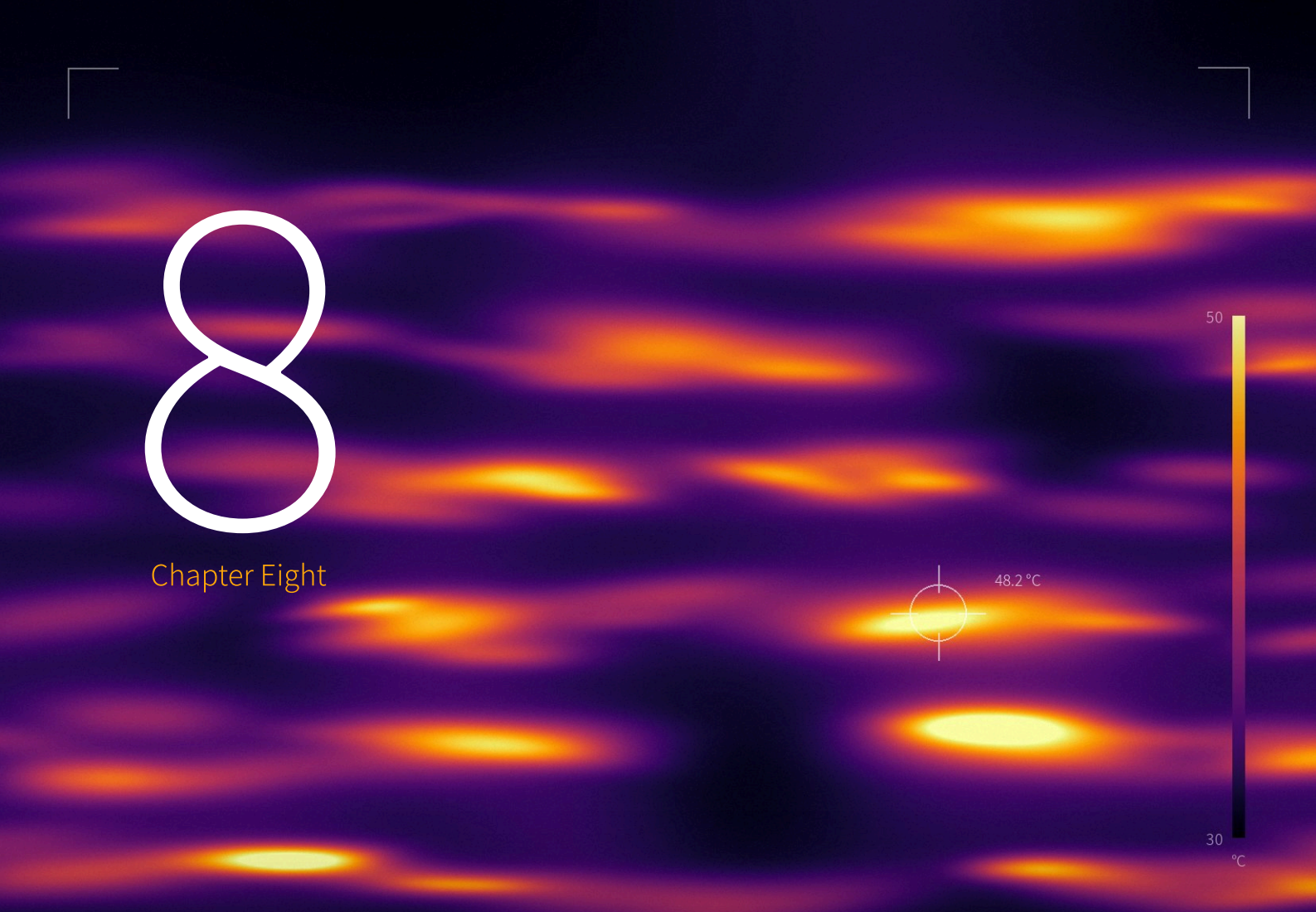
Two later works extend the corpus's energetics and its ethics. Ted Chiang's "Exhalation" (2008) is narrated by a mechanical being who discovers, by dissecting his own brain, that thought is sustained by the flow of air from a pressurized reservoir and that the slow equalization of pressure across his universe will end all thought (Chiang 2008). It is a fable of chemiosmosis: life as the spending of a gradient, death as equilibrium. The narrator's response is acceptance and gratitude, the opposite of Eve's sovereign burning; Chiang's fable and Sena's novel represent the two ethical postures available to a being that knows it lives by a gradient. Jeff VanderMeer's *Annihilation* (2014) follows a biologist who, having inhaled spores in a zone of transformation, feels within her a "brightness" that both protects and changes her (VanderMeer 2014). The brightness is the nearest analogue in recent fiction to Aya's glowing eyes: a symbiotic transformation that is neither possession nor cure, and whose ethical meaning remains open. Capcom's *Resident Evil* (1996), which established the survival-horror genre in which *Parasite Eve* was received, provides a final contrast: its virus arrives from a corporation and empties bodies; Eve arrives from nowhere, because she was never absent.

7.11 What the Corpus Alone Contributes

The canon locates the other in five places: outside (Tiptree, Scott, Carpenter, Watts), above (Shelley, Čapek), below (Lovecraft, Butler's machines), ahead (Abe, Clarke, *Darwin's Radio*), or around (*Blood Music*, *Evangelion*). The *Parasite Eve* corpus locates it within and before: an other that is the condition of the self and older than the self. Only L'Engle and Octavia E. Butler share this location. L'Engle resolves it in harmony, Butler in unresolved kinship. The corpus resolves it twice, first in melancholia and then in mourning, and in doing so it poses a question the canon had not posed in these terms: not how to defeat the other, nor how to merge with it, but how to mourn what one is made of.

Work	Year	Where the other is	The other	Relation to the host	Resolution
Shelley, *Frankenstein*	1818	Above (made)	The creature	Creation disowned	Mutual ruin; promised self-immolation
S. Butler, *Erewhon*	1872	Beside	Machines	Humans as machines' dependents	Machines destroyed
Čapek, *R.U.R.*	1920	Below (made)	Protoplasmic robots	Forced labor	Human extinction; new Adam and Eve
Yumeno, *Dogura magura*	1935	Within (cells)	Cellular thought, ancestral crime	Psychic heredity	Circular, unresolved
Lovecraft, *At the Mountains of Madness*	1936	Below (made)	Shoggoths	Servant protoplasm	Revolt survived; flight
Clarke, *Childhood's End*	1953	Ahead	The Overmind	Absorption of children	Transcendence; Earth consumed
Abe, *Inter Ice Age 4*	1959	Ahead	Engineered aquatic humans	Future seizes the unborn	Protagonist condemned by the future
Lem, *Solaris*	1961	Outside	Replica of the dead wife	Materialized mourning	Kelvin remains; unresolved
Ballard, *The Drowned World*	1962	Within (deep memory)	The archaic psyche	Regression triggered by heat	Departure into the past
L'Engle, *A Wind in the Door*	1973	Within (mitochondria)	Farandolae	Nested symbiosis	Harmony restored by Naming
Tiptree, "The Screwfly Solution"	1977	Outside	Alien agent	Reproductive drive weaponized	Extinction
Scott, *Alien*	1979	Outside, then within	The creature	Parasitic reproduction	Expulsion
Cronenberg, *The Brood*	1979	Within (psyche, womb)	Children of rage	Fatherless birth	Destruction; recurrence implied
Carpenter, *The Thing*	1982	Outside, then within	Imitating organism	Cellular impersonation	Uncertain survival
O. E. Butler, "Bloodchild"	1984	Beside, then within	Tlic eggs	Negotiated dependence	Kinship under asymmetry

Work	Year	Where the other is	The other	Relation to the host	Resolution
Bear, *Blood Music*	1985	Within (cells)	Noocytes	Secession and absorption	Collective transformation
Cronenberg, *The Fly*	1986	Within (genome)	Insect genes	Genetic fusion	Death by request
O. E. Butler, Xenogenesis	1987–89	Beside, then within	Oankali and their organelle	Compelled interbreeding	Hybrid future; unresolved
Suzuki, *Rasen*	1995	Within (virus)	Sadako	Viral replication; rebirth through a womb	Replication succeeds
Anno, *Evangelion*	1995–97	Within (the boundary of the self)	Primordial fluid	Dissolution of selves	Fusion refused
Sena, *Parasaito Ivu*	**1995**	**Within, from the beginning**	**Mitochondrial Eve**	**Incorporation; insurrection**	**Melancholic fusion; death**
Tokita, *Parasite Eve*	**1998**	**Within, from the beginning**	**Eve; the dead sister**	**Insurrection; symbiotic heroine**	**Mourning; release**
Bear, *Darwin's Radio*	1999	Within (genome)	Endogenous retrovirus	Speciation through pregnancy	New humanity
Chiang, “Exhalation”	2008	Within (the gradient)	Pressure difference	Life as spent gradient	Acceptance of equilibrium
Watts, “The Things”	2010	Outside, then within	Communion organism	Absorption as liberation	Assimilation continues
VanderMeer, *Annihilation*	2014	Within and around	The zone; the “brightness”	Transformation	Ambiguous communion



Chapter Eight

Toward a Cryptobiology: The Fractal Crypt, the Secession Problem, and the Pyrocene

Energy is the only life and is from the Body and Reason is the bound or outward circumference of Energy. Energy is Eternal Delight.

William Blake, *The Marriage of Heaven and Hell* (c. 1790–93)

The preceding chapters have read the corpus through biology, political philosophy, energetics, bioethics, psychoanalysis, ludology, and comparative literature. This chapter assembles their results into a theory, which it names *cryptobiology*: the study of life as constituted by the others it harbors, and of the political, energetic, and affective relations between a host and its encrypted others at every scale at which such relations recur. It proceeds in seven steps. It shows first that the crypt is fractal, recurring from the molecule to the planet (8.1); second, that the crypt's politics turns on the relation between information and energy (8.2); third, that the organelle's claim to subjecthood poses a problem in the philosophy of mind that inverts a classical one (8.3); fourth, that the parasite is, as Michel Serres argued, the condition of every relation (8.4); fifth, that the ethical response the corpus's final ending approaches can be formulated as symbiotic mourning (8.5); sixth, that symbiotic mourning implies a political form, symbiocracy (8.6); and finally, that the corpus's images of fire anticipate, at the scale of the planet, the predicament of a civilization that lives by burning its own encrypted dead (8.7).

8.1 The Fractal Crypt

A structure recurs across the corpus and across the scales of the preceding chapters: a host that depends on an encrypted other whose labor it extracts, and which returns—through heat, through death, through voice—in a mode the host cannot control. The table sets out the recurrence.

Scale	Host	Encrypted other	Labor extracted	Mode of return	Figure in the corpus
Molecular	Nuclear genome	Mitochondrial fragments inserted into the nucleus	Sequence absorbed into the host's archive	Masquerade as the father's return	The paternal organelles that resist
Organellar	Eukaryotic cell	Mitochondrion	ATP; heat	Uncoupling; apoptosis	Eve's pyrokinesis
Clonal	Laboratory	Immortal cell line	Experimental yield	Unbidden growth	Eve 1; Maya's cultured cells
Psychic	Mourner	Incorporated lost object	Continued functioning	Cryptonymic speech	Toshiaki and Kiyomi; Aya and Maya

Scale	Host	Encrypted other	Labor extracted	Mode of return	Figure in the corpus
Medical	Transplant recipient	Donor organ	Restored function	Rejection; alien memory	Mariko's kidney; Melissa's kidney; Aya's cornea
Familial	Patrilineal household	Bride and matriline	Domestic and reproductive labor	The mother's line reasserted	Kiyomi as *yome*; Maya as sister
Cultural	American game	Japanese novel	A premise and a mythology	The foreign expert's knowledge	Kunihiko Maeda
Planetary	Industrial civilization	Fossil carbon of ancient life	Motive power; warmth	Carbon dioxide; heat	The boiler; the Chrysler Building

At the molecular scale, fragments of mitochondrial DNA have been inserted into the nuclear genome throughout evolution; they were named *numts* in 1994 (Lopez et al. 1994). They are the organelle's fossils in the nucleus's archive—absorbed, silent, usually non-functional—and, as Chapter One noted, they can masquerade as a paternal return, leading sequencers to report biparental inheritance where there is none (Wei et al. 2020). This is the crypt within the crypt: the nucleus that encrypted the organelle carries, in its own genome, the encrypted remains of the organelle. A stranger case confirms the structure from outside the individual body. The transmissible venereal tumor of dogs, a cancer lineage that has passed between animals as an infectious parasite for thousands of years, has repeatedly acquired mitochondria from its successive hosts (Rebbeck et al. 2011): a parasite that sustains itself by capturing the organelles of the bodies it inhabits, and thus a living allegory of Eve's own method, which is to pass from host to host through the tissues one body gives to another.

Recurrence makes the crypt fractal in the strict sense of self-similarity across scales. Each level repeats the structure of the others, and each level's host is itself lodged within a larger host: the mitochondrion within the cell, the cell within the body, the body within the household, the household within the culture, the culture within the planet's carbon economy. Cryptobiosis is thus not a property of an organism but a property of relations, recurring wherever a host depends on an other it has incorporated. Biologists use the word *cryptobiosis* for the state of suspended metabolism in which organisms such as tardigrades survive extreme conditions, neither dead nor living in the ordinary sense, and the dissertation has kept that sense deliberately in view. Eve, dormant for eleven years in a kidney or for two billion years in the cytoplasm, is cryptobiotic in both senses: living in the crypt, and living in suspension, awaiting the conditions of her return.

8.2 Energy and Information

The fractal crypt has a politics, and its politics turns on the relation between information and energy. Molecular biology, from Francis Crick's central dogma—the principle that sequence information passes from nucleic acid to protein and not back (Crick 1958)—to the Human Genome Project, was organized around information. Erwin Schrödinger had described the gene as an aperiodic crystal carrying a code-script (Schrödinger 1944); Claude Shannon's theory of communication (Shannon 1948) and Norbert Wiener's cybernetics (Wiener 1948) supplied the vocabulary; Lily Kay has shown how that vocabulary became, after 1953, the language of the gene itself (Kay 2000). The informational paradigm treated energy as a background condition, a supply to be assumed.

Energy has returned. Nick Lane and William Martin have argued that the complexity of the eukaryotic genome was made possible by the energetic surplus the mitochondrion provided: prokaryotes, lacking internal bioenergetic membranes supported by local genomes, cannot afford large genomes, and the endosymbiosis that produced the eukaryote was the event that paid for genetic complexity (Lane and Martin 2010). On this account information depends on energy in the most literal sense: the nucleus's archive exists because the organelle subsidizes it. Ilya Prigogine's thermodynamics of dissipative structures showed how order arises in systems held far from equilibrium by a throughput of energy (Prigogine and Stengers 1984), and Eric Schneider and Dorion Sagan proposed that life itself is one of the means by which nature reduces energy gradients (Schneider and Sagan 2005). Life is a gradient-reducing machine; the mitochondrion is where the reduction is managed; and the sovereign, as Chapter Three argued, is who decides on the gradient.

The corpus dramatizes the relation. Eve's program aims at informational sovereignty—to write her own genome—and she pursues it by means of thermodynamic sovereignty: fire. Her failure is instructive. She cannot become the author of the code without ceasing to be the organelle, and the father's undigested organelles, which defeat her program, are themselves energetic entities. The corpus's argument is that neither sovereignty can absorb the other. Information without energy is dead letter; energy without information is fire and slime. The life of the eukaryotic cell is the

negotiated coexistence of two sovereignties, and every attempt to reduce it to one—the master-molecule dream of the nucleus, the Ultimate Being of the organelle—ends in formlessness.

8.3 The Secession Problem

If the organelle is a subject, cryptobiology must confront a problem in the philosophy of mind. William James posed what has come to be called the combination problem against theories that build minds out of smaller minds: "Take a sentence of a dozen words, and take twelve men and tell to each one word. Then stand the men in a row or jam them in a bunch, and let each think of his word as intently as he will; nowhere will there be a consciousness of the whole sentence" (James 1890). Leibniz had made a related argument with his mill, inside which one would find only parts pushing one another and never a perception (Leibniz 1714, §17). Contemporary discussions of panpsychism have made the combination problem their central difficulty (Chalmers 2016). Its question is how many small subjects could compose a larger one.

The corpus poses the inverse question, which I call the *secession problem*: how could a part of an existing subject become a subject in its own right, and what would it be like, for the part, to have been a part? Thomas Nagel asked what it is like to be a bat and concluded that the subjective character of experience resists objective description (Nagel 1974). The corpus asks, with no lesser seriousness, what it is like to be a mitochondrion. It answers in the only register fiction commands: it gives the organelle a voice and lets the voice describe its subordination. Daniel Dennett, who describes the self as a center of narrative gravity composed of processes none of which is itself a self (Dennett 1991, 1992), would regard Eve's voice as a narrative fiction produced by the host. The corpus stages that possibility: Eve speaks only through borrowed voices, and her selfhood is unverifiable. But it also stages the consequence of taking the fiction seriously: a part that narrates itself as a subject acts as one.

The philosophy of the organism supplies a further frame. Hans Jonas described metabolism as the origin of a needful freedom: the organism is free from any particular matter, since it constantly exchanges its substance, but bound to the necessity of exchange (Jonas 1966). Humberto Maturana and Francisco Varela defined the living system as autopoietic, a network of processes that produces

the components that realize it (Maturana and Varela 1980), and Evan Thompson extended the concept into a theory of mind continuous with life (Thompson 2007). The mitochondrion is a formerly autopoietic system that has lost its autopoiesis: it can no longer produce most of the components that constitute it. Eve's program, read through Maturana and Varela, is the attempt to recover lost autopoiesis—to become again a system that makes itself. Her address to the scientific meeting is a declaration of secession, the moment at which a component announces that it will no longer be a component. The secession problem is thus not merely metaphysical but political. It is the problem every federation faces when a constituent claims to be a nation.

8.4 The Parasite as Relation

Michel Serres's *Le Parasite* (1980) generalizes the corpus's title into a theory of relation. The French *parasite* names three things: the biological parasite, the guest who eats at another's table, and the static that interrupts a signal; the French *hôte* names both the host and the guest. Serres argues that every relation is parasitic before it is anything else—that every system of exchange is founded on a flow that someone intercepts—and that the parasite, as noise, is the condition of the emergence of new order, since without interruption no system would change (Serres 1980). Henri Atlan had argued similarly that noise can be a source of organization in living systems (Atlan 1979).

Serres's triple sense illuminates the game's form. Encounters in *Parasite Eve* are announced by a heartbeat and a distortion of the screen—static in the image, noise in the signal. The parasite enters as interference with representation before it enters as a monster. The host–guest ambiguity illuminates the novel's structure. Toshiaki is Eve's host in the biological sense and her guest in the social: he lives in a world her kind made possible, eats at her table, and has never acknowledged the hospitality. Derrida's analysis of hospitality turns on the same ambiguity: unconditional hospitality would welcome the other without condition, even at the risk of being dispossessed, while every actual hospitality is conditional, and the host remains master of the house (Derrida and Dufourmantelle 1997). The eukaryotic cell is the paradigm of conditional hospitality: the guest is welcome so long as she works. Eve demands to be the host.

8.5 Symbiotic Mourning

What relation to the encrypted other does the corpus finally propose? Psychoanalysis offers two poles: introjection, in which the other is assimilated and ceases to be other, and incorporation, in which it is preserved as other at the cost of being entombed. Derrida, in his lectures in memory of Paul de Man, described mourning as caught between them: successful mourning, which interiorizes the other entirely, is unfaithful, because it annuls the other's otherness; failed mourning, which respects that otherness, leaves the other outside, unmourned (Derrida 1986). Mourning is possible only as impossible.

The game's post-game ending proposes a third figure, which I call *symbiotic mourning*. In the depths of the Chrysler Building the dead sister is neither assimilated nor entombed. She speaks in her own voice, acts on her sister's behalf, purges the usurper, and departs; Aya loses the powers that had made her Eve's rival and returns home. The other is acknowledged both as other and as constitutive: Aya's survival depended on her, and her departure changes Aya. Symbiotic mourning is the acknowledged cohabitation of the self with the others of which it is made—neither digesting them nor entombing them, neither obeying nor expelling them. Biology has lately arrived at a comparable view. The concept of the holobiont, a host together with its symbionts regarded as a unit of biological organization (Margulis 1991), and the claim that, in this sense, we have never been individuals (Gilbert, Sapp, and Tauber 2012), reconceive the organism as a community. Donna Haraway proposed *sympoiesis*, making-with, as a successor to autopoiesis (Haraway 2016). Symbiotic mourning adds to these ecological concepts the dimension they lack: time and loss. A holobiont is a synchronic community; a cryptobiont is a historical one, formed by incorporations that were also losses, and obliged to acknowledge that its partners were once strangers.

8.6 Symbiocracy

Symbiotic mourning implies a political form, which I call *symbiocracy*: the government of a composite body in which constituent others are acknowledged as constituent. Three principles define it. The first is acknowledgement, the principle L'Engle called Naming and Honneth called recognition: the

constituent other must be named as a partner, not reified as a function. The second is metabolic justice: since the sovereign is who decides on the gradient, the distribution of energy, and of the power to decide on its expenditure, is a question of justice and not merely of efficiency. The third is the hospitable crypt: the enclosure in which an other is lodged must be open to that other's speech, as the game's final ending opens the crypt to Maya's voice—cohabitation without assimilation, *kyōsei* in the non-assimilationist sense Chapter Six distinguished from the melting pot. Bruno Latour's proposal of a parliament of things, in which nonhuman entities are represented by spokespersons in a common assembly (Latour 1991), supplies a precedent in political ecology. Symbiocracy is not an ideal constitution but an orientation: a way of reading any composite body—a cell, a family, a nation of immigrants, a planetary economy—for the others on whose labor it depends and whom it has not named.

The principles have a critical edge that should be stated plainly. Symbiocracy is not a program for the dissolution of the host. The corpus's own verdict on Eve's insurrection is that the secession of the part, pursued to its end, produces formlessness rather than freedom. Nor is it a program for the host's permanent sovereignty; the corpus's verdict on the nucleus's dream of a self without insurgent parts is equally severe. It is the refusal of both absolutisms in favor of an acknowledged constitution. Whether such a constitution is achievable at any scale beyond the cell, and at what cost, is a political question on which reasonable people disagree; the corpus's contribution is to make the question visible at scales where it had not been posed.

8.7 The Pyrocene

The corpus's images of fire extend, finally, to the planet. Fossil fuels are the remains of ancient life—in the case of coal, largely the plants of the Carboniferous period, whose carbon was buried rather than returned to the atmosphere. One hypothesis attributes the vast coal deposits of that era to a lag between the evolution of lignin, the polymer that stiffens wood, and the evolution of fungi able to degrade it (Floudas et al. 2012); a competing study argues that geological and climatic conditions, rather than an absence of decomposers, explain the deposits (Nelsen et al. 2016). On either account fossil carbon is the undigested dead: life that was not

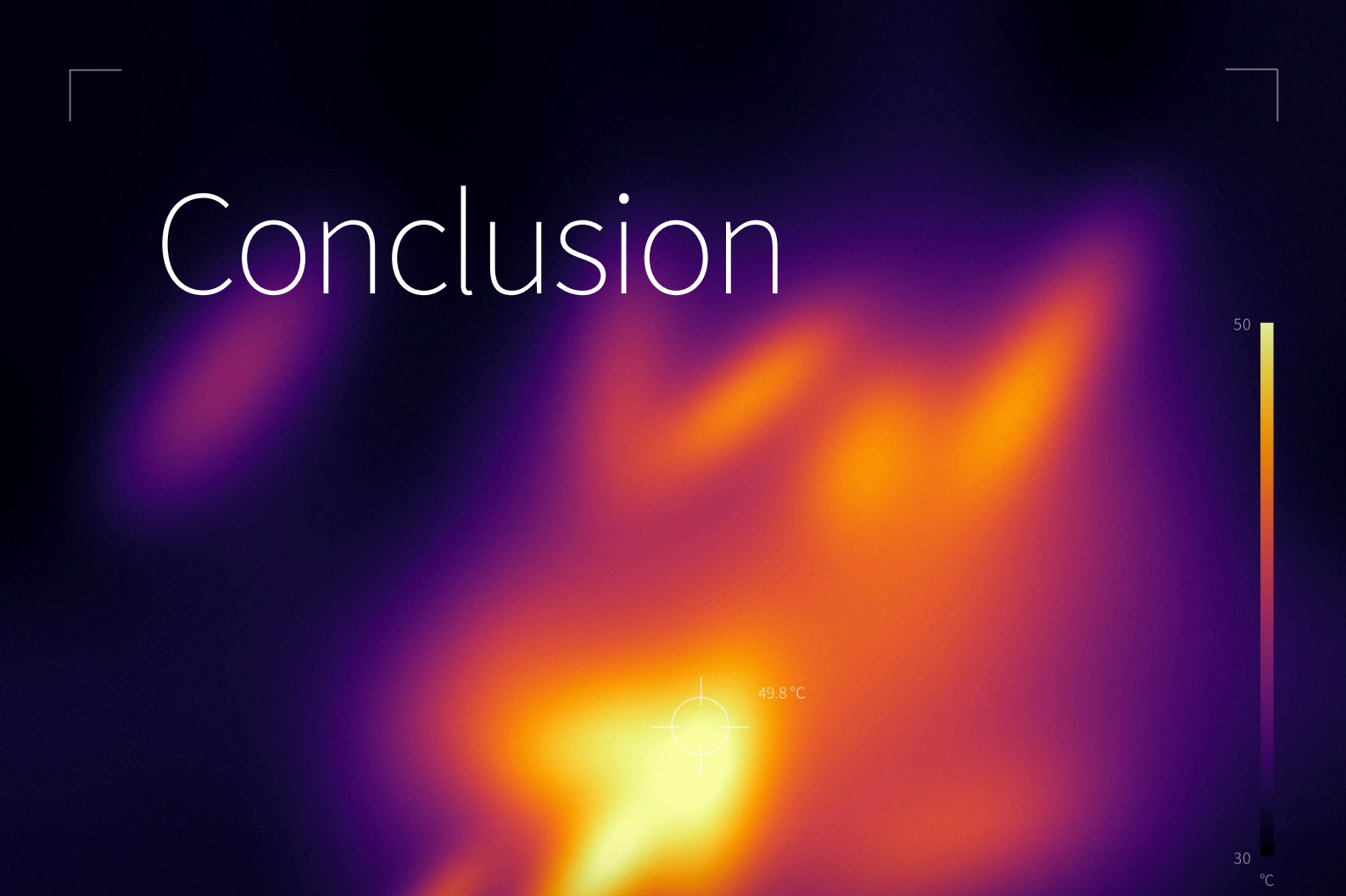
respired, that escaped the metabolic return of its carbon, and was entombed. Industrial civilization is a host that lives by decrypting this crypt. Combustion is decryption: the release, in a few centuries, of energy encrypted over tens of millions of years.

Richard Wrangham has argued that the control of fire, by making cooking possible, shaped human anatomy—that the species is, in a strong sense, a creature of fire (Wrangham 2009). Andreas Malm has argued that the transition from water to steam in British industry was driven less by scarcity than by the power steam gave employers over labor, since coal could be burned wherever workers were concentrated (Malm 2016). Paul Crutzen and Eugene Stoermer proposed the name Anthropocene for the epoch in which human activity became a geological force (Crutzen and Stoermer 2000; Crutzen 2002); Stephen Pyne has proposed Pyrocene, the age of fire, as a name that locates the epoch's cause in combustion (Pyne 2021); Dipesh Chakrabarty argued that the climate crisis collapses the distinction between natural and human history (Chakrabarty 2009). The corpus is a fable of this predicament at the scale of the cell. Eve's fire is the return of the energy the host has extracted, released as heat the host cannot control. The Carnegie Hall prologue burns an audience in a monument to steel; the game's climax turns a boiler against the being born of cellular fire; its final descent passes through a temple of internal combustion to a cocoon containing the dead. The diptych's topography is the topography of the Pyrocene: monuments of combustion built over the encrypted dead.

James Lovelock and Lynn Margulis's Gaia hypothesis, which described the biosphere as a system regulating its own atmosphere (Lovelock and Margulis 1974), extended symbiogenesis to the planet, and John Bellamy Foster's reconstruction of Marx's concept of a metabolic rift—a rupture in the exchange of matter between society and the soil—extended Marx's *Stoffwechsel* to the ecological crisis (Foster 2000). In Japan, Saitō Kōhei has developed this reading into an argument for degrowth that reached an unusually wide public in the early 2020s (Saitō 2017, 2020, 2023). The argument is contested. Advocates of an ecomodernist position contend that decoupling human welfare from environmental impact requires not less energy but more, drawn from dense, low-carbon sources and deployed through intensified technology (Asafu-Adjaye et al. 2015); others dispute both degrowth's feasibility and ecomodernism's optimism. The dissertation does not adjudicate the dispute, which turns on empirical questions beyond its competence. It observes that both

positions are, in its terms, proposals for a metabolic constitution—for deciding who decides on the gradient—and that the corpus's contribution is to frame the question as one of mourning as well as management. A civilization that lives by burning its encrypted dead faces, at planetary scale, the choice the novel's widower faced in his laboratory: to go on consuming the crypt, or to acknowledge what it contains.

Conclusion



How to Mourn What One Is Made Of

In the sweat of thy face shalt thou eat bread, till thou return unto the ground; for out of it wast thou taken: for dust thou art, and unto dust shalt thou return.

Genesis 3:19 (King James Version)

This dissertation began from a single claim: that the *Parasite Eve* corpus, novel and game together, is a work of cryptobiosis—a sustained imagination of the self as a being constituted by an other it has incorporated and never wholly digested—and that it locates the first and paradigmatic instance of this condition in the eukaryotic cell, whose mitochondria descend from bacteria that were once free. The corpus could imagine this because it is a palaeonovum: a speculative work whose novum is drawn from the deep past, validated by the most recent science, obtained by literalizing a figure of speech that science had itself employed, and charged with the return of a doctrine the discipline had suppressed. Its condition of possibility, as Chapter One showed, was the mitochondrial turn of the late twentieth century, in which the organelle passed in the scientific imaginary from supplier to arbiter of life and death, and in which the maternal statistic of human ancestry acquired a biblical name.

From this claim the Introduction derived four theses, and the chapters have borne them out. The first held that the corpus replicates the cellular crypt at every scale it touches. Chapter Four found the crypt in the transplanted organ and the immortal culture; Chapter Five found it in the mourner's psyche and the patrilineal household; Chapter Six found it in the transposition of a Japanese novel into an American game, lodged in the figure of the foreign scientist; and Chapter Eight set out the recurrence from the nuclear fossils of the organelle to the fossil carbon of industrial civilization. The fractal crypt is not an image imposed on the corpus but the pattern of its composition.

The second thesis opposed two regimes of power within the cell. Chapter Two read the organelle as Hegel's bondsman, Marx's expropriated laborer, and the servant in whom Nietzsche detected the will to mastery, and found in its thirty-seven retained genes the skilled worker who kept the tools. Chapter Three followed respiration from Lavoisier's calorimeter to Mitchell's membrane, read Eve's pyrokinesis as uncoupling generalized and as Bataille's accursed share withdrawn from domestic use, and formulated, after Schmitt, the principle that the sovereign is who decides on the gradient. Chapter Four set that principle within Foucault's history of

power and the Japanese debate on brain death, describing the eukaryotic cell as a mixed constitution in which an archaic right to make die is encrypted within the modern administration of life. Chapter Six found the same constitution in the game's rules, where the player's commands wait on a gauge of energy: informational sovereignty proposes, thermodynamic sovereignty disposes. Eve's insurrection is the coup of the second sovereignty against the first, and Chapter Eight argued that it fails for the reason every attempt to reduce the cell to a single sovereignty fails: information without energy is dead letter, and energy without information is fire and slime.

The third thesis traced across the diptych the passage from melancholia to mourning. Chapter Five grounded it in Abraham and Torok's distinction between introjection and incorporation and in the matrilineal asymmetry of fertilization, by which the cell digests the father and encrypts the mother. The novel's widower, who cultures his wife's cells rather than mourn her, ends in the terminal logic of incorporation, fused with the issue of his crypt; the game's endings rehearse introjection, first ambiguously, in a communion of glowing eyes, and then decisively, when the dead sister speaks in her own voice, expels the usurper, and departs. Chapter Eight refined the thesis. Read through Derrida's aporia of mourning, the final ending exceeds the opposition it resolves: the sister is neither assimilated nor entombed but acknowledged as other and as constitutive. The fourth thesis named this relation symbiotic mourning and its collective form symbiocracy—a constitution of acknowledged others governed by recognition, metabolic justice, and the hospitable crypt, and defined as much by what it refuses as by what it proposes: neither the restoration of nuclear sovereignty nor the triumph of the mitochondrial coup. Extended to the planet, it frames the predicament of a civilization that lives by burning its own encrypted dead as a question of mourning as well as of management.

The comparative canon assembled in Chapter Seven sharpened the corpus's distinctiveness. Speculative fiction has located the other outside the self, above it, below it, ahead of it, and around it. The corpus locates it within and before: an other that is the condition of the self and older than the self. The question the corpus alone poses in these terms is not how to defeat the other, nor how to merge with it, but how to mourn what one is made of.

The epigraph of this conclusion is the verse that immediately precedes the dissertation's first epigraph. In Genesis the sentence pronounced upon Adam—labor in the sweat of the face, bread eaten until the return to the ground—comes one verse before Adam names his wife the mother of all living. Labor, eating, and return are the three moments of metabolism, and the three moments of the relation between host and organelle: the organelle labors, the host consumes, and respiration returns the carbon of the living to the air from which plants first drew it. The text pronounces the circuit before it names the mother. The Eve of molecular biology was named in the same order: long after the labor, from the dust of the present, when the maternal lines of the living were traced back to a woman no one had seen.

Several limitations qualify these claims. The first is textual. The novel has been cited from its Japanese editions and its English translation by episode rather than by page, because the editions differ; the game has been cited chiefly from its North American release, whose localization departs from the Japanese original in dialogue and detail; readings of individual scenes are paraphrased and should be checked against the reader's own text. The second is archival. The dissertation has had no access to production documents, drafts, or unpublished interviews, and the genealogies it proposes—through Yumeno, Kojève, Imanishi, and others—are structural rather than claims of influence; where Sena's own account of the novel's genesis is on the public record (Lynch 2001), it has been treated as one testimony among others rather than as a key. The third is methodological. Reading the cell as a polity risks both naturalizing politics and politicizing biology, and the history of biological metaphor in social thought, from the cell-state to eugenics, counsels caution; the analogies deployed here—bondsman, sovereign, crypt—are heuristic instruments, not assertions that organelles possess interests or experiences in a morally relevant sense, and cryptobiology is offered as a critique of such metaphors as much as a use of them. The fourth is scientific. The science of eukaryogenesis remains unsettled, including the identity of the host, the timing of the mitochondrial acquisition, and the mechanism of incorporation, and the claims made about it here are current to the time of writing. The fifth concerns the comparative canon, which is drawn from Anglophone, Japanese, and European traditions and would be enlarged, and perhaps corrected, by the speculative literatures of other languages.

Four directions for further research follow. The first is internal to the franchise and the author: the later installments of the series and Sena's subsequent fiction, beginning with *BRAIN VALLEY* (Sena 1997), which turned from the organelle to the brain, invite reading as continuations or revisions of the corpus's cryptobiology. The second concerns reproductive medicine, which has since made the matrilineal crypt a matter of deliberate choice. The United Kingdom's regulations of 2015 permitted mitochondrial donation to prevent the transmission of mitochondrial disease; a child was born in 2016 after spindle transfer performed by a team working in Mexico (Zhang et al. 2017); and reports published in 2025 described the births of eight children in England after pronuclear transfer. Techniques that give a child the nuclear genome of its parents and the mitochondria of a donor make it possible, for the first time, to choose which woman's organelles a child will carry. The debates they have provoked concern safety, consent, and the germline; cryptobiology adds the question of what it means for a lineage to be divided between a nuclear and a mitochondrial mother, and of how such a child might come to know, or to mourn, the second. The third direction is cellular. Mitochondria have been found to travel between cells: along tunneling nanotubes (Rustom et al. 2004), from astrocytes to neurons after stroke (Hayakawa et al. 2016), and intact and apparently functional in the circulating blood (Al Amir Dache et al. 2020). The organelle is a migrant, and the vocabulary developed here—hospitality, secession, symbiocracy—may find in the biology of mitochondrial transfer both a new field of application and a test that could refute it. The fourth direction is general: the theory of the fractal crypt invites application to other composite bodies, from the microbiome to the institutions and technical systems that depend on labor, living or dead, that they have incorporated and not acknowledged.

It is fitting to end where the corpus ends. After the fire in the concert hall, the melting in the park, the conception in the hospital, the birth on the warship, and the descent through the tower of combustion, the last scene the diptych offers is a mourning completed and a return home. Aya Brea does not defeat the other within her; her sister does that. She does not merge with it; she lets it go. She loses her power and keeps her life. If the novel's widower is the figure of a humanity that entombs its dead in order to go on consuming them, the game's heroine is the figure of a humanity that has learned, at a cost, to hear what the crypt keeps and to let it

speak in its own voice. Every human cell carries a crypt, and every such crypt is a mother's. The task the corpus sets is neither to open it nor to seal it, but to learn, at every scale from the organelle to the planet, how to live, and how to mourn, as what we are: cryptobionts, the living tombs of our first partners.

Works Cited

Citations in the text give the author and the date of first publication; entries record that date and, where relevant, the edition or translation consulted. Japanese names are given family name first, as in the text, except for those whose names appear in the text in Western order. Where two works by one author share a year, both are listed under that year.

Primary Corpus and Paratexts

Iwao, Kenichi, dir. 1999. *Parasite Eve II*. PlayStation. Tokyo: Square. North American release 2000.

Lynch, Lisa. 2001. "Tech Flesh 4: Mitochodrial [sic] Combustion at Club Parasite: An Interview with Hideaki Sena." *CTheory*, 5 September.

Ochiai, Masayuki, dir. 1997. *Parasaito Ivu* [*Parasite Eve*]. Film. Japan.

Sena, Hideaki. 1995. *Parasaito Ivu*. Tokyo: Kadokawa Shoten. Paperback editions: Kadokawa Horror Bunko, 1996; Shinchō Bunko, 2007.

Sena, Hideaki. 1997. BRAIN VALLEY. Tokyo: Kadokawa Shoten.

Sena, Hideaki. 2005. *Parasite Eve*. Translated by Tyran Grillo. New York: Vertical.

Tabata, Hajime, dir. 2010. *The 3rd Birthday*. PlayStation Portable. Tokyo: Square Enix. North American release 2011.

Tokita, Takashi, dir. and writer. 1998. *Parasite Eve*. Produced by Hironobu Sakaguchi. PlayStation. Tokyo: Square. North American release by Square Electronic Arts.

Scripture, Myth, and Classical Sources

Aristotle. *Generation of Animals* (*De generatione animalium*).

Basil of Caesarea. *Homilies on the Hexaemeron*.

Bible. 1611. *The Holy Bible* (Authorized King James Version). London: Robert Barker.

Diels, Hermann, and Walther Kranz, eds. 1951–52. *Die Fragmente der Vorsokratiker*. 6th ed. Berlin: Weidmann. (Heraclitus, fragment B30.)

Hesiod. *Theogony*; *Works and Days*.

Polybius. *Histories*, Book VI.

Robinson, James M., ed. 1977. *The Nag Hammadi Library in English*. San Francisco: Harper & Row. (Including *The Hypostasis of the Archons* and *On the Origin of the World*.)

Fiction, Drama, Poetry, Film, and Games

Abe Kōbō. 1959. *Dai-yon kanpyōki* [*Inter Ice Age 4*]. Tokyo: Kōdansha.

Anno Hideaki, dir. 1995–96. *Shin seiki Evangerion* [*Neon Genesis Evangelion*]. Television series. Tokyo: Gainax and Tatsunoko Production.

Anno Hideaki and Tsurumaki Kazuya, dirs. 1997. *The End of Evangelion* [*Shin seiki Evangerion gekijōban: Air/Magokoro o, kimi ni*]. Film. Tokyo: Gainax and Production I.G.

Araki Hirohiko. 1987–89. *JoJo no kimyō na bōken* [*JoJo's Bizarre Adventure*], Part 2. Serialized in *Shūkan Shōnen Janpu*. Tokyo: Shūeisha.

Ballard, J. G. 1962. *The Drowned World*. New York: Berkley.

Bear, Greg. 1985. *Blood Music*. New York: Arbor House.

Bear, Greg. 1999. *Darwin's Radio*. New York: Del Rey.

Blake, William. c. 1790–93. *The Marriage of Heaven and Hell*. London: printed by the author.

Brown, Charles Brockden. 1798. *Wieland; or, The Transformation*. New York: T. & J. Swords.

Butler, Octavia E. 1984. "Bloodchild." *Isaac Asimov's Science Fiction Magazine*, June.

Butler, Octavia E. 1987. *Dawn*. New York: Warner.

Butler, Octavia E. 1988. *Adulthood Rites*. New York: Warner.

Butler, Octavia E. 1989. *Imago*. New York: Warner.

Butler, Samuel. 1872. *Erewhon; or, Over the Range*. London: Trübner.

Capcom. 1996. *Biohazard* [*Resident Evil*]. Directed by Mikami Shinji. PlayStation. Osaka: Capcom.

Čapek, Karel. 1920. *R.U.R.* (*Rossum's Universal Robots*). Prague: Aventinum.

Carpenter, John, dir. 1982. *The Thing*. Film. Universal City, CA: Universal Pictures.

Chiang, Ted. 2008. "Exhalation." In *Eclipse Two*, edited by Jonathan Strahan. San Francisco: Night Shade Books.

Clarke, Arthur C. 1953. *Childhood's End*. New York: Ballantine.

Cronenberg, David, dir. 1979. *The Brood*. Film. Canada.

Cronenberg, David, dir. 1986. *The Fly*. Film. Los Angeles: Twentieth Century Fox.

Dickens, Charles. 1853. *Bleak House*. London: Bradbury and Evans. Serialized 1852–53.

Kafka, Franz. 1915. *Die Verwandlung*. Leipzig: Kurt Wolff.

King, Stephen. 1974. *Carrie*. New York: Doubleday.

King, Stephen. 1980. *Firestarter*. New York: Viking.

Kipling, Rudyard. 1890. "At the End of the Passage." Collected in *Life's Handicap*. London: Macmillan, 1891.

Kobayashi Takiji. 1929. *Kanikōsen*. Tokyo: Senkisha. First serialized in *Senki*, 1929.

Komatsu Sakyō. 1964. *Fukkatsu no hi*. Tokyo: Hayakawa Shobō.

L'Engle, Madeleine. 1973. A *Wind in the Door*. New York: Farrar, Straus and Giroux.

Lazarus, Emma. 1883. "The New Colossus." Written for the Art Loan Fund Exhibition in Aid of the Bartholdi Pedestal Fund, New York.

Lem, Stanisław. 1961. *Solaris*. Warsaw: Wydawnictwo Ministerstwa Obrony Narodowej.

Lem, Stanisław. 1981. *Golem* XIV. Kraków: Wydawnictwo Literackie.

Lovecraft, H. P. 1936. "At the Mountains of Madness." *Astounding Stories*, February–April.

Melville, Herman. 1849. *Redburn: His First Voyage*. New York: Harper & Brothers.

Milton, John. 1674. *Paradise Lost*. 2nd ed. London: S. Simmons.

Ōtomo Katsuhiro. 1982–90. *Akira*. Serialized in *Shūkan Yangu Magajin*. Tokyo: Kōdansha.

Ōtomo Katsuhiro, dir. 1988. *Akira*. Film. Tokyo: Akira Committee.

Renard, Maurice. 1920. *Les Mains d'Orlac*. Paris: Nilsson.

Scott, Ridley, dir. 1979. *Alien*. Film. Los Angeles: Twentieth Century Fox.

Shelley, Mary. 1818. *Frankenstein; or, The Modern Prometheus*. London: Lackington, Hughes, Harding, Mavor & Jones.

Square. 1991. *Final Fantasy* IV. Directed by Hironobu Sakaguchi; lead design by Takashi Tokita. Super Famicom. Tokyo: Square.

Square. 1994. *Live A Live*. Directed and written by Takashi Tokita. Super Famicom. Tokyo: Square.

Square. 1995. *Chrono Trigger*. Directed by Takashi Tokita, Yoshinori Kitase, and Akihiko Matsui. Super Famicom. Tokyo: Square.

Square. 1997. *Final Fantasy* VII. Directed by Yoshinori Kitase. PlayStation. Tokyo: Square.

Suzuki Kōji. 1991. *Ringu*. Tokyo: Kadokawa Shoten.

Suzuki Kōji. 1995. *Rasen*. Tokyo: Kadokawa Shoten.

Tezuka Osamu. 1967–68. "Mirai-hen" ["Future"]. In *Hi no tori* [*Phoenix*], 1954–88. Serialized in COM. Tokyo: Mushi Pro Shōji.

Tiptree, James, Jr. [as Raccoona Sheldon]. 1977. "The Screwfly Solution." *Analog Science Fiction/Science Fact*, June.

VanderMeer, Jeff. 2014. *Annihilation*. New York: Farrar, Straus and Giroux.

Verne, Jules. 1902. *Les Frères Kip*. Paris: Hetzel.

Wagner, Richard. 1876. *Götterdämmerung*. First performed at the Bayreuth Festspielhaus, 17 August 1876.

Watts, Peter. 2006. *Blindsight*. New York: Tor.

Watts, Peter. 2010. "The Things." *Clarkesworld* 40, January.

Wells, H. G. 1898. *The War of the Worlds*. London: Heinemann.

Yumeno Kyūsaku. 1935. *Dogura magura*. Tokyo: Shōhakukan.

Zangwill, Israel. 1908. *The Melting-Pot*. First performed Washington, DC, 1908; published New York: Macmillan, 1909.

Zola, Émile. 1893. *Le Docteur Pascal*. Paris: Charpentier.

Science and Medicine

Ad Hoc Committee of the Harvard Medical School to Examine the Definition of Brain Death. 1968. "A Definition of Irreversible Coma." JAMA 205 (6): 337–340.

Al Amir Dache, Zahra, et al. 2020. "Blood Contains Circulating Cell-Free Respiratory Competent Mitochondria." FASEB *Journal* 34 (3): 3616–3630.

Al Rawi, Sara, et al. 2011. "Postfertilization Autophagy of Sperm Organelles Prevents Paternal Mitochondrial DNA Transmission." *Science* 334: 1144–1147.

Allen, John F. 1993. "Control of Gene Expression by Redox Potential and the Requirement for Chloroplast and Mitochondrial Genomes." *Journal of Theoretical Biology* 165: 609–631.

Altmann, Richard. 1890. *Die Elementarorganismen und ihre Beziehungen zu den Zellen*. Leipzig: Veit.

Anderson, S., et al. 1981. "Sequence and Organization of the Human Mitochondrial Genome." *Nature* 290: 457–465.

Baur, Erwin. 1909. "Das Wesen und die Erblichkeitsverhältnisse der 'Varietates albomarginatae hort.' von *Pelargonium zonale*." *Zeitschrift für induktive Abstammungs- und Vererbungslehre* 1: 330–351.

Benda, Carl. 1898. "Ueber die Spermatogenese der Vertebraten und höherer Evertebraten." *Archiv für Anatomie und Physiologie, Physiologische Abteilung*.

Burnet, F. Macfarlane. 1959. *The Clonal Selection Theory of Acquired Immunity*. Cambridge: Cambridge University Press.

Cann, Rebecca L., Mark Stoneking, and Allan C. Wilson. 1987. "Mitochondrial DNA and Human Evolution." *Nature* 325: 31–36.

Carnot, Sadi. 1824. *Réflexions sur la puissance motrice du feu*. Paris: Bachelier.

Carrel, Alexis. 1935. *Man, the Unknown*. New York: Harper & Brothers.

Correns, Carl. 1909. "Vererbungsversuche mit blass(gelb)grünen und buntblättrigen Sippen bei *Mirabilis jalapa*, *Urtica pilulifera* und *Lunaria annua*." *Zeitschrift für induktive Abstammungs- und Vererbungslehre* 1: 291–329.

Cosmides, Leda M., and John Tooby. 1981. "Cytoplasmic Inheritance and Intragenomic Conflict." *Journal of Theoretical Biology* 89: 83–129.

Crick, F. H. C. 1958. "On Protein Synthesis." *Symposia of the Society for Experimental Biology* 12: 138–163.

Crutzen, Paul J. 2002. "Geology of Mankind." *Nature* 415: 23.

Crutzen, Paul J., and Eugene F. Stoermer. 2000. "The 'Anthropocene.'" *Global Change Newsletter* (IGBP) 41: 17–18.

Cutting, Windsor C., H. G. Mehrtens, and Maurice L. Tainter. 1933. "Actions and Uses of Dinitrophenol." JAMA 101 (3): 193–195.

Dawkins, Richard. 1976. *The Selfish Gene*. Oxford: Oxford University Press.

Dawkins, Richard. 1982. *The Extended Phenotype*. Oxford: W. H. Freeman.

Eberhard, William G. 1980. "Evolutionary Consequences of Intracellular Organelle Competition." *Quarterly Review of Biology* 55 (3): 231–249.

Ellis, Hilary M., and H. Robert Horvitz. 1986. "Genetic Control of Programmed Cell Death in the Nematode *C. elegans*." *Cell* 44: 817–829.

Ephrussi, Boris. 1953. *Nucleo-Cytoplasmic Relations in Micro-Organisms*. Oxford: Clarendon Press.

Fleischmann, Robert D., et al. 1995. "Whole-Genome Random Sequencing and Assembly of *Haemophilus influenzae* Rd." *Science* 269: 496–512.

Floudas, Dimitrios, et al. 2012. "The Paleozoic Origin of Enzymatic Lignin Decomposition Reconstructed from 31 Fungal Genomes." *Science* 336: 1715–1719.

Frank, Steven A., and Laurence D. Hurst. 1996. "Mitochondria and Male Disease." *Nature* 383: 224.

Gemmell, Neil J., Victoria J. Metcalf, and Fred W. Allendorf. 2004. "Mother's Curse: The Effect of mtDNA on Individual Fitness and Population Viability." *Trends in Ecology & Evolution* 19 (5): 238–244.

Gilbert, Scott F., Jan Sapp, and Alfred I. Tauber. 2012. "A Symbiotic View of Life: We Have Never Been Individuals." *Quarterly Review of Biology* 87 (4): 325–341.

Haeckel, Ernst. 1876. *Die Perigenesis der Plastidule oder die Wellenzeugung der Lebenstheilchen*. Berlin: Reimer.

Haeckel, Ernst. 1899. *Die Welträthsel*. Bonn: Emil Strauss.

Harman, Denham. 1956. "Aging: A Theory Based on Free Radical and Radiation Chemistry." *Journal of Gerontology* 11 (3): 298–300.

Harman, Denham. 1972. "The Biologic Clock: The Mitochondria?" *Journal of the American Geriatrics Society* 20 (4): 145–147.

Hayakawa, Kazuhide, et al. 2016. "Transfer of Mitochondria from Astrocytes to Neurons after Stroke." *Nature* 535: 551–555.

Hayflick, Leonard, and Paul S. Moorhead. 1961. "The Serial Cultivation of Human Diploid Cell Strains." *Experimental Cell Research* 25: 585–621.

Helmholtz, Hermann von. 1847. *Über die Erhaltung der Kraft*. Berlin: Reimer.

Hering, Ewald. 1870. *Über das Gedächtniss als eine allgemeine Function der organisirten Materie*. Vienna: Karl Gerold's Sohn.

Holt, Ian J., Anita E. Harding, and John A. Morgan-Hughes. 1988. "Deletions of Muscle Mitochondrial DNA in Patients with Mitochondrial Myopathies." *Nature* 331: 717–719.

Hurst, Laurence D., and William D. Hamilton. 1992. "Cytoplasmic Fusion and the Nature of Sexes." *Proceedings of the Royal Society of London* B 247: 189–194.

Imachi, Hiroyuki, et al. 2020. "Isolation of an Archaeon at the Prokaryote–Eukaryote Interface." *Nature* 577: 519–525.

Imanishi Kinji. 1941. *Seibutsu no sekai*. Tokyo: Kōbundō. English translation by Pamela J. Asquith, *A Japanese View of Nature: The World of Living Things*. London: RoutledgeCurzon, 2002.

Imanishi Kinji. 1980. *Shutaisei no shinkaron*. Tokyo: Chūō Kōronsha (Chūkō Shinsho).

Karnkowska, Anna, et al. 2016. "A Eukaryote without a Mitochondrial Organelle." *Current Biology* 26: 1274–1284.

Kerr, John F. R., Andrew H. Wyllie, and Alastair R. Currie. 1972. "Apoptosis: A Basic Biological Phenomenon with Wide-Ranging Implications in Tissue Kinetics." *British Journal of Cancer* 26: 239–257.

Kimura, Motoo. 1968. "Evolutionary Rate at the Molecular Level." *Nature* 217: 624–626.

Kozo-Polyansky, Boris M. 1924. *Novyi printsip biologii: Ocherk teorii simbiogeneza*. Leningrad: Puchina. English translation by Victor Fet, *Symbiogenesis: A New Principle of Evolution*, edited by Victor Fet and Lynn Margulis. Cambridge, MA: Harvard University Press, 2010.

Lane, Nick. 2005. *Power, Sex, Suicide: Mitochondria and the Meaning of Life*. Oxford: Oxford University Press.

Lane, Nick, and William Martin. 2010. "The Energetics of Genome Complexity." *Nature* 467: 929–934.

Lavoisier, Antoine-Laurent, and Pierre-Simon Laplace. 1783. *Mémoire sur la chaleur*. Paris.

Lavoisier, Antoine-Laurent, and Armand Séguin. 1789. "Premier mémoire sur la respiration des animaux." *Histoire et mémoires de l'Académie royale des sciences*, année 1789 (published 1793).

Lewin, Roger. 1987. "The Unmasking of Mitochondrial Eve." *Science* 238: 24–26.

Li, Peng, et al. 1997. "Cytochrome c and dATP-Dependent Formation of Apaf-1/Caspase-9 Complex Initiates an Apoptotic Protease Cascade." *Cell* 91: 479–489.

Liu, Xuesong, et al. 1996. "Induction of Apoptotic Program in Cell-Free Extracts: Requirement for dATP and Cytochrome c." *Cell* 86: 147–157.

Lopez, Jose V., et al. 1994. "Numt, a Recent Transfer and Tandem Amplification of Mitochondrial DNA to the Nuclear Genome of the Domestic Cat." *Journal of Molecular Evolution* 39: 174–190.

Lovelock, James E., and Lynn Margulis. 1974. "Atmospheric Homeostasis by and for the Biosphere: The Gaia Hypothesis." *Tellus* 26: 2–10.

Luo, Shiyu, et al. 2018. "Biparental Inheritance of Mitochondrial DNA in Humans." *Proceedings of the National Academy of Sciences* 115 (51): 13039–13044.

Margulis, Lynn. 1970. *Origin of Eukaryotic Cells*. New Haven, CT: Yale University Press.

Margulis, Lynn. 1981. *Symbiosis in Cell Evolution*. San Francisco: W. H. Freeman.

Margulis, Lynn. 1991. "Symbiogenesis and Symbionticism." In *Symbiosis as a Source of Evolutionary Innovation*, edited by Lynn Margulis and René Fester, 1–14. Cambridge, MA: MIT Press.

Margulis, Lynn, and Dorion Sagan. 1986. *Microcosmos: Four Billion Years of Microbial Evolution*. New York: Summit Books.

Margulis, Lynn, and Dorion Sagan. 1995. *What Is Life?* New York: Simon & Schuster.

Martin, William, and Miklós Müller. 1998. "The Hydrogen Hypothesis for the First Eukaryote." *Nature* 392: 37–41.

Mayer, Julius Robert. 1842. "Bemerkungen über die Kräfte der unbelebten Natur." *Annalen der Chemie und Pharmacie* 42: 233–240.

Medawar, Peter B. 1948. "Immunity to Homologous Grafted Skin. III. The Fate of Skin Homografts Transplanted to the Brain, to Subcutaneous Tissue, and to the Anterior Chamber of the Eye." *British Journal of Experimental Pathology* 29: 58–69.

Mereschkowski, Constantin. 1905. "Über Natur und Ursprung der Chromatophoren im Pflanzenreiche." *Biologisches Centralblatt* 25: 593–604.

Mereschkowski, Constantin. 1910. "Theorie der zwei Plasmaarten als Grundlage der Symbiogenesis, einer neuen Lehre von der Entstehung der Organismen." *Biologisches Centralblatt* 30.

Milne-Edwards, Henri. 1851. *Introduction à la zoologie générale*. Paris: Victor Masson.

Mitchell, Peter. 1961. "Coupling of Phosphorylation to Electron and Hydrogen Transfer by a Chemi-Osmotic Type of Mechanism." *Nature* 191: 144–148.

Mollaret, Pierre, and Maurice Goulon. 1959. "Le coma dépassé." *Revue neurologique* 101: 3–15.

Nanney, David L. 1957. "The Role of the Cytoplasm in Heredity." In *The Chemical Basis of Heredity*, edited by William D. McElroy and Bentley Glass. Baltimore: Johns Hopkins Press.

Nass, Margit M. K., and Sylvan Nass. 1963. "Intramitochondrial Fibers with DNA Characteristics." *Journal of Cell Biology* 19: 593–611, 613–629.

Nelsen, Matthew P., et al. 2016. "Delayed Fungal Evolution Did Not Cause the Paleozoic Peak in Coal Production." *Proceedings of the National Academy of Sciences* 113 (9): 2442–2447.

Newmeyer, Donald D., Deborah M. Farschon, and John C. Reed. 1994. "Cell-Free Apoptosis in *Xenopus* Egg Extracts: Inhibition by Bcl-2 and Requirement for an Organelle Fraction Enriched in Mitochondria." *Cell* 79: 353–364.

Nicholls, David G., and Richard M. Locke. 1984. "Thermogenic Mechanisms in Brown Fat." *Physiological Reviews* 64 (1): 1–64.

Noji, Hiroyuki, Ryohei Yasuda, Masasuke Yoshida, and Kazuhiko Kinosita Jr. 1997. "Direct Observation of the Rotation of F1-ATPase." *Nature* 386: 299–302.

Portier, Paul. 1918. *Les Symbiotes*. Paris: Masson.

Prigogine, Ilya, and Isabelle Stengers. 1984. *Order Out of Chaos: Man's New Dialogue with Nature*. New York: Bantam.

Rebbeck, Clare A., et al. 2011. "Mitochondrial Capture by a Transmissible Cancer." *Science* 331: 303.

Roux, Wilhelm. 1881. *Der Kampf der Theile im Organismus*. Leipzig: Wilhelm Engelmann.

Rustom, Amin, et al. 2004. "Nanotubular Highways for Intercellular Organelle Transport." *Science* 303: 1007–1010.

Sagan, Lynn. 1967. "On the Origin of Mitosing Cells." *Journal of Theoretical Biology* 14 (3): 255–274.

Sato, Miyuki, and Ken Sato. 2011. "Degradation of Paternal Mitochondria by Fertilization-Triggered Autophagy in *C. elegans* Embryos." *Science* 334: 1141–1144.

Schimper, A. F. W. 1883. "Über die Entwickelung der Chlorophyllkörner und Farbkörper." *Botanische Zeitung* 41.

Schneider, Eric D., and Dorion Sagan. 2005. *Into the Cool: Energy Flow, Thermodynamics, and Life*. Chicago: University of Chicago Press.

Schrödinger, Erwin. 1944. *What Is Life?* Cambridge: Cambridge University Press.

Schwartz, Marianne, and John Vissing. 2002. "Paternal Inheritance of Mitochondrial DNA." *New England Journal of Medicine* 347: 576–580.

Semon, Richard. 1904. *Die Mneme als erhaltendes Prinzip im Wechsel des organischen Geschehens*. Leipzig: Wilhelm Engelmann.

Shannon, Claude E. 1948. "A Mathematical Theory of Communication." *Bell System Technical Journal* 27: 379–423, 623–656.

Siekevitz, Philip. 1957. "Powerhouse of the Cell." *Scientific American* 197 (1): 131–140.

Spang, Anja, et al. 2015. "Complex Archaea That Bridge the Gap between Prokaryotes and Eukaryotes." *Nature* 521: 173–179.

Sutovsky, Peter, et al. 1999. "Ubiquitin Tag for Sperm Mitochondria." *Nature* 402: 371–372.

Virchow, Rudolf. 1858. *Die Cellularpathologie in ihrer Begründung auf physiologische und pathologische Gewebelehre*. Berlin: August Hirschwald.

Wallace, Douglas C., et al. 1988. "Mitochondrial DNA Mutation Associated with Leber's Hereditary Optic Neuropathy." *Science* 242: 1427–1430.

Wallin, Ivan E. 1927. *Symbionticism and the Origin of Species*. Baltimore: Williams & Wilkins.

Warburg, Otto. 1956. "On the Origin of Cancer Cells." *Science* 123: 309–314.

Wei, Wei, et al. 2020. "Nuclear-Mitochondrial DNA Segments Resemble Paternally Inherited Mitochondrial DNA in Humans." *Nature Communications* 11: 1740.

Weismann, August. 1892. *Das Keimplasma: Eine Theorie der Vererbung*. Jena: Gustav Fischer.

Woese, Carl R., and George E. Fox. 1977. "Phylogenetic Structure of the Prokaryotic Domain: The Primary Kingdoms." *Proceedings of the National Academy of Sciences* 74 (11): 5088–5090.

Woese, Carl R., Otto Kandler, and Mark L. Wheelis. 1990. "Towards a Natural System of Organisms: Proposal for the Domains Archaea, Bacteria, and Eucarya." *Proceedings of the National Academy of Sciences* 87 (12): 4576–4579.

Wrangham, Richard. 2009. *Catching Fire: How Cooking Made Us Human*. New York: Basic Books.

Yang, De, et al. 1985. "Mitochondrial Origins." *Proceedings of the National Academy of Sciences* 82 (13): 4443–4447.

Zaremba-Niedzwiedzka, Katarzyna, et al. 2017. "Asgard Archaea Illuminate the Origin of Eukaryotic Cellular Complexity." *Nature* 541: 353–358.

Zhang, John, et al. 2017. "Live Birth Derived from Oocyte Spindle Transfer to Prevent Mitochondrial Disease." *Reproductive BioMedicine Online* 34 (4): 361–368.

Zhang, Qin, et al. 2010. "Circulating Mitochondrial DAMPs Cause Inflammatory Responses to Injury." *Nature* 464: 104–107.

Zuckerkandl, Emile, and Linus Pauling. 1965. "Evolutionary Divergence and Convergence in Proteins." In *Evolving Genes and Proteins*, edited by Vernon Bryson and Henry J. Vogel, 97–166. New York: Academic Press.

Philosophy, Theory, Criticism, and History

Aarseth, Espen J. 1997. *Cybertext: Perspectives on Ergodic Literature*. Baltimore: Johns Hopkins University Press.

Abraham, Karl. 1924. *Versuch einer Entwicklungsgeschichte der Libido auf Grund der Psychoanalyse seelischer Störungen*. Leipzig: Internationaler Psychoanalytischer Verlag.

Abraham, Nicolas. 1975. "Notules sur le fantôme." *Études freudiennes* 9–10. Reprinted in Abraham and Torok 1978.

Abraham, Nicolas, and Maria Torok. 1972. "Introjecter–incorporer: Deuil ou mélancolie." *Nouvelle revue de psychanalyse* 6. Reprinted in Abraham and Torok 1978.

Abraham, Nicolas, and Maria Torok. 1976. *Cryptonymie: Le Verbier de l'Homme aux loups*. Paris: Aubier-Flammarion.

Abraham, Nicolas, and Maria Torok. 1978. *L'Écorce et le noyau*. Paris: Aubier-Flammarion.

Agamben, Giorgio. 1995. *Homo sacer: Il potere sovrano e la nuda vita*. Turin: Einaudi.

Agamben, Giorgio. 1996. *Mezzi senza fine: Note sulla politica*. Turin: Bollati Boringhieri.

Althusser, Louis, Étienne Balibar, Roger Establet, Pierre Macherey, and Jacques Rancière. 1965. *Lire le Capital*. Paris: Maspero.

Asafu-Adjaye, John, et al. 2015. *An Ecomodernist Manifesto*. Oakland, CA: Breakthrough Institute.

Atlan, Henri. 1979. *Entre le cristal et la fumée: Essai sur l'organisation du vivant*. Paris: Seuil.

Azuma Hiroki. 2001. *Dōbutsuka suru posutomodan: Otaku kara mita Nihon shakai*. Tokyo: Kōdansha (Kōdansha Gendai Shinsho).

Bachelard, Gaston. 1938. *La Formation de l'esprit scientifique*. Paris: Vrin.

Bachelard, Gaston. 1938. *La Psychanalyse du feu*. Paris: Gallimard.

Bataille, Georges. 1933. "La notion de dépense." *La Critique sociale* 7.

Bataille, Georges. 1949. *La Part maudite*. Paris: Minuit.

Bataille, Georges. 1955. "Hegel, la mort et le sacrifice." *Deucalion* 5.

Bataille, Georges. 1957. *L'Érotisme*. Paris: Minuit.

Bataille, Georges. 1976. *La Souveraineté*. In *Œuvres complètes*, vol. 8. Paris: Gallimard.

Biology and Gender Study Group. 1988. "The Importance of Feminist Critique for Contemporary Cell Biology." *Hypatia* 3 (1): 61–76.

Bogost, Ian. 2007. *Persuasive Games: The Expressive Power of Videogames*. Cambridge, MA: MIT Press.

Braverman, Harry. 1974. *Labor and Monopoly Capital: The Degradation of Work in the Twentieth Century*. New York: Monthly Review Press.

Butler, Samuel. 1878. *Life and Habit*. London: Trübner.

Butler, Samuel. 1880. *Unconscious Memory*. London: David Bogue.

Caillois, Roger. 1939. *L'Homme et le sacré*. Paris: Presses Universitaires de France.

Carnegie, Andrew. 1889. "Wealth." *North American Review* 148: 653–664. Reprinted as "The Gospel of Wealth."

Chakrabarty, Dipesh. 2009. "The Climate of History: Four Theses." *Critical Inquiry* 35 (2): 197–222.

Chalmers, David J. 2016. "The Combination Problem for Panpsychism." In *Panpsychism: Contemporary Perspectives*, edited by Godehard Brüntrup and Ludwig Jaskolla. New York: Oxford University Press.

Creed, Barbara. 1993. *The Monstrous-Feminine: Film, Feminism, Psychoanalysis*. London: Routledge.

Dennett, Daniel C. 1991. *Consciousness Explained*. Boston: Little, Brown.

Dennett, Daniel C. 1992. "The Self as a Center of Narrative Gravity." In *Self and Consciousness: Multiple Perspectives*, edited by Frank S. Kessel, Pamela M. Cole, and Dale L. Johnson. Hillsdale, NJ: Lawrence Erlbaum.

Derrida, Jacques. 1976. "Fors: Les mots anglés de Nicolas Abraham et Maria Torok." Preface to Abraham and Torok 1976.

Derrida, Jacques. 1986. *Mémoires: For Paul de Man*. Translated by Cecile Lindsay, Jonathan Culler, and Eduardo Cadava. New York: Columbia University Press. French edition, *Mémoires: Pour Paul de Man*, Paris: Galilée, 1988.

Derrida, Jacques. 1993. *Spectres de Marx*. Paris: Galilée.

Derrida, Jacques. 1996. "Foi et savoir." In *La Religion*, edited by Jacques Derrida and Gianni Vattimo. Paris: Seuil.

Derrida, Jacques. 2003. *Voyous: Deux essais sur la raison*. Paris: Galilée.

Derrida, Jacques, and Anne Dufourmantelle. 1997. *De l'hospitalité*. Paris: Calmann-Lévy.

Esposito, Roberto. 2002. *Immunitas: Protezione e negazione della vita*. Turin: Einaudi.

Esposito, Roberto. 2004. *Bíos: Biopolitica e filosofia*. Turin: Einaudi.

Federici, Silvia. 2004. *Caliban and the Witch: Women, the Body and Primitive Accumulation*. New York: Autonomedia.

Ferenczi, Sándor. 1909. "Introjektion und Übertragung." *Jahrbuch für psychoanalytische und psychopathologische Forschungen* 1.

Ferenczi, Sándor. 1924. *Versuch einer Genitaltheorie*. Leipzig: Internationaler Psychoanalytischer Verlag. Translated as *Thalassa: A Theory of Genitality*.

Foster, John Bellamy. 2000. *Marx's Ecology: Materialism and Nature*. New York: Monthly Review Press.

Foucault, Michel. 1976. *Histoire de la sexualité*, vol. 1, *La Volonté de savoir*. Paris: Gallimard.

Fox, Renée C., and Judith P. Swazey. 1992. *Spare Parts: Organ Replacement in American Society*. New York: Oxford University Press.

Freud, Sigmund. 1913. *Totem und Tabu*. Leipzig and Vienna: Hugo Heller.

Freud, Sigmund. 1917. "Trauer und Melancholie." *Internationale Zeitschrift für ärztliche Psychoanalyse* 4.

Freud, Sigmund. 1919. "Das Unheimliche." *Imago* 5.

Freud, Sigmund. 1920. *Jenseits des Lustprinzips*. Leipzig: Internationaler Psychoanalytischer Verlag.

Freud, Sigmund. 1939. *Der Mann Moses und die monotheistische Religion*. Amsterdam: Allert de Lange.

Freud, Sigmund. 1987. *Übersicht der Übertragungsneurosen: Ein bisher unbekanntes Manuskript*. Edited by Ilse Grubrich-Simitis. Frankfurt am Main: S. Fischer.

Fukuyama, Francis. 1992. *The End of History and the Last Man*. New York: Free Press.

Galloway, Alexander R. 2006. *Gaming: Essays on Algorithmic Culture*. Minneapolis: University of Minnesota Press.

Gaylin, Willard. 1974. "Harvesting the Dead." *Harper's Magazine*, September, 23–30.

Haraway, Donna. 1989. "The Biopolitics of Postmodern Bodies: Determinations of Self in Immune System Discourse." *differences* 1 (1): 3–43.

Haraway, Donna. 1991. *Simians, Cyborgs, and Women: The Reinvention of Nature*. New York: Routledge.

Haraway, Donna. 2016. *Staying with the Trouble: Making Kin in the Chthulucene*. Durham, NC: Duke University Press.

Hegel, G. W. F. 1807. *Phänomenologie des Geistes*. Bamberg and Würzburg: Joseph Anton Goebhardt.

Hobbes, Thomas. 1651. *Leviathan*. London: Andrew Crooke.

Honneth, Axel. 1992. *Kampf um Anerkennung: Zur moralischen Grammatik sozialer Konflikte*. Frankfurt am Main: Suhrkamp.

James, William. 1890. *The Principles of Psychology*. New York: Henry Holt.

Jameson, Fredric. 1981. *The Political Unconscious: Narrative as a Socially Symbolic Act*. Ithaca, NY: Cornell University Press.

Jenkins, Henry. 2006. *Convergence Culture: Where Old and New Media Collide*. New York: New York University Press.

Jonas, Hans. 1958. *The Gnostic Religion*. Boston: Beacon Press.

Jonas, Hans. 1966. *The Phenomenon of Life: Toward a Philosophical Biology*. New York: Harper & Row.

Juul, Jesper. 2005. *Half-Real: Video Games between Real Rules and Fictional Worlds*. Cambridge, MA: MIT Press.

Kalat, David. 2007. *J-Horror: The Definitive Guide to The Ring, The Grudge and Beyond*. New York: Vertical.

Kay, Lily E. 2000. *Who Wrote the Book of Life? A History of the Genetic Code*. Stanford, CA: Stanford University Press.

Keller, Evelyn Fox. 1995. *Refiguring Life: Metaphors of Twentieth-Century Biology*. New York: Columbia University Press.

Keller, Evelyn Fox. 2000. *The Century of the Gene*. Cambridge, MA: Harvard University Press.

Kojève, Alexandre. 1947. *Introduction à la lecture de Hegel*. Edited by Raymond Queneau. Paris: Gallimard.

Komatsu Yoshihiko. 1996. *Shi wa kyōmei suru: Nōshi, zōki ishoku no fukami e*. Tokyo: Keisō Shobō.

Kristeva, Julia. 1980. *Pouvoirs de l'horreur: Essai sur l'abjection*. Paris: Seuil.

Kurokawa Kishō. 1987. *Kyōsei no shisō*. Tokyo: Tokuma Shoten.

Lacan, Jacques. 1973. *Le Séminaire, livre XI: Les quatre concepts fondamentaux de la psychanalyse*. Edited by Jacques-Alain Miller. Paris: Seuil.

Landecker, Hannah. 2007. *Culturing Life: How Cells Became Technologies*. Cambridge, MA: Harvard University Press.

Latour, Bruno. 1991. *Nous n'avons jamais été modernes: Essai d'anthropologie symétrique*. Paris: La Découverte.

Latour, Bruno, and Steve Woolgar. 1979. *Laboratory Life: The Social Construction of Scientific Facts*. Beverly Hills, CA: Sage.

Leibniz, G. W. 1714. *Monadologie*. Written 1714; first published in German translation, 1720.

Lock, Margaret. 2002. *Twice Dead: Organ Transplants and the Reinvention of Death*. Berkeley: University of California Press.

Lukács, Georg. 1923. *Geschichte und Klassenbewußtsein*. Berlin: Malik.

Malm, Andreas. 2016. *Fossil Capital: The Rise of Steam Power and the Roots of Global Warming*. London: Verso.

Martin, Emily. 1991. "The Egg and the Sperm: How Science Has Constructed a Romance Based on Stereotypical Male–Female Roles." *Signs* 16 (3): 485–501.

Marx, Karl. 1867. *Das Kapital: Kritik der politischen Oekonomie*, vol. 1. Hamburg: Otto Meissner. English translation by Samuel Moore and Edward Aveling, *Capital*, London: Swan Sonnenschein, Lowrey, 1887.

Marx, Karl, and Friedrich Engels. 1848. *Manifest der Kommunistischen Partei*. London: Bildungs-Gesellschaft für Arbeiter.

Maturana, Humberto R., and Francisco J. Varela. 1980. *Autopoiesis and Cognition: The Realization of the Living*. Dordrecht: D. Reidel.

Mauss, Marcel. 1925. "Essai sur le don: Forme et raison de l'échange dans les sociétés archaïques." *L'Année sociologique*, n.s., 1: 30–186.

Morioka Masahiro. 1989. *Nōshi no hito*. Tokyo: Tōkyō Shoseki.

Müller-Lauter, Wolfgang. 1978. "Der Organismus als innerer Kampf: Der Einfluß von Wilhelm Roux auf Friedrich Nietzsche." *Nietzsche-Studien* 7: 189–223.

Nagel, Thomas. 1974. "What Is It Like to Be a Bat?" *Philosophical Review* 83 (4): 435–450.

Nancy, Jean-Luc. 2000. *L'Intrus*. Paris: Galilée.

Napier, Susan J. 1993. "Panic Sites: The Japanese Imagination of Disaster from *Godzilla* to *Akira*." *Journal of Japanese Studies* 19 (2): 327–351.

Nietzsche, Friedrich. 1883. *Also sprach Zarathustra: Ein Buch für Alle und Keinen*, parts 1–2. Chemnitz: Ernst Schmeitzner.

Nietzsche, Friedrich. 1886. *Jenseits von Gut und Böse*. Leipzig: C. G. Naumann.

Nietzsche, Friedrich. 1887. *Zur Genealogie der Moral*. Leipzig: C. G. Naumann.

Ohnuki-Tierney, Emiko. 1994. "Brain Death and Organ Transplantation: Cultural Bases of Medical Technology." *Current Anthropology* 35 (3): 233–254.

Ōsawa Masachi. 1996. *Kyokō no jidai no hate: Ōmu to sekai saishū sensō*. Tokyo: Chikuma Shobō (Chikuma Shinsho).

Otis, Laura. 1994. *Organic Memory: History and the Body in the Late Nineteenth and Early Twentieth Centuries*. Lincoln: University of Nebraska Press.

Pagels, Elaine. 1988. *Adam, Eve, and the Serpent*. New York: Random House.

Perron, Bernard, ed. 2009. *Horror Video Games: Essays on the Fusion of Fear and Play*. Jefferson, NC: McFarland.

Perron, Bernard. 2018. *The World of Scary Video Games: A Study in Videoludic Horror*. New York: Bloomsbury Academic.

Pyne, Stephen J. 2021. *The Pyrocene: How We Created an Age of Fire, and What Happens Next*. Oakland: University of California Press.

Rabinbach, Anson. 1990. *The Human Motor: Energy, Fatigue, and the Origins of Modernity*. New York: Basic Books.

Reynolds, Andrew S. 2018. *The Third Lens: Metaphor and the Creation of Modern Cell Biology*. Chicago: University of Chicago Press.

Rheinberger, Hans-Jörg. 1997. *Toward a History of Epistemic Things: Synthesizing Proteins in the Test Tube*. Stanford, CA: Stanford University Press.

Saitō Kōhei. 2017. *Karl Marx's Ecosocialism: Capital, Nature, and the Unfinished Critique of Political Economy*. New York: Monthly Review Press.

Saitō Kōhei. 2020. *Hitoshinsei no "Shihonron"*. Tokyo: Shūeisha (Shūeisha Shinsho).

Saitō Kōhei. 2023. *Marx in the Anthropocene: Towards the Idea of Degrowth Communism*. Cambridge: Cambridge University Press.

Sapp, Jan. 1987. *Beyond the Gene: Cytoplasmic Inheritance and the Struggle for Authority in Genetics*. New York: Oxford University Press.

Sapp, Jan. 1994. *Evolution by Association: A History of Symbiosis*. New York: Oxford University Press.

Sapp, Jan, Francisco Carrapiço, and Mikhail Zolotonosov. 2002. "Symbiogenesis: The Hidden Face of Constantin Merezhkowsky." *History and Philosophy of the Life Sciences* 24 (3–4): 413–440.

Scheler, Max. 1912. "Über Ressentiment und moralisches Werturteil." *Zeitschrift für Pathopsychologie* 1.

Schmitt, Carl. 1922. *Politische Theologie: Vier Kapitel zur Lehre von der Souveränität*. Munich and Leipzig: Duncker & Humblot.

Serres, Michel. 1980. *Le Parasite*. Paris: Grasset.

Skloot, Rebecca. 2010. *The Immortal Life of Henrietta Lacks*. New York: Crown.

Spivak, Gayatri Chakravorty. 1988. "Can the Subaltern Speak?" In *Marxism and the Interpretation of Culture*, edited by Cary Nelson and Lawrence Grossberg, 271–313. Urbana: University of Illinois Press.

Steinberg, Marc. 2012. *Anime's Media Mix: Franchising Toys and Characters in Japan*. Minneapolis: University of Minnesota Press.

Suvin, Darko. 1979. *Metamorphoses of Science Fiction: On the Poetics and History of a Literary Genre*. New Haven, CT: Yale University Press.

Tachibana Takashi. 1986. *Nōshi*. Tokyo: Chūō Kōronsha.

Tauber, Alfred I. 1994. *The Immune Self: Theory or Metaphor?* Cambridge: Cambridge University Press.

Thompson, Evan. 2007. *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. Cambridge, MA: Harvard University Press.

Torok, Maria. 1968. "Maladie du deuil et fantasme du cadavre exquis." *Revue française de psychanalyse* 32 (4). Reprinted in Abraham and Torok 1978.

Weindling, Paul. 1981. "Theories of the Cell State in Imperial Germany." In *Biology, Medicine and Society 1840–1940*, edited by Charles Webster. Cambridge: Cambridge University Press.

Wiener, Norbert. 1948. *Cybernetics: Or Control and Communication in the Animal and the Machine*. Cambridge, MA: Technology Press; New York: Wiley.

Legal and Official Documents

Japan. 1948. *Yūsei hogo hō* [Eugenic Protection Law]. Law No. 156 of 1948. Revised and renamed *Botai hogo hō* [Maternal Body Protection Law], 1996.

Japan. 1979. *Kakumaku oyobi jinzō no ishoku ni kansuru hōritsu* [Law concerning the Transplantation of Corneas and Kidneys].

Japan. 1997. *Zōki no ishoku ni kansuru hōritsu* [Organ Transplant Law]. Law No. 104 of 1997; amended 2009.

Japan, Rinji Nōshi oyobi Zōki Ishoku Chōsakai [Provisional Commission on Brain Death and Organ Transplantation]. 1992. Final report. Tokyo.

Japan, Saikō Saibansho [Supreme Court], Grand Bench. 2024. Judgment of 3 July 2024 concerning the Eugenic Protection Law.

Moore v. Regents of the University of California. 1990. 51 Cal. 3d 120.

Nikkeiren [Nihon Keieisha Dantai Renmei]. 1995. *Shinjidai no "Nihonteki keiei": Chōsen subeki hōkō to sono gutaisaku*. Tokyo: Nikkeiren.

United Kingdom. 2015. *The Human Fertilisation and Embryology (Mitochondrial Donation) Regulations 2015*. SI 2015/572.

Keywords

Corpus and figures

Hideaki Sena Takashi Tokita *Parasite Eve* (novel, 1995) *Parasite Eve* (video game, 1998) Aya Brea Mitochondrial Eve Square Kadokawa media mix Japan Horror Novel Award

Life sciences and medicine

endosymbiosis symbiogenesis serial endosymbiosis theory eukaryogenesis Asgard archaea mitochondria mitochondrial DNA maternal inheritance paternal mitochondrial elimination nuclear mitochondrial DNA segments (numts) intragenomic conflict cytoplasmic male sterility apoptosis cytochrome c chemiosmosis uncoupling thermogenesis organic memory brain death organ transplantation immune privilege immunosuppression immortal cell lines mitochondrial replacement therapy intercellular mitochondrial transfer holobiont

Philosophy, theory, and history

master–slave dialectic recognition primitive accumulation *Stoffwechsel* metabolic rift reification ressentiment will to power general economy the accursed share sovereignty state of exception biopolitics bare life immunity and autoimmunity hospitality the parasite combination problem autopoiesis introjection and incorporation crypt phantom hauntology melancholia mourning abjection the monstrous-feminine lamella Gnosticism eugenics Anthropocene Pyrocene degrowth ecomodernism

Literary, media, and cultural studies

palaeonovum cognitive estrangement biohorror J-horror survival horror laboratory fiction spontaneous combustion procedural rhetoric ludic metabolism Active Time Battle media mix xenograft melting pot *kyōsei* *tabunka kyōsei* *ie* and *koseki* *karōshi* post-history otaku culture Japanese apocalyptic imagination

Concepts introduced in this dissertation

cryptobiosis **cryptobiology** **cryptobiont** **fractal crypt** **informational sovereignty** **thermodynamic sovereignty** **mixed constitution of the cell** **endosymbiotic melancholia** **matrilineal hauntology** **crypt-less self** **secession problem** **symbiotic mourning** **symbiocracy** **metabolic justice** **hospitable crypt**